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(12) **United States Patent**
Hennessy et al.(10) **Patent No.:** **US 12,398,101 B2**(45) **Date of Patent:** **Aug. 26, 2025**(54) **HERBICIDAL 2-AZASPIRO[3-5]NONANE COMPOUNDS**(71) Applicant: **SYNGENTA CROP PROTECTION AG**, Basel (CH)(72) Inventors: **Alan Joseph Hennessy**, Bracknell (GB); **Elizabeth Pearl Jones**, Bracknell (GB); **Suzanna Dale**, Bracknell (GB); **Alexander William Gregory**, Bracknell (GB); **Ian Thomas Tinmouth Houlshby**, Bracknell (GB); **Yunas Bhonoah**, Bracknell (GB); **Julia Comas-Barcelo**, Bracknell (GB); **Philip Michael Elves**, Bracknell (GB)(73) Assignee: **SYNGENTA CROP PROTECTION AG**, Basel (CH)

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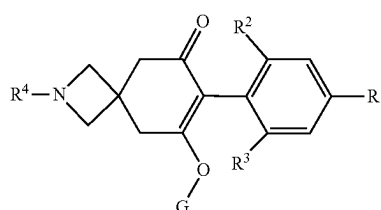
(51) **Int. Cl.****C07D 401/10** (2006.01)**A01N 25/32** (2006.01)**A01N 43/40** (2006.01)**A01N 43/44** (2006.01)**A01N 43/54** (2006.01)**A01N 43/56** (2006.01)**A01N 43/58** (2006.01)**A01N 43/78** (2006.01)**A01N 43/80** (2006.01)**A01P 13/00** (2006.01)**C07D 205/12** (2006.01)**C07D 401/12** (2006.01)**C07D 401/14** (2006.01)**C07D 403/10** (2006.01)**C07D 403/12** (2006.01)**C07D 405/12** (2006.01)**C07D 413/12** (2006.01)**C07D 417/12** (2006.01)(52) **U.S. Cl.**CPC **C07D 205/12** (2013.01); **A01N 25/32** (2013.01); **A01N 43/40** (2013.01); **A01N 43/44** (2013.01); **A01N 43/54** (2013.01); **A01N 43/56** (2013.01); **A01N 43/58** (2013.01); **A01N 43/78** (2013.01); **A01N 43/80** (2013.01); **A01P 13/00** (2021.08); **C07D 401/10** (2013.01); **C07D 401/12** (2013.01); **C07D 401/14** (2013.01); **C07D 403/10** (2013.01); **C07D 403/12** (2013.01); **C07D 405/12** (2013.01); **C07D 413/12** (2013.01); **C07D 417/12** (2013.01)(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited****FOREIGN PATENT DOCUMENTS**NO 2014096289 A2 6/2014
WO 2015197468 A1 12/2015**OTHER PUBLICATIONS**

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Primary Examiner — Alton N Pryor(74) *Attorney, Agent, or Firm* — BakerHostetler;
Toni-Junell Herbert(57) **ABSTRACT**The present invention relates to compounds of Formula (I), wherein R¹, R², R³, R⁴ and G are as defined herein. The invention further relates to herbicidal compositions which comprise a compound of Formula (I), to their use for controlling weeds, in particular in crops of useful plants.

(I)

14 Claims, No Drawings

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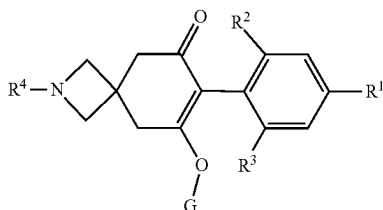
HERBICIDAL 2-AZASPIRO[3-5]NONANE
COMPOUNDSCROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a 371 National Stage application of International Application No. PCT/EP2019/077185 filed Oct. 8, 2019 which claims priority to GB 1816459.0, filed Oct. 9, 2018, the entire contents of these applications are hereby incorporated by reference.

The present invention relates to novel herbicidal cyclohexanedione compounds, to processes for their preparation, to herbicidal compositions which comprise the novel compounds, and to their use for controlling weeds.

Herbicidal cyclic dione compounds substituted by a phenyl which has an alkynyl-containing substituent are disclosed in, for example, WO2014/096289 and WO2015/197468. The present invention relates to novel herbicidal cyclohexanedione derivatives with improved properties.

Thus, according to the present invention there is provided a compound of Formula (I)



wherein

R¹ is selected from the group consisting of methyl, ethynyl, 1-propynyl, phenyl and a 5 or 6 membered heteroaryl which comprises one or two nitrogen heteroatoms, said phenyl and heteroaryl optionally substituted by one or two R¹⁵ substituents;

R² is selected from the group consisting of methyl, ethyl, methoxy and chloro;

R³ is selected from the group consisting of methyl, ethyl, methoxy and chloro;

R⁴ is selected from the group consisting of C₁-C₄alkyl, C₁-C₄alkoxy-, C₁-C₄haloalkyl, —C(O)C₁-C₄alkyl, —C(O)C₁-C₄haloalkyl, —S(O)_nC₁-C₆alkyl, —S(O)_nC₁-C₆haloalkyl, —S(O)_n—(CH₂)_n—C₃-C₆cycloalkyl, —S(O)_nC(R¹¹)R¹²R¹³, —C(O)H, —C(O)—(CH₂)_n—C₃-C₆cycloalkyl, —C(O)C(R¹¹)R¹²R¹³, —C(O)C₂-C₆alkenyl, —C(O)(CR⁹R¹⁰)CN, —C(O)(CR⁹R¹⁰)(CR⁹R¹⁰)CN, —C(O)CH₂C(O)—C₁-C₆alkyl, —C(O)CH₂OC(O)—C₁-C₆alkyl, —C(O)OC₁-C₆alkyl, —C(O)OC₁-C₆haloalkyl, —C(O)(CH₂)_nS(O)_nC₁-C₆alkyl, —C(O)C₁-C₃alkoxyC₁-C₆alkyl, —C(O)C₁-C₃alkoxyC₂-C₆alkenyl, —C(O)C₁-C₃alkoxyC₂-C₆haloalkyl, —C(O)C₁-C₃alkoxyC₃-C₆cycloalkyl, —C(O)OC₁-C₃alkoxyC₁-C₆alkyl, —C(O)C₁-C₃alkoxyC₁-C₆alkoxyC₁-C₆alkyl, —C(O)(CH₂)_nNR⁵R⁶, —C(O)—(CH₂)_n—NR⁷C(O)R⁸, —C(O)—(CH₂)_n—O—N=CR⁵R⁵, —CN, —S(O)₂NR¹⁶R¹⁷, —S(O)(=NR¹⁸)R¹⁹, —C(O)C(O)R²⁰, —C(O)C(R²³)=N—O—R²⁴, —C(O)C(R²³)=N—NR²⁵R²⁶, —(CH₂)_n-phenyl, —C(O)—(CH₂)_n-phenyl, —S(O)_n—(CH₂)_n-phenyl, —heterocyclyl, —C(O)—(CH₂)_n-heterocyclyl,

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—S(O)_n—(CH₂)_n-heterocyclyl, wherein each heterocyclyl is a 5- or 6-membered heterocyclyl which may be aromatic, saturated or partially saturated and can contain from 1 to 4 heteroatoms each independently selected from the group consisting of oxygen, nitrogen and sulphur, and wherein said heterocyclyl or phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R⁵ is selected from the group consisting of hydrogen and C₁-C₆alkyl;

R⁶ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₂-C₆alkenyl, C₂-C₆alkynyl, C₁-C₆haloalkyl, hydroxyl-, C₁-C₆alkoxy, C₃-C₆cycloalkyl, —C₁-C₄alkoxyC₁-C₆alkyl, —C₁-C₃alkoxyC₁-C₆haloalkyl, —(CR⁹R¹⁰)C₁-C₆haloalkyl, —(CR⁹R¹⁰)C(O)NR⁵R⁵, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro; or

R⁵ and R⁶ together form —CH₂CH₂OCH₂CH₂—; and

R⁷ is selected from the group consisting of hydrogen and C₁-C₆alkyl;

R⁸ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₁-C₆alkoxy, C₃-C₆cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R⁹ is hydrogen or methyl;

R¹⁰ is hydrogen or methyl; or

R⁹ and R¹⁰ together form —CH₂CH₂—; and

R¹¹ is hydrogen or methyl;

R¹² is selected from the group consisting of hydrogen, C₁-C₆alkyl, hydroxyl and C₁-C₆alkoxy;

R¹³ is selected from the group consisting of hydrogen, C₁-C₆alkyl, hydroxyl and C₁-C₆alkoxy; or

R¹² and R¹³ together form —CH₂—X—CH₂—; and

X is selected from the group consisting of O, S and N—R¹⁴;

R¹⁴ is selected from the group consisting of hydrogen, C₁-C₃alkyl and C₁-C₃alkoxy;

R¹⁵ is independently selected from the group consisting of C₁-C₄alkyl, C₁-C₄haloalkyl, cyano and halogen;

R¹⁶ is hydrogen or C₁-C₆alkyl; and

R¹⁷ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₃-C₆cycloalkyl, C₁-C₆alkoxy-C₁-C₃alkyl-, —C(O)C₁-C₆alkyl, —C(O)OC₁-C₆alkyl and CH₂CN; or

R¹⁶ and R¹⁷ together form —CH₂CH₂OCH₂CH₂—, —CH₂CH₂S(O)₂CH₂CH₂—;

R¹⁸ is hydrogen or C₁-C₆alkyl;

R¹⁹ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₁-C₆alkoxy, C₃-C₆cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R²⁰ is selected from the group consisting of C₁-C₆alkyl, C₁-C₆haloalkyl, C₁-C₆alkoxy-, C₁-C₆haloalkoxy, —NR²¹R²², phenyl and -pyridyl, wherein the phenyl

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and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro;

R²¹ is selected from the group consisting of hydrogen, C₁-C₆ alkyl, C₁-C₆ alkoxy, C₁-C₆alkoxyC₁-C₃alkyl-, C₃-C₆ cycloalkyl, C₁-C₆haloalkyl- and C₁-C₆haloalkoxy-, —C(O)C₁-C₆alkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro;

R²² is hydrogen or C₁-C₆alkyl; or

R²¹ and R²² together form —CH₂CH₂OCH₂CH₂—;

R²³ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₁-C₆haloalkyl, C₁-C₆alkoxy- and C₁-C₆haloalkoxy-;

R²⁴ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₁-C₆alkoxyC₁-C₃alkyl-, C₃-C₆cycloalkyl, —CH₂CN, tetrahydropyranyl-, phenyl and -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R²¹ is hydrogen or C₁-C₆ alkyl;

R²⁶ is hydrogen or C₁-C₆ alkyl;

G is selected from the group consisting of hydrogen, —(CH₂)_n—R^a, —C(O)—R^a, —C(O)—(CR^aR^a)_n—O—R^b, —C(O)NR^aR^a, —S(O)₂—R^a and C₁-C₈alkoxy-C₁-C₃alkyl-;

R^a is independently selected from the group consisting of hydrogen, C₁-C₈alkyl, C₁-C₃haloalkyl, C₂-C₈alkenyl, C₂-C₈alkynyl, C₃-C₆cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R^b is selected from the group consisting of C₁-C₈alkyl, C₁-C₃haloalkyl, C₂-C₈alkenyl, C₂-C₈alkynyl, C₃-C₆cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R^c is hydrogen or C₁-C₃ alkyl;

R^d is hydrogen or C₁-C₃ alkyl; and

n is independently 0, 1 or 2;

or an agriculturally acceptable salt thereof.

Alkyl groups (e.g C₁-C₆alkyl) include, for example, methyl (Me, CH₃), ethyl (Et, C₂H₅), n-propyl (n-Pr), isopropyl (i-Pr), n-butyl (n-Bu), isobutyl (i-Bu), sec-butyl (s-Bu) and tert-butyl (t-Bu).

Alkenyl and alkynyl moieties can be in the form of straight or branched chains, and the alkenyl moieties, where appropriate, can be of either the (E)- or (Z)-configuration. Examples are vinyl, allyl and propargyl. Alkenyl and alkynyl moieties can contain one or more double and/or triple bonds in any combination.

Halogen (or halo) encompasses fluorine, chlorine, bromine or iodine. The same correspondingly applies to halogen in the context of other definitions, such as haloalkyl.

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Haloalkyl groups (e.g C₁-C₄haloalkyl) are, for example, fluoromethyl, difluoromethyl, trifluoromethyl, chloromethyl, dichloromethyl, trichloromethyl, 2,2,2-trifluoroethyl, 2-fluoroethyl, 2-chloroethyl, pentafluoroethyl, 1,1-difluoro-2,2,2-trichloroethyl, 2,2,3,3-tetrafluoroethyl and 2,2,2-trichloroethyl.

Alkoxy groups (e.g C₁-C₄alkoxy-) are, for example, methoxy, ethoxy, propoxy, isopropoxy, n-butoxy, isobutoxy, sec-butoxy or tert-butoxy, preferably methoxy and ethoxy.

Alkoxyalkyl groups (e.g C₁-C₆alkoxy-C₁-C₃alkyl-) includes, for example, methoxymethyl, methoxyethyl, ethoxymethyl, ethoxyethyl, n-propoxymethyl, n-propoxyethyl, isopropoxymethyl or isopropoxyethyl.

Cycloalkyl groups (e.g C₃-C₆cycloalkyl-) include, for example cyclopropyl (c-propyl, c-Pr), cyclobutyl (c-butyl, c-Bu), cyclopentyl (c-pentyl) and cyclohexyl (c-hexyl) and may be substituted or unsubstituted as indicated.

C₁-C₆alkyl-S— (alkylthio) is, for example, methylthio, ethylthio, propylthio, isopropylthio, n-butylthio, isobutylthio, sec-butylthio or tert-butylthio, preferably methylthio or ethylthio.

C₁-C₆alkyl-S(O)— (alkylsulfinyl) is, for example, methylsulfinyl, ethylsulfinyl, propylsulfinyl, isopropylsulfinyl, n-butylsulfinyl, isobutylsulfinyl, sec-butylsulfinyl or tert-butylsulfinyl, preferably methylsulfinyl or ethylsulfinyl.

C₁-C₆alkyl-S(O)₂— (alkylsulfonyl) is, for example, methylsulfonyl, ethylsulfonyl, propylsulfonyl, isopropylsulfonyl, n-butylsulfonyl, isobutylsulfonyl, sec-butylsulfonyl or tert-butylsulfonyl, preferably methylsulfonyl or ethylsulfonyl.

Heterocyclyl is a 5- or 6-membered heterocyclyl which may be aromatic, saturated or partially saturated and can contain from 1 to 4 heteroatoms each independently selected from the group consisting of oxygen, nitrogen and sulphur.

The invention also relates agriculturally acceptable salts of the compounds of Formula (I). Such salts include those which are able to form with amines, alkali metal and alkaline earth metal bases or quaternary ammonium bases. Among the alkali metal and alkaline earth metal hydroxides as salt formers, special mention should be made of the hydroxides of lithium, sodium, potassium, magnesium and calcium, but especially the hydroxides of sodium and potassium. The compounds of Formula (I) according to the invention also include hydrates which may be formed during the salt formation.

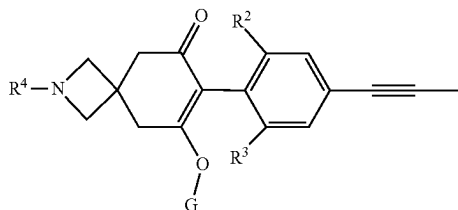
Examples of amines suitable for ammonium salt formation include ammonia as well as primary, secondary and tertiary C₁-C₁₈alkylamines, C₁-C₄hydroxyalkylamines and C₂-C₄alkoxyalkylamines, for example methylamine, ethylamine, n-propylamine, isopropylamine, the four butylamine isomers, n-ethylamine, isoethylamine, hexylamine, heptylamine, octylamine, nonylamine, decylamine, pentadecylamine, hexadecylamine, heptadecylamine, octadecylamine, methylethylamine, methylisopropylamine, methylhexylamine, methylnonylamine, methylpentadecylamine, methyloctadecylamine, ethylbutylamine, ethylheptylamine, ethyloctylamine, hexylheptylamine, hexyloctylamine, dimethylamine, diethylamine, di-n-propylamine, diisopropylamine, di-n-butylamine, di-n-ethylamine, diisopropylamine, dihexylamine, diheptylamine, dioctylamine, ethanolamine, n-propanolamine, isopropanolamine, N,N-diethanolamine, N-ethylpropanolamine, N-butylethanolamine, allylamine, n-but-2-enylamine, n-pent-2-enylamine, 2,3-dimethylbut-2-enylamine, dibut-2-enylamine, n-hex-2-enylamine, propylenediamine, trimethylamine, triethylamine, tri-n-propylamine, triisopropylamine, tri-n-butylamine, triisobutylamine, tri-sec-butylamine, tri-n-am-

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ylamine, methoxyethylamine and ethoxyethylamine; heterocyclic amines, for example pyridine, quinoline, isoquinoline, morpholine, piperidine, pyrrolidine, indoline, quinclidine and azepine; primary arylamines, for example anilines, methoxyanilines, ethoxyanilines, o-, m- and p-toluidines, phenylene-diamines, benzidines, naphthylamines and o-, m- and p-chloroanilines; but especially triethylamine, isopropylamine and diisopropylamine.

In one embodiment of the present invention R^1 is selected from the group consisting 1-propynyl, phenyl, pyridyl, pyrimidinyl and pyridazinyl wherein the phenyl, pyridyl, pyrimidinyl and pyridazinyl are optionally substituted by one or two R^{15} substituents each independently selected from the group consisting of C_1 - C_4 alkyl, C_1 - C_4 haloalkyl, cyano and halogen (preferably fluoro or chloro). In a preferred embodiment R^1 is 1-propynyl.

In one embodiment of the present invention, there is provided a compound of Formula (1a)



wherein

R^2 is methyl or methoxy;

R^3 is methyl or methoxy;

R^4 is selected from the group consisting of C_1 - C_4 alkyl, C_1 - C_4 alkoxy-, C_1 - C_4 haloalkyl, $-C(O)C_1$ - C_4 alkyl, $-C(O)C_1$ - C_4 haloalkyl, $-S(O)_n C_1$ - C_6 alkyl, $-S(O)_n C_1$ - C_6 haloalkyl, $-S(O)_n (CH_2)_n - C_3$ - C_6 cycloalkyl, $-S(O)_n C(R^{11})R^{12}R^{13}$, $-C(O)H$, $-C(O)-(CH_2)_n - C_3$ - C_6 cycloalkyl, $-C(O)C(R^{11})R^{12}R^{13}$, $-C(O)C_2$ - C_4 alkenyl, $-C(O)(CR^9R^{10})CN$, $-C(O)OC_1$ - C_6 alkyl, $-C(O)OC_1$ - C_6 haloalkyl, $-C(O)(CH_2)_n S(O)_n C_1$ - C_6 alkyl, $-C(O)C_1$ - C_3 alkoxy C_1 - C_6 alkyl, $-C(O)NR^5R^6$, $-C(O)-(CH_2)_n - NR^7C(O)R^8$, $-CN$, $-(CH_2)_n$ -phenyl, $-C(O)-(CH_2)_n$ -phenyl, $-S(O)_n -(CH_2)_n$ -phenyl, -heterocyclyl, $-C(O)-(CH_2)_n$ -heterocyclyl, $-S(O)_n -(CH_2)_n$ -heterocyclyl, wherein each heterocyclyl is a 5- or 6-membered heterocyclyl which may be aromatic, saturated or partially saturated and can contain from 1 to 4 heteroatoms each independently selected from the group consisting of oxygen, nitrogen and sulphur, and wherein said heterocyclyl or phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C_1 - C_3 alkyl, C_1 - C_3 haloalkyl, C_1 - C_3 alkoxy, C_2 - C_3 alkenyl, C_2 - C_3 alkynyl, halogen, cyano and nitro;

R^5 is selected from the group consisting of hydrogen and C_1 - C_6 alkyl;

R^6 is selected from the group consisting of hydrogen, C_1 - C_6 alkyl, C_1 - C_6 alkoxy, C_3 - C_6 cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C_1 - C_3 alkyl, C_1 - C_3 haloalkyl, C_1 - C_3 alkoxy, C_2 - C_3 alkenyl, C_2 - C_3 alkynyl, halogen, cyano and nitro; or

R^5 and R^6 together form $-CH_2CH_2OCH_2CH_2-$; and

R^7 is selected from the group consisting of hydrogen and C_1 - C_6 alkyl;

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R^8 is selected from the group consisting of hydrogen, C_1 - C alkyl, C_1 - C alkoxy, C_3 - C_6 cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C_1 - C_3 alkyl, C_1 - C_3 haloalkyl, C_1 - C_3 alkoxy, C_2 - C_3 alkenyl, C_2 - C_3 alkynyl, halogen, cyano and nitro;

R^9 is hydrogen or methyl;

R^{10} is hydrogen or methyl; or

R^9 and R^{10} together form $-CH_2CH_2-$; and

R^{11} is hydrogen or methyl;

R^{12} and R^{13} together form $-CH_2-X-CH_2-$;

X is selected from the group consisting of O, S and N- R^{14} ;

R^{14} is selected from the group consisting of hydrogen, C_1 - C_3 alkyl and C_1 - C_3 alkoxy-;

G is selected from the group consisting of hydrogen, $-(CH_2)_n-R^a$, $-C(O)-R^a$, $-C(O)-(CR^cR^d)_n-O-R^b$, $-C(O)NR^aR^a$, $-S(O)_2-R^a$ and C_1 - C_8 alkoxy- C_1 - C_3 alkyl-;

R^a is independently selected from the group consisting of hydrogen, C_1 - C_8 alkyl, C_1 - C_3 haloalkyl, C_2 - C_8 alkenyl, C_2 - C_8 alkynyl, C_3 - C_6 cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C_1 - C_3 alkyl, C_1 - C_3 haloalkyl, C_1 - C_3 alkoxy, C_2 - C_3 alkenyl, C_2 - C_3 alkynyl, halogen, cyano and nitro;

R^b is selected from the group consisting of C_1 - C_8 alkyl, C_1 - C_3 haloalkyl, C_2 - C_8 alkenyl, C_2 - C_8 alkynyl, C_3 - C_6 cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C_1 - C_3 alkyl, C_1 - C_3 haloalkyl, C_1 - C_3 alkoxy, C_2 - C_3 alkenyl, C_2 - C_3 alkynyl, halogen, cyano and nitro;

R^c is hydrogen or C_1 - C_3 alkyl;

R^d is hydrogen or C_1 - C_3 alkyl; and

n is independently 0, 1 or 2;

or an agriculturally acceptable salt thereof.

In one embodiment of the present invention R^2 is methyl.

In one embodiment of the present invention R^3 is methyl.

In another embodiment of the present invention R^3 is methoxy.

In one embodiment of the present invention R^2 is methyl and R^3 is methyl.

In one embodiment of the present invention R^2 is methyl and R^3 is methoxy.

In one embodiment of the present invention R^2 is methoxy and R^3 is methoxy.

In one embodiment of the present invention, R^4 is C_1 - C_2 alkoxy- (e.g methoxy or ethoxy).

In another embodiment of the present invention R^4 is $-C(O)C_1$ - C_3 alkyl (e.g $-C(O)$ methyl, $-C(O)$ ethyl, $-C(O)$ i-propyl) or $-C(O)$ t-butyl).

In another embodiment of the present invention, R^4 is $-C(O)C_1$ - C_3 haloalkyl, more preferably $-C(O)C_1$ - C_2 fluoroalkyl e.g $-C(O)CH_2F$, $-C(O)CHF_2$, $-C(O)CF_3$.

In one embodiment of the present invention, R^4 is $-S(O)_n C_1$ - C_6 alkyl especially $-S(O)_2$ methyl or $-S(O)_2$ ethyl

In another embodiment R^4 is $-S(O)_n C_1$ - C_6 haloalkyl, for example $-S(O)_2$ chloromethyl.

In another embodiment R^4 is $-S(O)_n -(CH_2)_n - C_3$ - C_6 cycloalkyl, for example $-S(O)_2 -(CH_2)_n$ -c-propyl or, where n is 0, $-C_3$ - C_6 cycloalkyl (for example cyclopropyl).

In another embodiment of the present invention, R^4 is $-C(O)OC_1-C_6$ alkyl, especially $-C(O)-O$ -methyl.

In another embodiment of the present invention, R^4 is $-S(O)_n C(R^{11})R^{12}R^{13}$ or $-C(O)C(R^{11})R^{12}R^{13}$ wherein R^{11} is hydrogen or methyl and $R^{12}R^{13}$ taken together are $-CH_2OCH_2-$ (oxetan-3-yl).

In another embodiment of the present invention, R^4 is $-C(O)-(CH_2)_n-C_3-C_6$ cycloalkyl, for example $-C(O)$ -c-propyl or $-C(O)-(CH_2)$ -c-propyl.

In another embodiment of the present invention, R^4 is $-C(O)(CR^9R^{10})CN$, for example $-C(O)CH_2CN$, $-C(O)CH(CH_3)CN$ or $-C(O)C(CH_3)_2CN$.

In another embodiment of the present invention, R^4 is $-C(O)(CH_2)_n S(O)_n C_1-C_6$ alkyl, for example $-C(O)CH_2S(O)_2$ methyl.

In another embodiment of the present invention, R^4 is $-C(O)C_1-C_3$ alkoxy C_1-C_6 alkyl, for example $-C(O)CH_2-O-CH_3$, $-C(O)CH_2CH_2-O-CH_3$ or $-C(O)CH(CH_3)-O-CH_3$.

In another embodiment of the present invention, R^4 is $-C(O)NR^5R^6$, especially wherein R^5 is hydrogen and R^6 is C_1-C_6 alkyl e.g t-butyl.

In another embodiment of the present invention, R^4 is $-C(O)-(CH_2)_n-NR^7C(O)R^8$, for example $-C(O)-(CH_2)_n-NR^7C(O)R^8$ or $-C(O)NR^7C(O)R^8$, for example $-C(O)NHC(O)$ -t-butyl or $-C(O)NHC(O)$ pyrid-2-yl.

In another embodiment of the present invention, R^4 is selected from the group consisting of -phenyl, $-C(O)$ -phenyl, $-S(O)_n$ phenyl wherein each phenyl is optionally substituted as defined previously.

In another embodiment of the present invention R^4 is heterocyclyl, $-C(O)$ - heterocyclyl or $-S(O)_n$ -heterocyclyl. In another embodiment, each aforementioned heterocyclyl is an aromatic heterocyclyl (i.e heteroaryl), more preferably selected from the group consisting of furanyl, pyrrolyl, thiophenyl, imidazolyl, pyrazolyl, oxazolyl, isoxazolyl, thiazolyl, pyranyl, pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl, and triazolyl more preferably selected from the group consisting of pyridyl, pyridazinyl, pyrimidinyl and pyrazinyl each of which is optionally substituted as defined previously. In another embodiment, each aforementioned heterocyclyl is a partially saturated heterocyclyl, more preferably selected from the group consisting of imidazolyl, isoxazolyl and thiazolyl each of which is optionally substituted as defined previously. In another embodiment, each aforementioned heterocyclyl is a saturated heterocyclyl more preferably selected from the group consisting of morpholinyl, tetrahydrofuryl and tetrahydropyranyl each of which is optionally substituted as defined previously.

In one embodiment of the present invention, G is selected from the group consisting of hydrogen, C_1-C_6 alkyl (e.g methyl, ethyl, n-propyl, i-propyl, n-butyl, t-butyl, $-C_2-C_8$ alkenyl (e.g vinyl), C_2-C_8 alkynyl (e.g propargyl), $-C(O)C_1-C_6$ alkyl (more preferably $-C(O)C_1-C_6$ alkyl e.g $-C(O)$ i-propyl and $-C(O)$ -t-butyl) and $-C(O)-O-C_1-C_6$ alkyl (more preferably $-C(O)-O-C_1-C_6$ alkyl e.g $-C(O)-O$ -methyl). In a preferred embodiment, G is hydrogen.

Depending on the nature of the substituents, compounds of Formula (I) may exist in different isomeric forms. When G is hydrogen, for example, compounds of Formula (I) may exist in different tautomeric forms.

This invention covers all such isomers and tautomers and mixtures thereof in all proportions. Also, when substituents contain double bonds, cis- and trans-isomers can exist. These isomers, too, are within the scope of the claimed compounds of the Formula (I). Compounds of Formula (I) may contain asymmetric centres and may be present as a

single enantiomer, pairs of enantiomers in any proportion or, where more than one asymmetric centre are present, contain diastereoisomers in all possible ratios. Typically one of the enantiomers has enhanced biological activity compared to the other possibilities.

The compounds of Formula (I) according to the invention can be used as herbicides by themselves, but they are generally formulated into herbicidal compositions using formulation adjuvants, such as carriers, solvents and surface-active agents (SFAs). Thus, the present invention further provides a herbicidal composition comprising a herbicidal compound according to any one of the previous claims and an agriculturally acceptable formulation adjuvant. The composition can be in the form of concentrates which are diluted prior to use, although ready-to-use compositions can also be made. The final dilution is usually made with water, but can be made instead of, or in addition to, water, with, for example, liquid fertilisers, micronutrients, biological organisms, oil or solvents.

The herbicidal compositions generally comprise from 0.1 to 99% by weight, especially from 0.1 to 95% by weight, compounds of Formula (I) and from 1 to 99.9% by weight of a formulation adjuvant which preferably includes from 0 to 25% by weight of a surface-active substance.

The compositions can be chosen from a number of formulation types, many of which are known from the Manual on Development and Use of FAO Specifications for Plant Protection Products, 5th Edition, 1999. These include dustable powders (DP), soluble powders (SP), water soluble granules (SG), water dispersible granules (WG), wettable powders (WP), granules (GR) (slow or fast release), soluble concentrates (SL), oil miscible liquids (OL), ultra low volume liquids (UL), emulsifiable concentrates (EC), dispersible concentrates (DC), emulsions (both oil in water (EW) and water in oil (EO)), micro-emulsions (ME), suspension concentrates (SC), aerosols, capsule suspensions (CS) and seed treatment formulations. The formulation type chosen in any instance will depend upon the particular purpose envisaged and the physical, chemical and biological properties of the compound of Formula (I).

Dustable powders (DP) may be prepared by mixing a compound of Formula (I) with one or more solid diluents (for example natural clays, kaolin, pyrophyllite, bentonite, alumina, montmorillonite, kieselguhr, chalk, diatomaceous earths, calcium phosphates, calcium and magnesium carbonates, sulphur, lime, flours, talc and other organic and inorganic solid carriers) and mechanically grinding the mixture to a fine powder.

Soluble powders (SP) may be prepared by mixing a compound of Formula (I) with one or more water-soluble inorganic salts (such as sodium bicarbonate, sodium carbonate or magnesium sulphate) or one or more water-soluble organic solids (such as a polysaccharide) and, optionally, one or more wetting agents, one or more dispersing agents or a mixture of said agents to improve water dispersibility/solubility. The mixture is then ground to a fine powder. Similar compositions may also be granulated to form water soluble granules (SG).

Wettable powders (WP) may be prepared by mixing a compound of Formula (I) with one or more solid diluents or carriers, one or more wetting agents and, preferably, one or more dispersing agents and, optionally, one or more suspending agents to facilitate the dispersion in liquids. The mixture is then ground to a fine powder. Similar compositions may also be granulated to form water dispersible granules (WG).

Granules (GR) may be formed either by granulating a mixture of a compound of Formula (I) and one or more powdered solid diluents or carriers, or from pre-formed blank granules by absorbing a compound of Formula (I) (or a solution thereof, in a suitable agent) in a porous granular material (such as pumice, attapulgite clays, fuller's earth, kieselguhr, diatomaceous earths or ground corn cobs) or by adsorbing a compound of Formula (I) (or a solution thereof, in a suitable agent) on to a hard core material (such as sands, silicates, mineral carbonates, sulphates or phosphates) and drying if necessary. Agents which are commonly used to aid absorption or adsorption include solvents (such as aliphatic and aromatic petroleum solvents, alcohols, ethers, ketones and esters) and sticking agents (such as polyvinyl acetates, polyvinyl alcohols, dextrans, sugars and vegetable oils). One or more other additives may also be included in granules (for example an emulsifying agent, wetting agent or dispersing agent).

Dispersible Concentrates (DC) may be prepared by dissolving a compound of Formula (I) in water or an organic solvent, such as a ketone, alcohol or glycol ether. These solutions may contain a surface active agent (for example to improve water dilution or prevent crystallisation in a spray tank).

Emulsifiable concentrates (EC) or oil-in-water emulsions (EW) may be prepared by dissolving a compound of Formula (I) in an organic solvent (optionally containing one or more wetting agents, one or more emulsifying agents or a mixture of said agents). Suitable organic solvents for use in ECs include aromatic hydrocarbons (such as alkylbenzenes or alkyl-naphthalenes, exemplified by SOLVESSO 100, SOLVESSO 150 and SOLVESSO 200; SOLVESSO is a Registered Trade Mark), ketones (such as cyclohexanone or methylcyclohexanone) and alcohols (such as benzyl alcohol, furfuryl alcohol or butanol), N-alkylpyrrolidones (such as N-methylpyrrolidone or N-octylpyrrolidone), dimethyl amides of fatty acids (such as C₈-C₁₀ fatty acid dimethylamide) and chlorinated hydrocarbons. An EC product may spontaneously emulsify on addition to water, to produce an emulsion with sufficient stability to allow spray application through appropriate equipment.

Preparation of an EW involves obtaining a compound of Formula (I) either as a liquid (if it is not a liquid at room temperature, it may be melted at a reasonable temperature, typically below 70° C.) or in solution (by dissolving it in an appropriate solvent) and then emulsifying the resultant liquid or solution into water containing one or more SFAs, under high shear, to produce an emulsion. Suitable solvents for use in EWs include vegetable oils, chlorinated hydrocarbons (such as chlorobenzenes), aromatic solvents (such as alkylbenzenes or alkyl-naphthalenes) and other appropriate organic solvents which have a low solubility in water.

Microemulsions (ME) may be prepared by mixing water with a blend of one or more solvents with one or more SFAs, to produce spontaneously a thermodynamically stable isotropic liquid formulation. A compound of Formula (I) is present initially in either the water or the solvent/SFA blend. Suitable solvents for use in MEs include those hereinbefore described for use in ECs or in EWs. An ME may be either an oil-in-water or a water-in-oil system (which system is present may be determined by conductivity measurements) and may be suitable for mixing water-soluble and oil-soluble pesticides in the same formulation. An ME is suitable for dilution into water, either remaining as a microemulsion or forming a conventional oil-in-water emulsion.

Suspension concentrates (SC) may comprise aqueous or non-aqueous suspensions of finely divided insoluble solid

particles of a compound of Formula (I). SCs may be prepared by ball or bead milling the solid compound of Formula (I) in a suitable medium, optionally with one or more dispersing agents, to produce a fine particle suspension of the compound. One or more wetting agents may be included in the composition and a suspending agent may be included to reduce the rate at which the particles settle. Alternatively, a compound of Formula (I) may be dry milled and added to water, containing agents hereinbefore described, to produce the desired end product.

Aerosol formulations comprise a compound of Formula (I) and a suitable propellant (for example n-butane). A compound of Formula (I) may also be dissolved or dispersed in a suitable medium (for example water or a water miscible liquid, such as n-propanol) to provide compositions for use in non-pressurised, hand-actuated spray pumps.

Capsule suspensions (CS) may be prepared in a manner similar to the preparation of EW formulations but with an additional polymerisation stage such that an aqueous dispersion of oil droplets is obtained, in which each oil droplet is encapsulated by a polymeric shell and contains a compound of Formula (I) and, optionally, a carrier or diluent therefor. The polymeric shell may be produced by either an interfacial polycondensation reaction or by a coacervation procedure. The compositions may provide for controlled release of the compound of Formula (I) and they may be used for seed treatment. A compound of Formula (I) may also be formulated in a biodegradable polymeric matrix to provide a slow, controlled release of the compound.

The composition may include one or more additives to improve the biological performance of the composition, for example by improving wetting, retention or distribution on surfaces; resistance to rain on treated surfaces; or uptake or mobility of a compound of Formula (I). Such additives include surface active agents (SFAs), spray additives based on oils, for example certain mineral oils or natural plant oils (such as soy bean and rape seed oil), and blends of these with other bio-enhancing adjuvants (ingredients which may aid or modify the action of a compound of Formula (I)).

Wetting agents, dispersing agents and emulsifying agents may be SFAs of the cationic, anionic, amphoteric or non-ionic type.

Suitable SFAs of the cationic type include quaternary ammonium compounds (for example cetyltrimethyl ammonium bromide), imidazolines and amine salts.

Suitable anionic SFAs include alkali metals salts of fatty acids, salts of aliphatic monoesters of sulphuric acid (for example sodium lauryl sulphate), salts of sulphonated aromatic compounds (for example sodium dodecylbenzenesulphonate, calcium dodecylbenzenesulphonate, butyl-naphthalene sulphonate and mixtures of sodium di-isopropyl- and tri-isopropyl-naphthalene sulphonates), ether sulphates, alcohol ether sulphates (for example sodium laureth-3-sulphate), ether carboxylates (for example sodium laureth-3-carboxylate), phosphate esters (products from the reaction between one or more fatty alcohols and phosphoric acid (predominately mono-esters) or phosphorus pentoxide (predominately di-esters), for example the reaction between lauryl alcohol and tetraphosphoric acid; additionally these products may be ethoxylated), sulposuccinamates, paraffin or olefine sulphonates, taurates and lignosulphonates.

Suitable SFAs of the amphoteric type include betaines, propionates and glycines.

Suitable SFAs of the non-ionic type include condensation products of alkylene oxides, such as ethylene oxide, propylene oxide, butylene oxide or mixtures thereof, with fatty alcohols (such as oleyl alcohol or cetyl alcohol) or with

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alkylphenols (such as octylphenol, nonylphenol or octyl-cresol); partial esters derived from long chain fatty acids or hexitol anhydrides; condensation products of said partial esters with ethylene oxide; block polymers (comprising ethylene oxide and propylene oxide); alkanolamides; simple esters (for example fatty acid polyethylene glycol esters); amine oxides (for example lauryl dimethyl amine oxide); and lecithins.

Suitable suspending agents include hydrophilic colloids (such as polysaccharides, polyvinylpyrrolidone or sodium carboxymethylcellulose) and swelling clays (such as bentonite or attapulgite).

The herbicidal compounds of present invention can also be used in mixture with one or more additional herbicides and/or plant growth regulators. Examples of such additional herbicides or plant growth regulators include acetochlor, acifluorfen (including acifluorfen-sodium), aclonifen, ametryn, amicarbazone, aminopyralid, aminotriazole, atrazine, bensulfuron (including bensulfuron-methyl), bentazone, bicyclopyrone, bilanafos, bispyribac-sodium, bixlozone, bromacil, bromoxynil, butachlor, butafenacil, carfentrazone (including carfentrazone-ethyl), cloransulam (including cloransulam-methyl), chlorimuron (including chlorimuron-ethyl), chlorotoluron, chlorsulfuron, cinmethylin, clacyfos, clethodim, clodinafop (including clodinafop-propargyl), clomazone, clopyralid, cyclopyranil, cyclopyrimorate, cyclosulfamuron, cyhalofop (including cyhalofop-butyl), 2,4-D (including the choline salt and 2-ethylhexyl ester thereof), 2,4-DB, desmedipham, dicamba (including the aluminium, aminopropyl, bis-aminopropylmethyl, choline, dichloroprop, diglycolamine, dimethylamine, dimethylammonium, potassium and sodium salts thereof) diclosulam, diflufenican, diflufenzopyr, dimethachlor, dimethenamid-P, diquat dibromide, diuron, ethalfuralin, ethofumesate, fenoxaprop (including fenoxaprop-P-ethyl), fenoxasulfone, fenquinotrine, fentrazamide, flazasulfuron, florasulam, florypyrauxifen (including florypyrauxifen-benzyl), fluazifop (including fluazifop-P-butyl), flucarbazone (including flucarbazone-sodium), flufenacet, flumetsulam, flumioxazin, flupyralsulfuron (including flupyralsulfuron-methyl-sodium), fluroxypyr (including fluroxypyr-meptyl), fomesafen, foramsulfuron, glufosinate (including the ammonium salt thereof), glyphosate (including the diammonium, isopropylammonium and potassium salts thereof), halauxifen (including halauxifen-methyl), haloxyfop (including haloxyfop-methyl), hexazinone, hydantocidin, imazamox, imazapic, imazapyr, imazethapyr, indaziflam, iodosulfuron (including iodosulfuron-methyl-sodium), iofensulfuron (including iofensulfuron-sodium), ioxynil, isoproturon, isoxaflutole, lancotrione, MCPA, MCPB, mecoprop-P, mesosulfuron (including mesosulfuron-methyl), mesotrione, metamitron, metazachlor, methiozolin, metolachlor, metosulam, metribuzin, metsulfuron, napropamide, nicosulfuron, norflurazon, oxadiazon, oxasulfuron, oxyfluorfen, paraquat dichloride, pendimethalin, penoxsulam, phenmedipham, picloram, pinoxaden, pretilachlor, primisulfuron-methyl, propanil, propaquizafop, propyrisulfuron, propyzamide, prosulfocarb, prosulfuron, pyraclonil, pyraflufen (including pyraflufen-ethyl), pyrasulfotole, pyridate, pyriftalid, pyrimisulfan, pyroxasulfone, pyroxulam, quinclorac, quinmerac, quizalofop (including quizalofop-P-ethyl and quizalofop-P-tefuryl), rimsulfuron, saflufenacil, sethoxydim, simazine, S-metolachlor, sulfentrazone, sulfosulfuron, tebuthiuron, terbutyltrione, tembotrione, terbutylazine, terbutryn, thiencarbazone, thifensulfuron, tiafenacil, tolpyralate, topramezone, tralkoxydim, triafamone, triallate, triasulfuron, tribenuron (including tribenuron-methyl), tric-

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lopyr, trifloxysulfuron (including trifloxysulfuron-sodium), trifludimoxazin, trifluralin, triflusaluron, 4-hydroxy-1-methoxy-5-methyl-3-[4-(trifluoromethyl)-2-pyridyl]imidazolidin-2-one, 4-hydroxy-1,5-dimethyl-3-[4-(trifluoromethyl)-2-pyridyl]imidazolidin-2-one, 5-ethoxy-4-hydroxy-1-methyl-3-[4-(trifluoromethyl)-2-pyridyl]imidazolidin-2-one, 4-hydroxy-1-methyl-3-[4-(trifluoromethyl)-2-pyridyl]imidazolidin-2-one, 4-hydroxy-1,5-dimethyl-3-[1-methyl-5-(trifluoromethyl)pyrazol-3-yl]imidazolidin-2-one, (4R)-1-(5-tert-butylisoxazol-3-yl)-4-ethoxy-5-hydroxy-3-methylimidazolidin-2-one, 3-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]bicyclo[3.2.1]octane-2,4-dione, 2-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]-5-methyl-cyclohexane-1,3-dione, 2-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]cyclohexane-1,3-dione, 2-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]-5,5-dimethyl-cyclohexane-1,3-dione, 6-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]-2,2,4,4-tetramethyl-cyclohexane-1,3,5-trione, 2-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]-5-ethyl-cyclohexane-1,3-dione, 2-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]-4,4,6,6-tetramethyl-cyclohexane-1,3-dione, 2-[6-cyclopropyl-2-(3,4-dimethoxyphenyl)-3-oxo-pyridazine-4-carbonyl]-5-methyl-cyclohexane-1,3-dione, 3-[6-cyclopropyl-2-(3,4-dimethoxyphenyl)-3-oxo-pyridazine-4-carbonyl]bicyclo[3.2.1]octane-2,4-dione, 2-[6-cyclopropyl-2-(3,4-dimethoxyphenyl)-3-oxo-pyridazine-4-carbonyl]-5,5-dimethyl-cyclohexane-1,3-dione, 6-[6-cyclopropyl-2-(3,4-dimethoxyphenyl)-3-oxo-pyridazine-4-carbonyl]-2,2,4,4-tetramethyl-cyclohexane-1,3,5-trione, 2-[6-cyclopropyl-2-(3,4-dimethoxyphenyl)-3-oxo-pyridazine-4-carbonyl]cyclohexane-1,3-dione, 4-[2-(3,4-dimethoxyphenyl)-6-methyl-3-oxo-pyridazine-4-carbonyl]-2,2,6,6-tetramethyl-tetrahydropyran-3,5-dione and 4-[6-cyclopropyl-2-(3,4-dimethoxyphenyl)-3-oxo-pyridazine-4-carbonyl]-2,2,6,6-tetramethyl-tetrahydropyran-3,5-dione.

The mixing partners of the compound of Formula (I) may also be in the form of esters or salts, as mentioned e.g. in The Pesticide Manual, Sixteenth Edition, British Crop Protection Council, 2012.

The compound of Formula (I) can also be used in mixtures with other agrochemicals such as fungicides, nematocides or insecticides, examples of which are given in The Pesticide Manual.

The mixing ratio of the compound of Formula (I) to the mixing partner is preferably from 1:100 to 1000:1.

The mixtures can advantageously be used in the above-mentioned formulations (in which case "active ingredient" relates to the respective mixture of compound of Formula (I) with the mixing partner).

The compounds or mixtures of the present invention can also be used in combination with one or more herbicide safeners. Examples of such safeners include benoxacor, cloquintocet (including cloquintocet-mexyl), cyprosulfamide, dichlorimide, fenclorazole (including fenclorazole-ethyl), fenclorim, fluxofenim, furilazole, isoxadifen (including isoxadifen-ethyl), mefenpyr (including mefenpyr-diethyl), metcamifen and oxabetrinil.

Particularly preferred are mixtures of a compound of Formula (I) with cyprosulfamide, isoxadifen-ethyl, cloquintocet-mexyl and/or N-(2-methoxybenzoyl)-4-[(methyl-aminocarbonyl)amino]benzenesulfonamide.

The safeners of the compound of Formula (I) may also be in the form of esters or salts, as mentioned e.g. in The Pesticide Manual, 16th Edition (BCPC), 2012. The reference

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to cloquintocet-mexyl also applies to a lithium, sodium, potassium, calcium, magnesium, aluminium, iron, ammonium, quaternary ammonium, sulfonium or phosphonium salt thereof as disclosed in WO 02/34048.

Preferably the mixing ratio of compound of Formula (I) to safener is from 100:1 to 1:10, especially from 20:1 to 1:1.

The mixtures can advantageously be used in the above-mentioned formulations (in which case "active ingredient" relates to the respective mixture of compound of Formula (I) with the safener).

The present invention still further provides a method of controlling weeds at a locus comprising crop plants and weeds, wherein the method comprises application to the locus of a weed controlling amount of a composition according to the present invention. 'Controlling' means killing, reducing or retarding growth or preventing or reducing germination. Generally the plants to be controlled are unwanted plants (weeds). 'Locus' means the area in which the plants are growing or will grow.

The rates of application of compounds of Formula (I) may vary within wide limits and depend on the nature of the soil, the method of application (pre- or post-emergence; seed dressing; application to the seed furrow; no tillage application etc.), the crop plant, the weed(s) to be controlled, the prevailing climatic conditions, and other factors governed by the method of application, the time of application and the target crop. The compounds of Formula (I) according to the invention are generally applied at a rate of from 10 to 2000 g/ha, especially from 50 to 1000 g/ha.

The application is generally made by spraying the composition, typically by tractor mounted sprayer for large areas, but other methods such as dusting (for powders), drip or drench can also be used.

Useful plants in which the composition according to the invention can be used include crops such as cereals, for example barley and wheat, cotton, oilseed rape, sunflower, maize, rice, soybeans, sugar beet, sugar cane and turf.

Crop plants can also include trees, such as fruit trees, palm trees, coconut trees or other nuts. Also included are vines such as grapes, fruit bushes, fruit plants and vegetables.

Crops are to be understood as also including those crops which have been rendered tolerant to herbicides or classes of herbicides (e.g. ALS-, GS-, EPSPS-, PPO-, ACCase- and HPPD-inhibitors) by conventional methods of breeding or by genetic engineering. An example of a crop that has been rendered tolerant to imidazolinones, e.g. imazamox, by conventional methods of breeding is Clearfield® summer rape (canola). Examples of crops that have been rendered tolerant to herbicides by genetic engineering methods include e.g. glyphosate- and glufosinate-resistant maize varieties commercially available under the trade names RoundupReady® and LibertyLink®.

Crops are also to be understood as being those which have been rendered resistant to harmful insects by genetic engineering methods, for example Bt maize (resistant to European corn borer), Bt cotton (resistant to cotton boll weevil) and also Bt potatoes (resistant to Colorado beetle). Examples of Bt maize are the Bt 176 maize hybrids of NK® (Syngenta Seeds). The Bt toxin is a protein that is formed naturally by *Bacillus thuringiensis* soil bacteria. Examples of toxins, or transgenic plants able to synthesise such toxins, are described in EP-A-451 878, EP-A-374 753, WO 93/07278, WO 95/34656, WO 03/052073 and EP-A-427 529. Examples of transgenic plants comprising one or more genes that code for an insecticidal resistance and express one or more toxins are KnockOut® (maize), Yield Gard® (maize), NuCOTIN33B® (cotton), Bollgard® (cotton),

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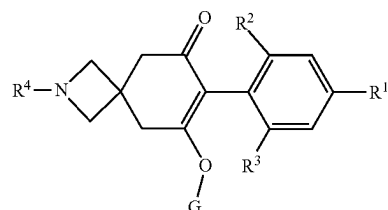
NewLeaf® (potatoes), NatureGard® and Protexcta®. Plant crops or seed material thereof can be both resistant to herbicides and, at the same time, resistant to insect feeding ("stacked" transgenic events). For example, seed can have the ability to express an insecticidal Cry3 protein while at the same time being tolerant to glyphosate.

Crops are also to be understood to include those which are obtained by conventional methods of breeding or genetic engineering and contain so-called output traits (e.g. improved storage stability, higher nutritional value and improved flavour).

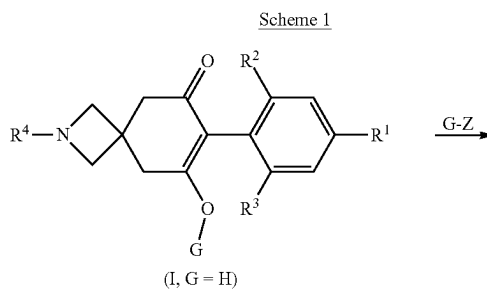
Other useful plants include turf grass for example in golf-courses, lawns, parks and roadsides, or grown commercially for sod, and ornamental plants such as flowers or bushes.

The compositions can be used to control unwanted plants (collectively, 'weeds'). The weeds to be controlled may be both monocotyledonous species, for example *Agrostis*, *Alopecurus*, *Avena*, *Brachiaria*, *Bromus*, *Cenchrus*, *Cyperus*, *Digitaria*, *Echinochloa*, *Eleusine*, *Lolium*, *Monochoria*, *Rottboellia*, *Sagittaria*, *Scirpus*, *Setaria* and *Sorghum*, and dicotyledonous species, for example *Abutilon*, *Amaranthus*, *Ambrosia*, *Chenopodium*, *Chrysanthemum*, *Conyza*, *Galium*, *Ipomoea*, *Nasturtium*, *Sida*, *Sinapis*, *Solanum*, *Stellaria*, *Veronica*, *Viola* and *Xanthium*. The compounds of the present invention have been shown to exhibit particularly good activity against certain grass weed species, especially *Lolium perenne*. Weeds can also include plants which may be considered crop plants but which are growing outside a crop area ('escapes'), or which grow from seed left over from a previous planting of a different crop ('volunteers'). Such volunteers or escapes may be tolerant to certain other herbicides.

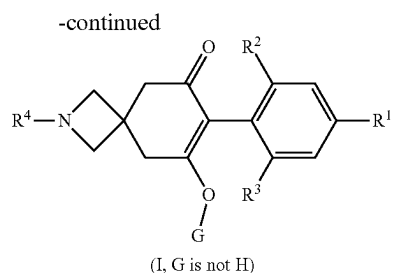
The compounds of the present invention can be prepared according to the following schemes.



Compounds of formula (I) wherein G is other than hydrogen may be prepared by treating a compound of formula (I) wherein G is hydrogen, with a reagent G-Z, wherein G-Z is an alkylating agent such as an alkyl halide, acylating agent such as an acid chloride or anhydride, sulfonylating agent such as a sulfonyl chloride, carbamylating agent such as a carbamoyl chloride, or carbonating agent such as a chloroformate, using known methods.



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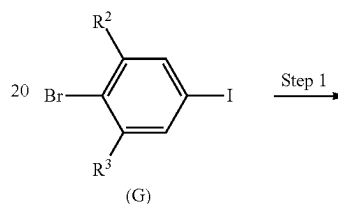


Compounds of formula (I) can be prepared via Pb coupling as shown in the scheme below by reacting a compound of formula (B), to form an organolead reagent of formula (C) and subsequent reaction with 1,3 dione (D) under conditions described, for example, by J. Pinhey, *Pure and Appl. Chem.*, (1996), 68 (4), 819 and by M. Moloney et al., *Tetrahedron Lett.*, (2002), 43, 3407 to give halide of formula (E). R¹ can then be added by a range of metal catalyzed cross-coupling reactions from intermediates of type (E) or (F) using standard literature procedures.

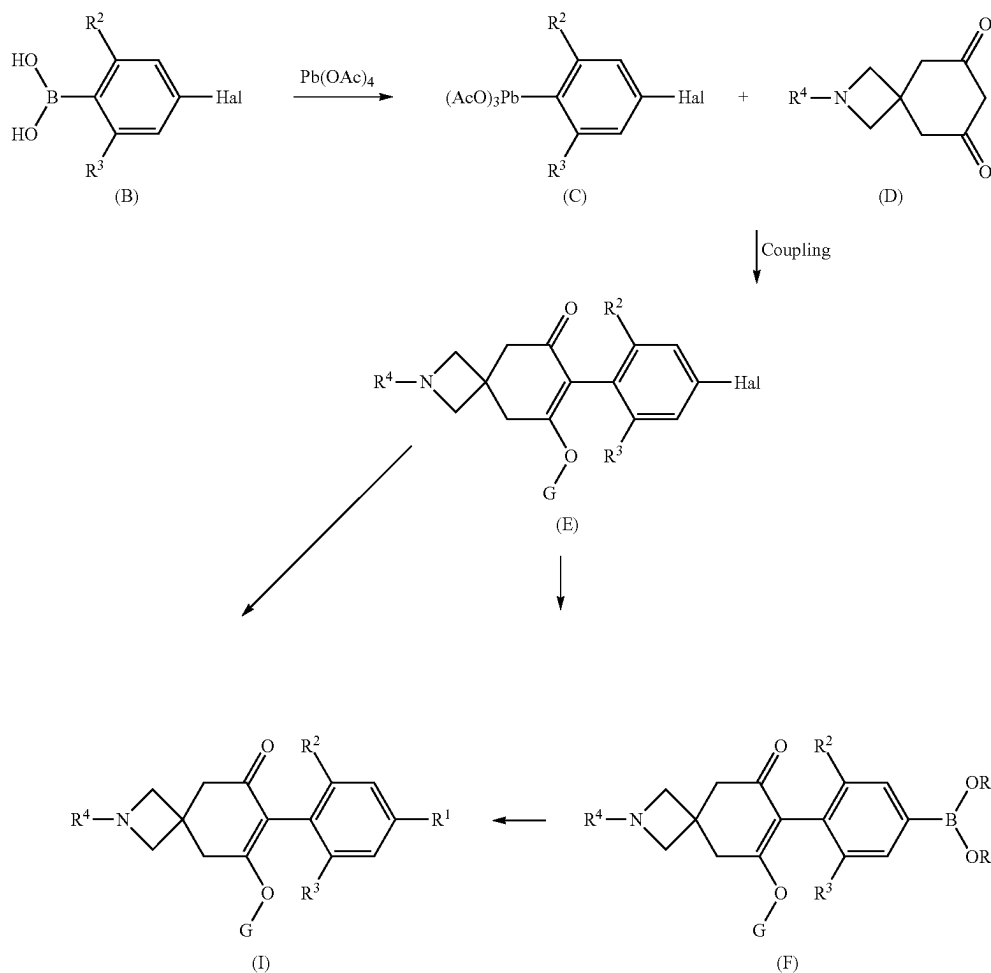
16

Boronic acids can be prepared by methods such as below in Scheme 3. For example, a compound of formula (B) may be prepared from an aryl halide of formula (A) by known methods. For example, an aryl halide of formula (A) may be treated with an alkyl lithium or alkyl magnesium halide in a suitable solvent, preferably diethyl ether or tetrahydrofuran, at a temperature of between -80° C. and 30° C., and the aryl magnesium or aryl lithium reagent obtained may then be reacted with a trialkyl borate (preferably trimethylborate) to give an aryl dialkylboronate which may be hydrolysed to provide a boronic acid of formula (B) under acidic conditions.

Scheme 3

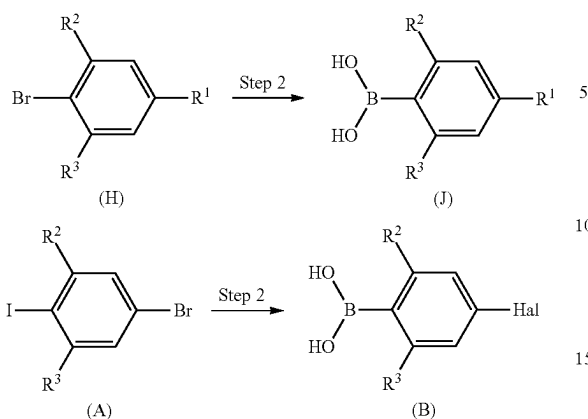


Scheme 2



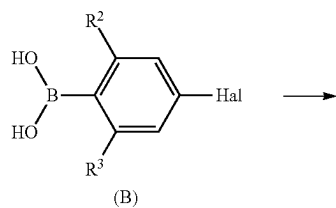
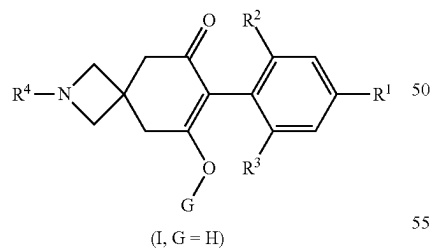
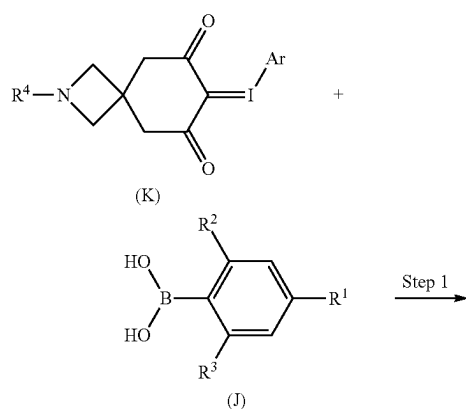
17

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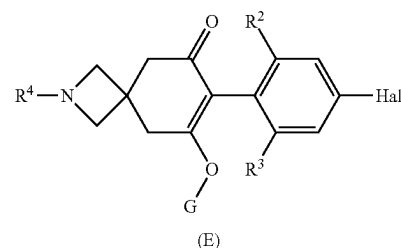
Intermediates of formula (I) or (E) may be prepared by reacting an iodonium ylide of formula (K), wherein Ar is an optionally substituted phenyl group, and an aryl boronic acid of formula (B) or (J), in the presence of a suitable palladium catalyst, a base and in a suitable solvent.

Scheme 4



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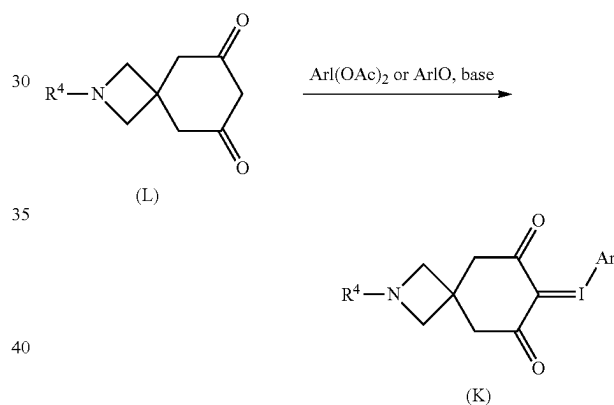
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Preferably the palladium catalyst is palladium acetate, the base is lithium hydroxide and the solvent is aqueous 1,2-dimethoxyethane.

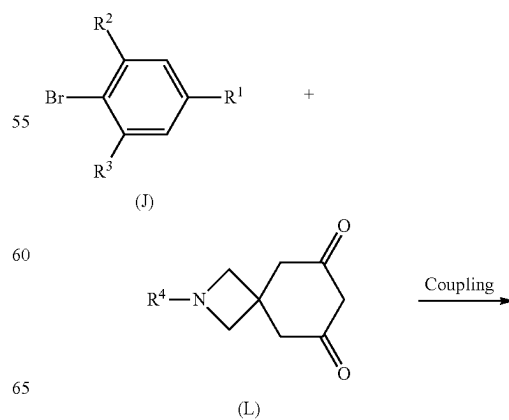
A compound of formula (K) may be prepared from a 1,3 dione compound of formula (L) by treatment with a hypervalent iodine reagent such as a (diacetoxy)iodobenzene or an iodosylbenzene and a base such as aqueous sodium carbonate, lithium hydroxide or sodium hydroxide in a solvent such as water or an aqueous alcohol such as aqueous ethanol using known procedures.

Scheme 5



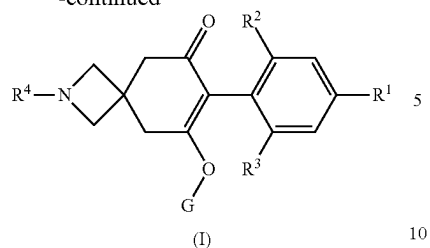
With suitable conditions, a suitable 1,3 dione may also be directly coupled to a Halo-compound (for example of formula (J)) with palladium catalysis).

Scheme 6



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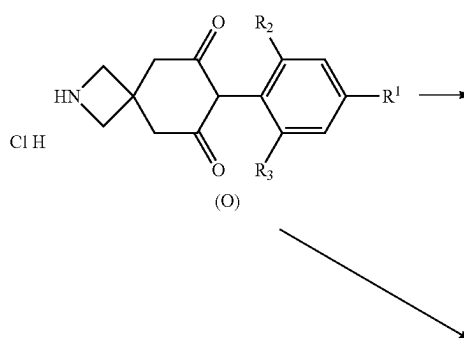
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**20**

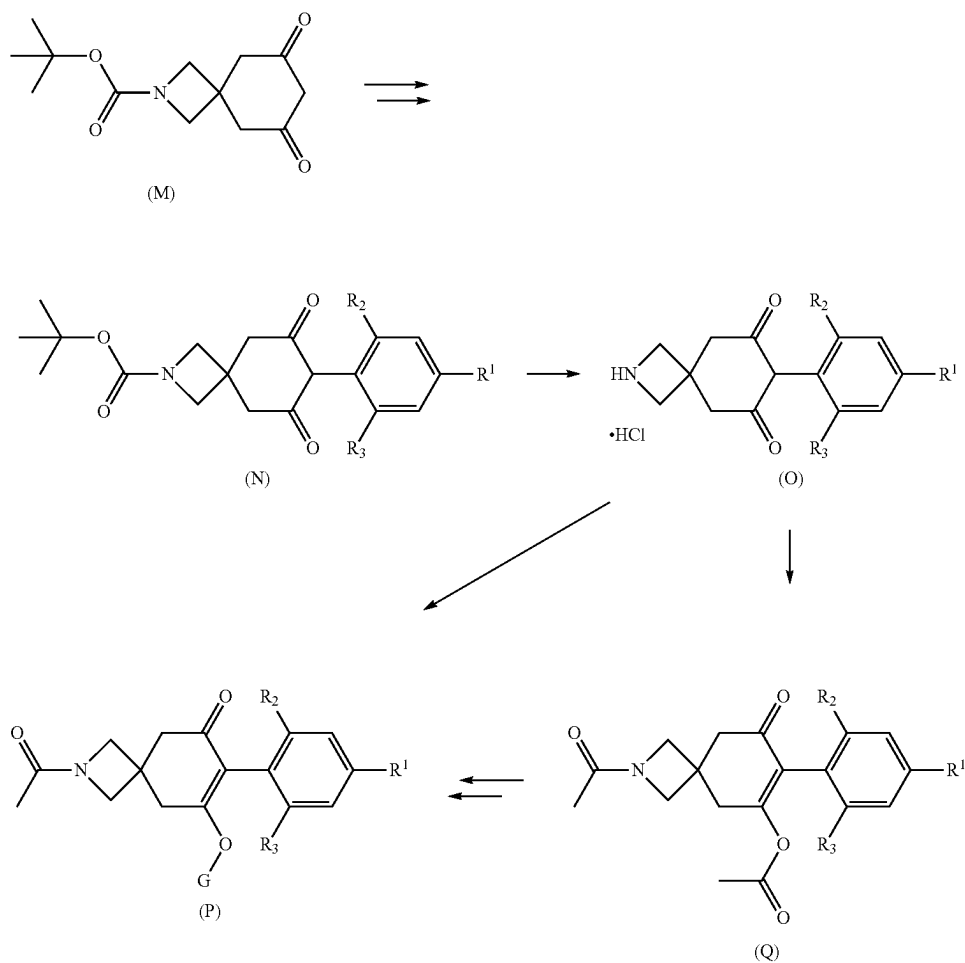
Alternatively, the sequence can be performed with sulfonylation rather than acylation to form sulfonamides such as (S).

Compounds of type (I) can also be made by late stage functionalisation with use of a suitable protecting group as shown in scheme 4 below. Compound (M) can be converted to intermediate (N) by the methods described above and then the protecting group (such as the BOC group shown) can be removed (under acidic conditions in this example). Intermediate (O) can then be directly converted to compounds (for example (P) by standard amide coupling conditions or doubly reacted on both oxygen and nitrogen atoms to give compounds of type (Q). Compounds of type (Q) can readily be converted to any compound of type (I)—for example the enol-ester of (Q) can be selectively hydrolysed to give (P, G=H), which can be then converted to (P, G is other than H) by the methods described earlier.

Scheme 8

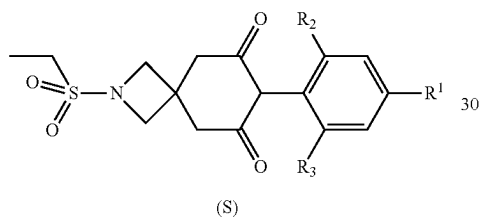
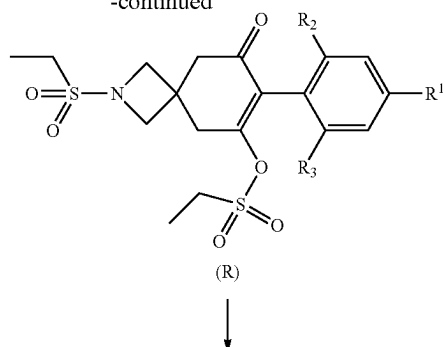


Scheme 7



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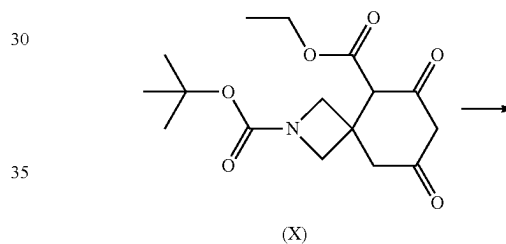
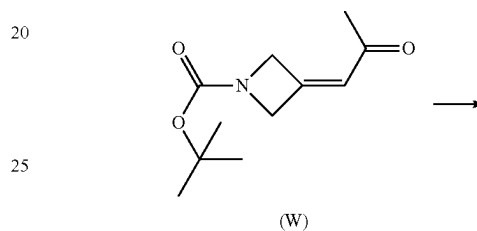
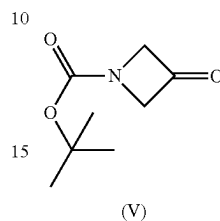


Ureas (T) and acyl ureas (U) can also be made from intermediate (O) by standard literature methods, for example as shown in scheme 9.

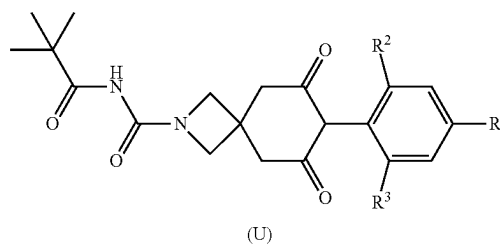
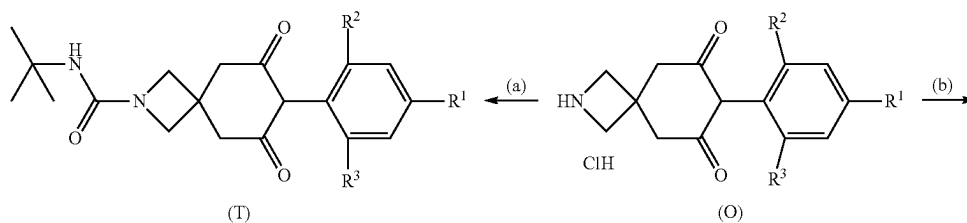
22

1,3 Diones such as these may be prepared using methods such as that shown below. So commercially available ketones (for example of type (V)) can be converted into intermediate (W) and then converted to intermediate (X) and finally decarboxylation gives intermediate (M) (these methods are described in WO2008/110308).

Scheme 7

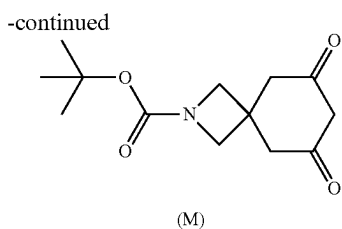


Scheme 9



(a) 2-isocyanato-2-methyl-propane, NEt_3 , DCM; (b) 2,2-dimethylpropanoyl isocyanate, NEt_3 , DCM

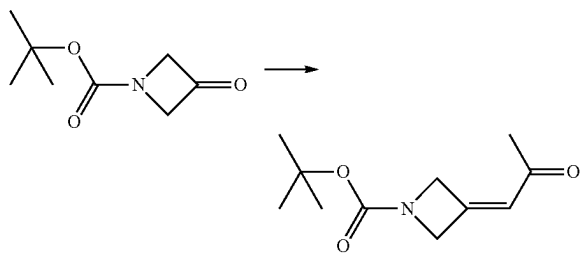
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The following non-limiting examples provide specific synthesis methods for representative compounds of the present invention, as referred to in Tables 1 & 2 below.

EXAMPLE 1: SYNTHESIS OF 2-ACETYL-7-(2,6-DIMETHYL-4-PROP-1-YNYL-PHENYL)-2-AZASPIRO[3.5]NONANE-6,8-DIONE (COMPOUND 1.001)

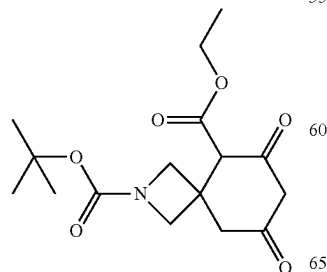
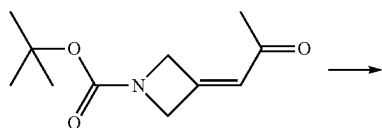
Step 1: Synthesis of tert-butyl 3-acetylidenesazetidine-1-carboxylate



tert-butyl 3-oxoazetidine-1-carboxylate (15.0 g, 87.6 mmol) and 1-(triphenyl-lambda5-phosphanylidene)propan-2-one (27.9 g, 87.6 mmol) were dissolved in anhydrous toluene (240 mL) and heated at 70° C. for 2 h. The reaction mixture was allowed to cool to RT, then concentrated in vacuo. The crude material was purified by flash column chromatography, eluting with 0-20% EtOAc in hexane to obtain tert-butyl 3-acetylidenesazetidine-1-carboxylate as a colourless oil (18.4 g, 99% yield).

¹H NMR (CDCl₃, 400 MHz): δ 6.14 (s, 1H), 4.85-4.82 (m, 2H), 4.59 (s, 2H), 2.19 (s, 3H), 1.45 (s, 9H).

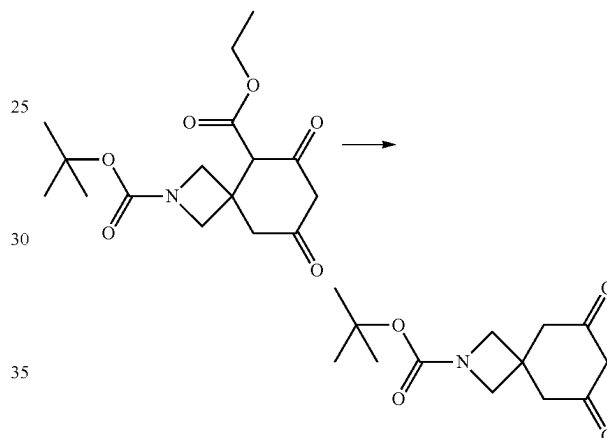
Step 2: Synthesis of O2-tert-butyl O5-ethyl 6,8-dioxo-2-azaspiro[3.5]nonane-2,5-dicarboxylate



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To a stirred solution of tert-butyl 3-acetylidenesazetidine-1-carboxylate (18.4 g, 87.1 mmol) in anhydrous ethanol (120 mL) was added diethyl propanedioate (13.3 mL, 87.1 mmol) at RT followed by drop wise addition of sodium ethoxide (21.0%, 42.3 mL, 113 mmol). The reaction mixture was stirred at RT for 3 h and then heated to reflux for 2 h. The reaction was allowed to cool to RT, and then concentrated in vacuo. The reaction mixture was diluted with water and washed with ethyl acetate. The aqueous layer was then acidified with 10% aq. citric acid and extracted with DCM (x3). The combined organics were washed with brine, dried over Na₂SO₄, filtered and concentrated in vacuo to obtain O2-tert-butyl O5-ethyl 6,8-dioxo-2-azaspiro[3.5]nonane-2,5-dicarboxylate as a brown gum which was used in the next step without further purification.

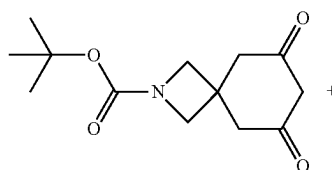
Step 3: Synthesis of tert-butyl 6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate



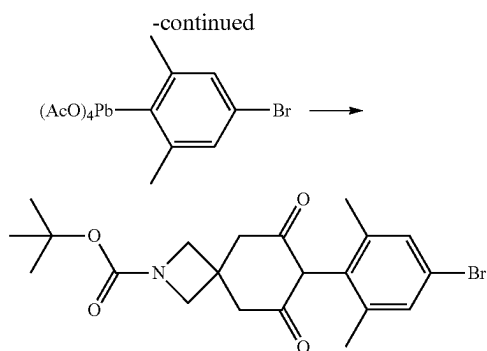
Crude O2-tert-butyl O5-ethyl 6,8-dioxo-2-azaspiro[3.5]nonane-2,5-dicarboxylate from step 2 (15.6 g, 47.9 mmol) was dissolved in ethanol (48 mL) and aqueous NaOH (12M, 84 mL) was added at RT. The reaction was stirred for 2 days at RT. Ethanol was removed in vacuo and the reaction mixture was acidified with 10% citric acid to pH-3 and saturated with brine. The mixture was then extracted with ethyl acetate (x3). The combined organics were dried over Na₂SO₄, filtered and concentrated in vacuo. The crude material was triturated with pentane and diethyl ether to give tert-butyl 6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate as a brown solid (6 g, 49% yield over 2 steps).

¹H NMR (400 MHz, MeOD) δ=3.67 (s, 4H), 2.63 (s, 4H), 1.43 (s, 9H).

Step 4: Synthesis of tert-butyl 7-(4-bromo-2,6-dimethyl-phenyl)-6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate



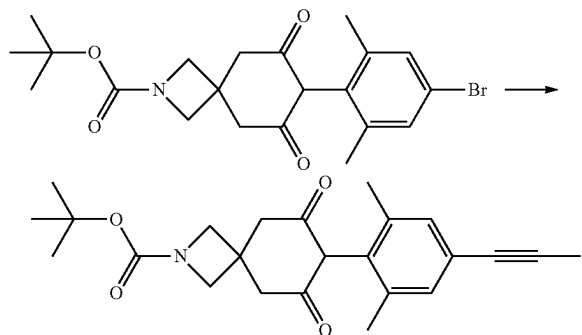
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tert-butyl 6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate (3.57 g, 14.1 mmol) and DMAP (10.7 g, 88.0 mmol) were dissolved in chloroform (80 mL) and stirred for 15 minutes under nitrogen before anhydrous toluene (50 mL) was added, followed by the addition of a solution of diacetoxy-(4-bromo-2,6-dimethyl-phenyl)plumblyl acetate (10.0 g, 17.6 mmol) in chloroform (80 mL). The reaction mixture was stirred at 80° C. for 3 h and then allowed to cool to RT. The reaction mixture was acidified with 10% citric acid solution, a white precipitate was formed, which was filtered and washed with chloroform. The layers of the filtrate were separated and the aqueous layer extracted with chloroform (x2). The combined organics were washed with brine, dried over Na₂SO₄, filtered and concentrated in vacuo. The crude material was purified by flash column chromatography, eluting with 10-50% EtOAc in hexane to obtain tert-butyl 7-(4-bromo-2,6-dimethyl-phenyl)-6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate as a white foam (3.3 g, 37% yield).

¹H NMR (MeOD, 400 MHz): δ 7.19 (s, 2H), 3.78 (s, 4H), 2.83 (s, 4H), 1.99 (s, 6H), 1.44 (s, 9H).

Step 5: Synthesis of tert-butyl 7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate



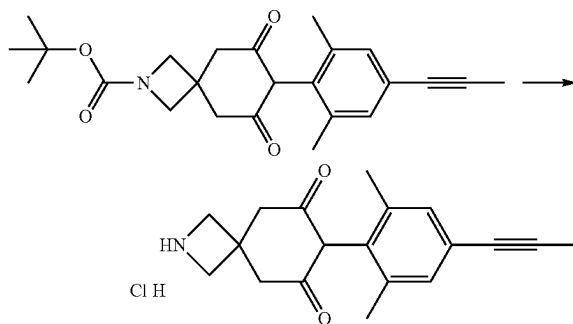
tert-butyl 7-(4-bromo-2,6-dimethyl-phenyl)-6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate (3.60 g, 8.25 mmol), 2-butyneic acid (2.08 g, 24.8 mmol) and 4-diphenylphosphanylbutyl(diphenyl)phosphane (0.703 g, 1.65 mmol) were taken up in anhydrous DMSO (50 mL) and de-gassed before adding bis(triphenylphosphine)palladium(II) chloride (0.579 g, 0.825 mmol) and 1,8-diazabicyclo[5.4.0]undec-7-ene (7.54 g, 49.5 mmol). The reaction was then heated at 120° C. for 20 h. The reaction was allowed to cool to RT and then acidified with 10% citric acid solution and extracted

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with EtOAc (x3). The combined organics were washed with brine solution, dried over Na₂SO₄, filtered and concentrated in vacuo. The crude material was purified by flash column chromatography, eluting with 10-80% EtOAc in Hexane to obtain tert-butyl 7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate (2.48 g, 68% yield).

¹H NMR (MeOD, 400 MHz): δ 7.02 (s, 2H), 3.77 (s, 4H), 2.82 (s, 4H), 2.00-1.96 (9H), 1.44 (s, 9H).

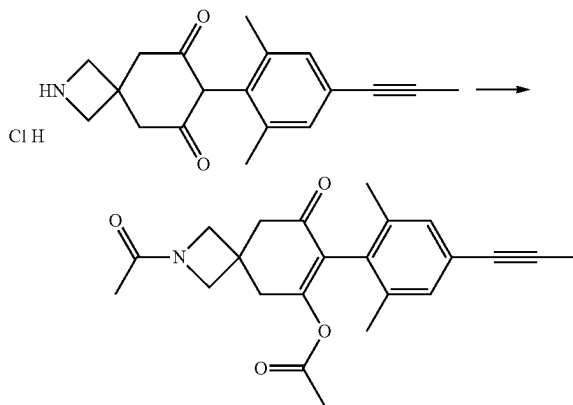
Step 6: Synthesis of 7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-2-azaspiro[3.5]nonane-6,8-dione; hydrochloride



tert-butyl 7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-8-hydroxy-6-oxo-2-azaspiro[3.5]non-7-ene-2-carboxylate (2.38 g, 6.02 mmol) was stirred for 2 h at RT in a solution of HCl in Dioxane (4.00 M, 48.0 mL, 192 mmol). The RM was concentrated in vacuo to leave a brown solid. This solid was washed with a 50% ethyl acetate/hexane mixture and then dried under reduced pressure to give 7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-2-azaspiro[3.5]nonane-6,8-dione; hydrochloride (1.98 g, quant. yield) as an off-white solid.

¹H NMR (MeOD, 400 MHz): δ 7.04 (s, 2H), 4.00 (s, 4H), 2.97 (s, 4H), 2.02-1.96 (9H).

Step 7: Synthesis of [2-acetyl-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-6-oxo-2-azaspiro[3.5]non-7-en-8-yl] acetate



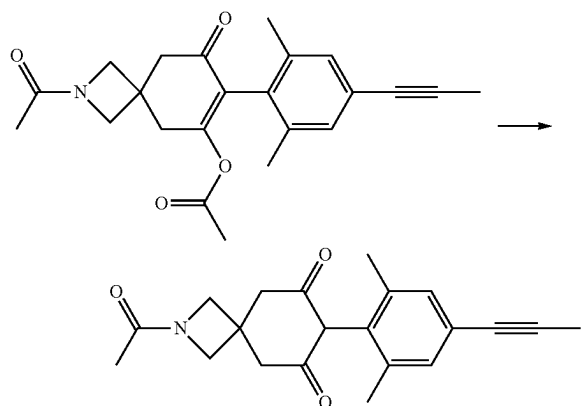
7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-2-azaspiro[3.5]nonane-6,8-dione; hydrochloride (342 mg, 1.03 mmol) was taken up in dichloromethane (20 mL) and acetyl chloride

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(0.324 g, 4.12 mmol) was added at 0° C. followed by triethylamine (1.15 mL, 8.25 mmol). The reaction was stirred at RT for 2 h before being diluted with DCM, washed with water and brine solution. The organic layer was dried over Na₂SO₄, filtered and concentrated in vacuo. The crude material was purified by flash column chromatography, eluting with 10% to 100% EtOAc in Hexane to obtain [2-acetyl-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-6-oxo-2-azaspiro[3.5]non-7-en-8-yl] acetate (237 mg, 61% yield) as an off white foam.

¹HNMR (MeOD, 400 MHz): δ 7.01 (s, 2H), 4.18 (d, 1H), 4.09 (d, 1H), 3.95 (d, 1H), 3.84 (d, 1H), 3.10 (s, 2H), 2.91 (s, 2H), 2.0-1.97 (9H), 1.89 (s, 6H).

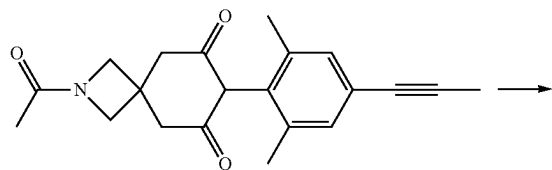
Step 8: Synthesis of 2-acetyl-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-2-azaspiro[3.5]nonane-6,8-dione (Compound 1.001)



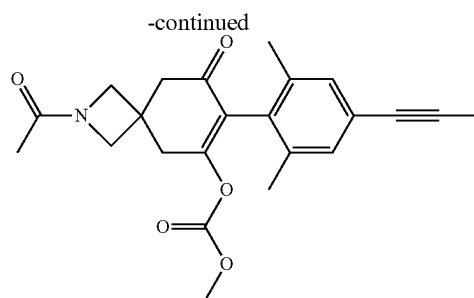
[2-acetyl-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-6-oxo-2-azaspiro[3.5]non-7-en-8-yl]acetate (237 mg, 0.625 mmol) was dissolved in methanol (7 mL) and to this was added K₂CO₃ (0.172 g, 1.25 mmol). The reaction mixture was stirred at room temperature for 1 h before being concentrated in vacuo. The residue was acidified with 10% citric acid solution and the solid that was formed, was filtered and dried to obtain 2-acetyl-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-2-azaspiro[3.5]nonane-6,8-dione (125 mg, 59% yield) as a white solid.

¹HNMR (MeOD, 400 MHz): δ 7.03 (s, 2H), 4.07 (s, 2H), 3.83 (s, 2H), 2.86 (s, 4H), 1.99-1.98 (9H), 1.89 (s, 3H).

EXAMPLE 2: SYNTHESIS OF [2-ACETYL-7-(2,6-DIMETHYL-4-PROP-1-YNYL-PHENYL)-6-OXO-2-AZASPIRO[3.5]NON-7-EN-8-YL] METHYL CARBONATE (COMPOUND 2.001)



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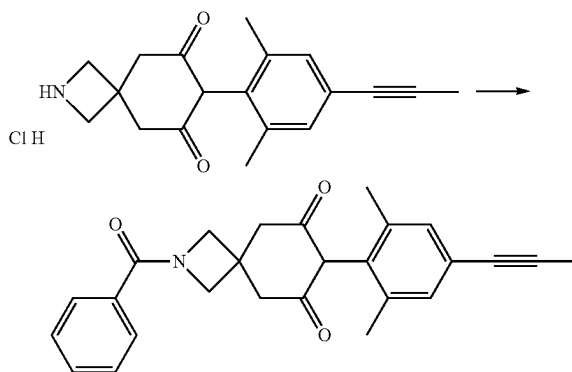


2-acetyl-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-2-

azaspiro[3.5]nonane-6,8-dione (89 mg, 0.237 mmol) was dissolved in anhydrous DCM (2.4 mL) then triethylamine (99.8 μL, 0.712 mmol) added with stirring. The reaction was cooled in a salt/ice bath to 0° C. and then methylchloroformate (44 μL, 0.564 mmol) was added slowly with stirring and cooling. The reaction was then stirred at RT for 60 mins. The reaction mixture was concentrated in vacuo, partitioned between water and DCM and then the organic phase passed through a phase separating cartridge before being concentrated in vacuo. The crude material was purified by reverse-phase flash column chromatography, eluting with 55-75% acetonitrile in water to obtain 2-acetyl-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-6-oxo-2-azaspiro[3.5]non-7-en-8-yl] methyl carbonate (32 mg, 34.% yield) as a colourless gum.

¹H NMR (400 MHz, chloroform) 6 ppm 1.90 (s, 3H) 1.98 (s, 6H) 2.03 (s, 3H) 2.87 (d, J=2.08 Hz, 2H) 3.07 (s, 2H) 3.71 (s, 3H) 3.87-3.99 (m, 2H) 4.01 (d, J=8.44 Hz, 2H) 7.08 (s, 2H).

EXAMPLE 3: SYNTHESIS OF 2-BENZOYL-7-(2,6-DIMETHYL-4-PROP-1-YNYL-PHENYL)-2-AZASPIRO[3.5]NONANE-6,8-DIONE (COMPOUND 1.003)



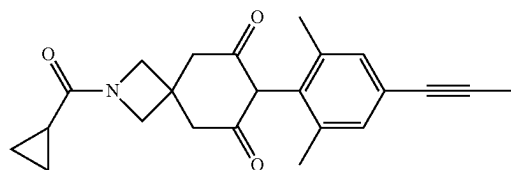
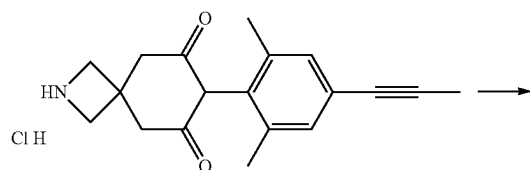
7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-8-hydroxy-2-azaspiro[3.5]non-7-en-6-one; hydrochloride (203 mg, 0.612 mmol) was taken up in DMF (5.00 mL) and cooled in an ice/water bath to 0° C. Benzoic acid (0.0700 g, 0.573 mmol) and HATU (0.262 g, 0.688 mmol) were added followed by the addition of N,N-diethylethanamine (0.174 g, 1.72 mmol). The reaction mixture was left to stir for 1 h at RT and then diluted with water and acidified with 10% citric acid solution. The mixture was extracted with EtOAc (x4) and the combined organics dried over Na₂SO₄, filtered and concentrated in vacuo. The crude material was purified by preparative HPLC to give 2-benzoyl-7-(2,6-dimethyl-4-

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prop-1-ynyl-phenyl)-2-azaspiro[3.5]nonane-6,8-dione (74 mg, 31% yield) as a white solid.

¹H NMR (400 MHz, MeOD) δ =7.67-7.65 (m, 2H), 7.52-7.44 (m, 3H), 7.02 (d, 2H), 4.21 (s, 2H), 4.03 (s, 2H), 2.86 (s, 4H), 1.99-1.94 (9H).

EXAMPLE 4: SYNTHESIS OF 2-(CYCLOPROPANECARBONYL)-7-(2,6-DIMETHYL-4-PROP-1-YNYL-PHENYL)-2-AZASPIRO[3.5]NONANE-6,8-DIONE (COMPOUND 1.008)



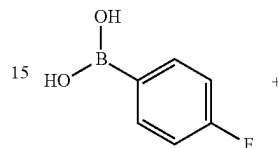
7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-8-hydroxy-2-azaspiro[3.5]non-7-en-6-one; hydrochloride (206 mg, 0.620 mmol) was taken up in DMF (5.00 mL) and cooled in an ice/water bath to 0° C. Cyclopropanecarboxylic acid (0.050 g, 0.581 mmol) and HATU (0.265 g, 0.697 mmol) were added followed by the addition of N,N-diethylethanamine (0.176 g, 1.74 mmol). The reaction mixture was left to stir for 1 h at RT and then diluted with water and acidified with 10% citric acid solution. The mixture was extracted with EtOAc (×4) and the combined organics dried over Na₂SO₄, filtered and concentrated in vacuo. The crude material was purified by preparative HPLC to give 2-(cyclopropanecarbonyl)-7-(2,6-dimethyl-4-prop-1-ynyl-phenyl)-2-azaspiro[3.5]nonane-6,8-dione (75 mg, 35% yield) as a white solid.

¹H NMR (400 MHz, MeOD) δ =7.02 (s, 2H), 4.18 (s, 2H), 3.82 (s, 2H), 2.84 (s, 4H), 2.00-1.98 (9H) 1.59-1.56 (m, 1H), 0.87-0.81 (m, 4H).

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EXAMPLE 4: SYNTHESIS OF 2-ACETYL-7-[4-(4-FLUOROPHENYL)-2,6-DIMETHYL-PHENYL]-2-AZASPIRO[3.5]NONANE-6,8-DIONE (COMPOUND 1.085)

Step 1: Synthesis of tert-butyl 7-[4-(4-fluorophenyl)-2,6-dimethyl-phenyl]-6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate



20

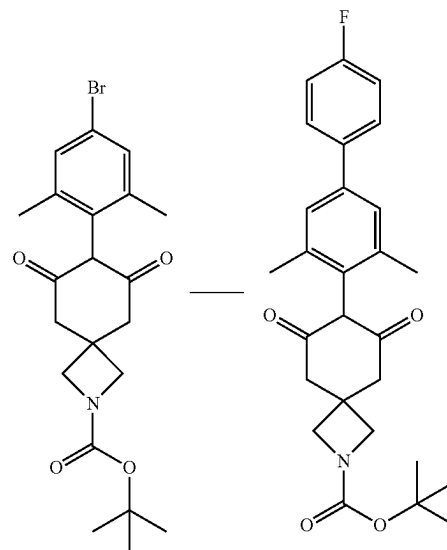
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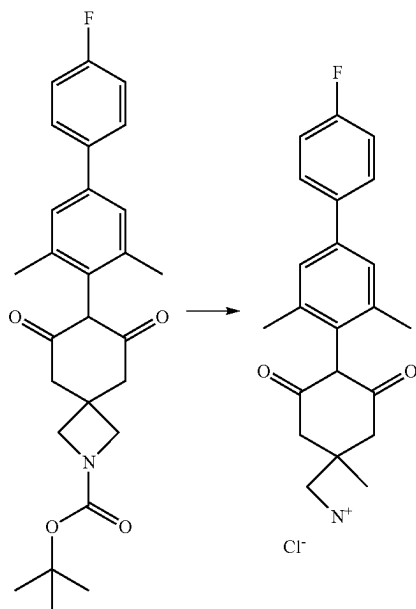
To a mixture of tert-butyl 7-(4-bromo-2,6-dimethyl-phenyl)-6,8-dioxo-2-azaspiro[3.5]nonane-2-carboxylate (10 g, 22.92 mmol), (4-fluorophenyl)boronic acid (1.5 equiv., 34.38 mmol) and [1,1'-BIS(Diphenylphosphino)]ferrocene] palladium (II) dichloride dichloromethane adduct (0.1 equiv., 2.292 mmol), 1,2-dimethoxyethane (4 mL/mmol) was added potassium phosphate (6 equiv., 137.5 mmol) and water (1 mL/mmol). The resulting mixture was heated to 100° C. overnight. The reaction was allowed to cool to RT and then quenched with 10% citric acid, extracted with EtOAc (×2), washed with brine (×1) and dried over Na₂SO₄.

The crude was purified using column chromatography (30-100% EtOAc in cyclohexane) to afford the title compound as a pale yellow solid (7.0 g, 54% yield).

¹H NMR (400 MHz, methanol) δ =7.63-7.57 (m, 2H), 7.26 (s, 2H), 7.17-7.09 (m, 2H), 3.81 (s, 4H), 2.86 (s, 4H), 2.08 (s, 6H), 1.46 (s, 9H)

31

Step 2: Synthesis of 7-[4-(4-fluorophenyl)-2,6-dimethyl-phenyl]-2-azoniaspiro[3.5]nonane-6,8-dione; chloride

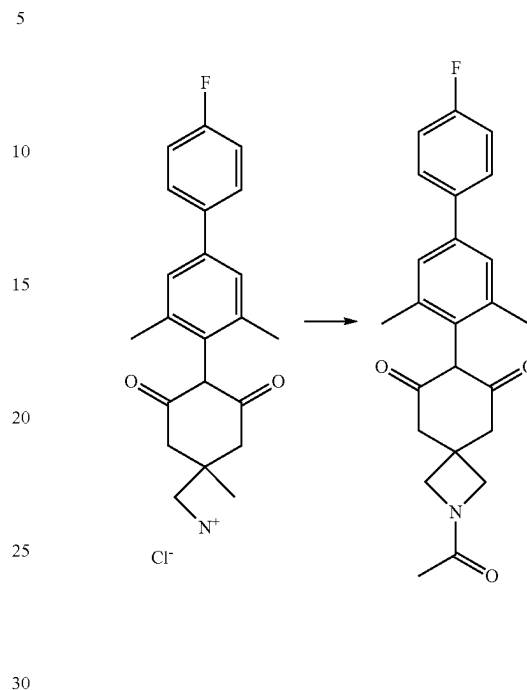


tert-butyl 7-[4-(4-fluorophenyl)-2,6-dimethyl-phenyl]-6,8-dioxo-2-azoniaspiro[3.5]nonane-2-carboxylate (5.295) was suspended in dichloromethane (39.09 mL, 0.3 M) at room temperature and then HCl (4M in Dioxane) (23 mL, 8 equiv.) was added dropwise and the resulting solution was left to stir at rt under air for 3 h. The mixture was concentrated in vacuo to give 7-[4-(4-fluorophenyl)-2,6-dimethyl-phenyl]-2-azoniaspiro[3.5]nonane-6,8-dione; chloride as a white solid (4.5 g, quant.)

¹H NMR (400 MHz, methanol) δ =7.63-7.56 (m, 2H), 7.28 (s, 2H), 7.14 (t, J=8.8 Hz, 2H), 4.04 (s, 4H), 3.01 (s, 4H), 2.08 (s, 6H)

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Step 3: Synthesis of 2-acetyl-7-[4-(4-fluorophenyl)-2,6-dimethyl-phenyl]-2-azaspiro[3.5]nonane-6,8-dione



To a stirred solution of 7-[4-(4-fluorophenyl)-2,6-dimethyl-phenyl]-2-azoniaspiro[3.5]nonane-6,8-dione; chloride (100 mg, 0.1933 mmol, 75 mass %) in DCM (1.933 mL, 0.1 M), N,N-diethylethanamine (0.119 mL, 0.08609 g, 4.4 equiv.) was added followed by HOAt (0.0357 g, 1.33 equiv.) and acetic acid (B, 0.0147 mL, 0.01544 g, 1.33 equiv.) at room temperature. After 5 min, EDCI (0.05931 g, 1.6 equiv.) was added and the reaction was left to stir at room temperature under air o/n.

Dil Aq citric acid was added and the resultant mixture was extracted with DCM. Combined organic extracts were dried and concentrated in vacuo.

Purification via preparative HPLC gave 2-acetyl-7-[4-(4-fluorophenyl)-2,6-dimethyl-phenyl]-2-azaspiro[3.5]nonane-6,8-dione (60.48% Yield, 0.046 g) as a pale brown solid.

¹H NMR (400 MHz, methanol) δ =7.61 (dd, J=5.5, 8.2 Hz, 2H), 7.27 (s, 2H), 7.13 (t, J=8.7 Hz, 2H), 4.11 (s, 2H), 3.86 (s, 2H), 2.92-2.85 (m, 4H), 2.10 (s, 3H), 2.09 (s, 3H), 1.91 (s, 3H)

Examples of herbicidal compounds of the present invention.

TABLE 1

COM- POUND	STRUCTURE	NMR
1.001		¹ H NMR (400 MHz, Methanol-d ₄): δ 7.03 (s, 2H), 4.07 (s, 2H), 3.83 (s, 2H), 2.86 (s, 4H), 1.99-1.98 (9H), 1.89 (s, 3H).

TABLE 1-continued

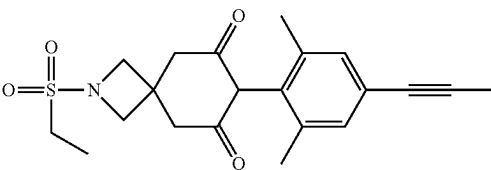
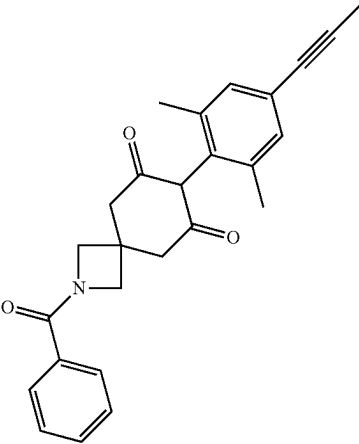
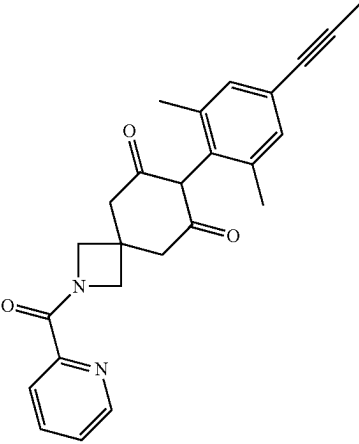
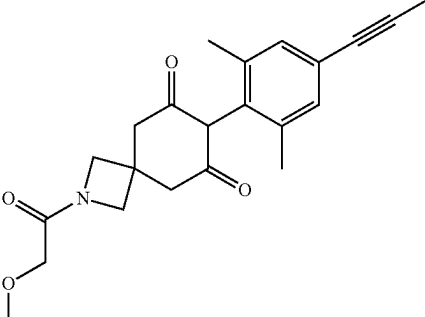
COM- POUND	STRUCTURE	NMR
1.002		^1H NMR (400 MHz, Methanol- d_4): δ = 6.96 (s, 2H), 3.75 (s, 4H), 3.09-3.04 (s, 2H), 2.66 (4H), 1.98-1.96 (9H), 1.32 (t, 3H)
1.003		^1H NMR (400 MHz, Methanol- d_4): δ = 7.67-7.65 (m, 2H), 7.52-7.44 (m, 3H), 7.02 (d, 2H), 4.21 (s, 2H), 4.03 (s, 2H), 2.86 (s, 4H), 1.99-1.94 (9H)
1.004		^1H NMR (400 MHz, Methanol- d_4): δ = 8.61 (d, 1H), 8.01 (d, 1H), 7.92 (t, 1H), 7.50-7.47 (m, 1H), 7.03 (s, 2H), 4.57 (s, 2H), 4.05 (s, 2H), 2.89 (s, 4H), 2.00 (s, 9H)
1.005		^1H NMR (400 MHz, Methanol- d_4): δ = 7.03 (s, 2H), 4.12 (s, 2H), 4.00 (s, 2H), 3.87 (s, 2H), 3.37 (s, 3H), 2.85 (s, 4H), 1.98-1.97 (9H)

TABLE 1-continued

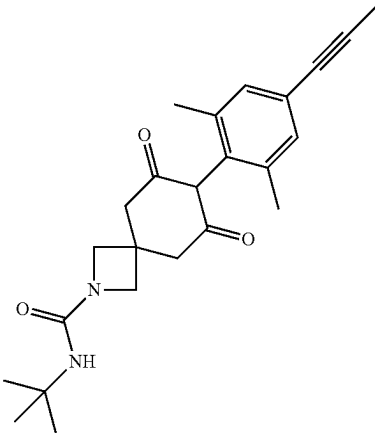
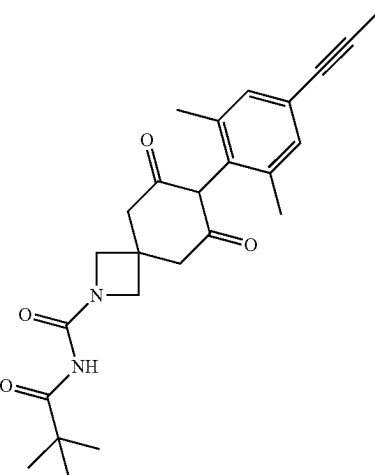
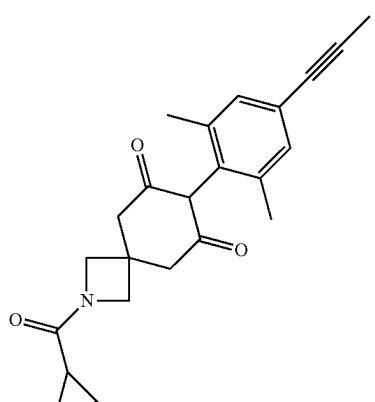
COM- POUND	STRUCTURE	NMR
1.006		^1H NMR (400 MHz, Methanol- d_4): δ = 7.02 (s, 2H), 3.74 (s, 4H), 2.79 (s, 4H), 1.97 (s, 9H), 1.31 (s, 9H)
1.007		^1H NMR (400 MHz, Methanol- d_4): δ = 7.02 (s, 2H), 3.92 (s, 4H), 2.85 (s, 4H), 1.98-1.96 (9H), 1.22 (s, 9H)
1.008		^1H NMR (400 MHz, Methanol- d_4): δ = 7.02 (s, 2H), 4.18 (s, 2H), 3.82 (s, 2H), 2.84 (s, 4H), 2.00-1.98 (9H), 1.59-1.56 (m, 1H), 0.87-0.81 (m, 4H)

TABLE 1-continued

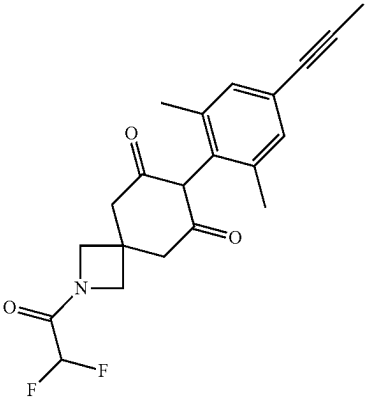
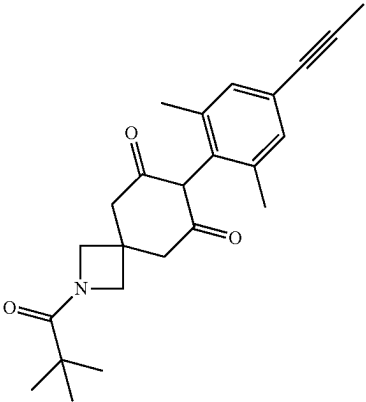
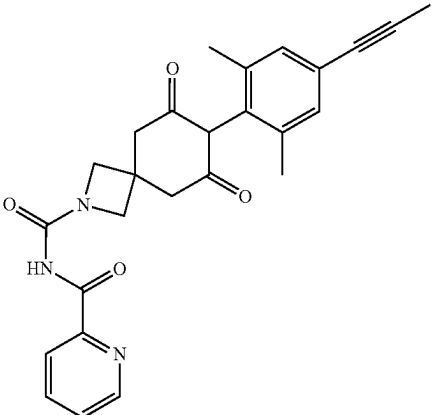
COM- POUND	STRUCTURE	NMR
1.009		^1H NMR (400 MHz, Methanol- d_4): δ = 7.02 (s, 2H), 6.16 (t, 1H), 4.26 (s, 2H), 3.95 (s, 2H), 2.86 (s, 4H), 2.02-1.98 (s, 9H)
1.010		^1H NMR (400 MHz, Methanol- d_4): δ = 7.02 (s, 2H), 4.30 (brs, 2H), 3.80 (brs, 2H), 2.82 (s, 4H), 1.98 (s, 9H), 1.20 (s, 9H)
1.011		^1H NMR (400 MHz, Methanol- d_4): δ = 8.66 (d, 1H), 8.18 (d, 1H), 8.01 (t, 1H), 7.64-7.61 (m, 1H), 7.00 (s, 2H), 4.13-4.03 (brs, 4H), 2.80 (s, 4H), 2.00-1.97 (s, 9H)

TABLE 1-continued

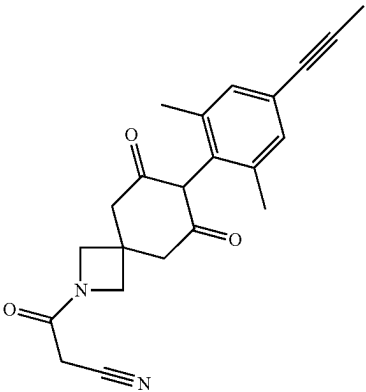
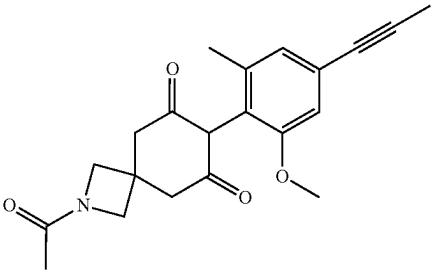
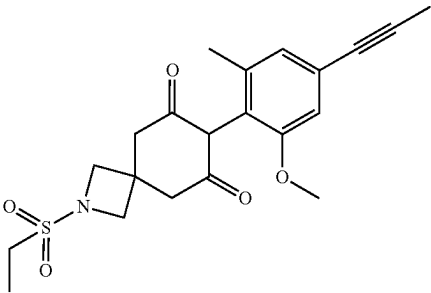
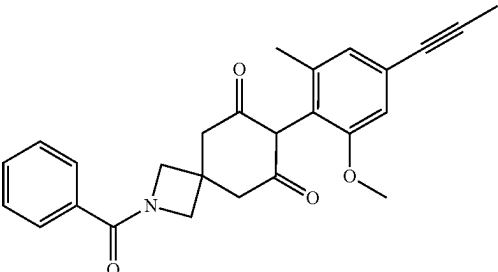
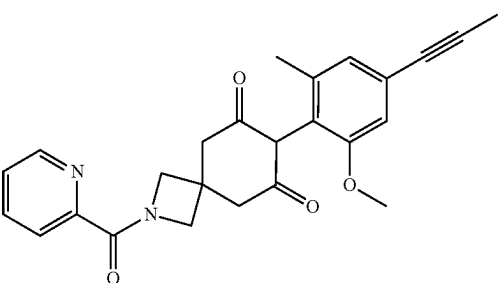
COM- POUND	STRUCTURE	NMR
1.012		^1H NMR (400 MHz, Methanol- d_4): δ = 6.99 (s, 2H), 3.96 (s, 2H), 3.74-3.70 (4H), 2.73 (brs, 4H), 2.01 (s, 3H), 1.90 (s, 6H)
1.013		^1H NMR (400 MHz, Methanol- d_4): δ = 6.84 (s, 1H), 6.76 (s, 1H), 4.04 (s, 2H), 3.81-3.80 (d, 2H), 3.66 (s, 3H), 2.82 (s, 4H), 1.99-1.97 (6H), 1.89-1.88 (3H).
1.014		^1H NMR (400 MHz, Methanol- d_4): δ = 6.83 (s, 1H), 6.75 (s, 1H), 3.79-3.78 (d, 4H), 3.65 (s, 3H), 3.10-3.05 (q, 2H), 2.82 (s, 4H), 1.99-1.97 (6H), 1.32 (t, 3H).
1.015		^1H NMR (400 MHz, Methanol- d_4): δ = 7.67-7.65 (d, 2H), 7.52-7.44 (m, 3H), 6.84-6.82 (d, 1H), 6.76-6.75 (d, 1H), 4.19 (s, 2H), 4.02-4.01 (d, 2H), 3.67-3.61 (3H), 2.84 (s, 4H), 1.99-1.94 (6H).
1.016		^1H NMR (400 MHz, Methanol- d_4): δ = 8.61 (t, 1H), 8.01-7.99 (d, 1H), 7.92 (t, 1H), 7.50-7.48 (m, 1H), 6.84 (s, 1H), 6.76 (s, 1H), 4.55-4.54 (d, 2H), 4.04-4.03 (d, 2H), 3.68-3.67 (3H), 2.84 (s, 4H), 2.02-1.97 (6H).

TABLE 1-continued

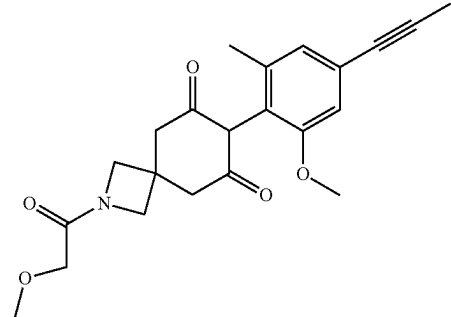
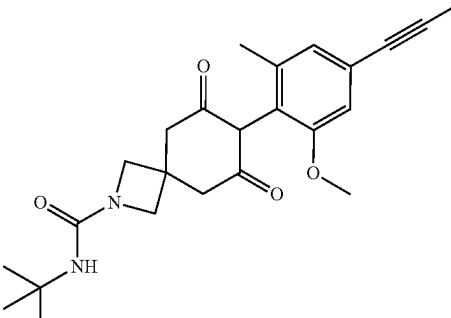
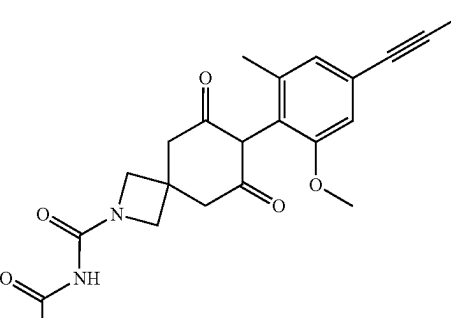
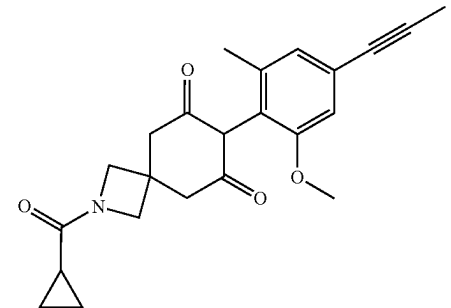
COM- POUND	STRUCTURE	NMR
1.017		^1H NMR (400 MHz, Methanol- d_4): δ = 6.83 (s, 1H), 6.75 (s, 1H), 4.10-4.09 (d, 2H), 4.00-3.99 (d, 2H), 3.85-3.84 (d, 2H), 3.65 (s, 3H), 3.37-3.36 (3H), 2.78 (s, 4H), 1.99-1.98 (6H).
1.018		^1H NMR (400 MHz, Methanol- d_4): δ = 6.83 (s, 1H), 6.76 (s, 1H), 3.73-3.72 (d, 4H), 3.65 (s, 3H), 2.77-2.76 (d, 4H), 1.99-1.98 (6H), 1.31 (s, 9H).
1.019		^1H NMR (400 MHz, Methanol- d_4): δ = 6.83 (s, 1H), 6.75 (s, 1H), 3.90 (s, 4H), 3.65 (s, 3H), 2.79 (s, 4H), 1.99-1.97 (6H), 1.21 (s, 9H).
1.020		^1H NMR (400 MHz, Methanol- d_4): δ = 6.84 (s, 1H), 6.76 (s, 1H), 4.16 (s, 2H), 3.81-3.80 (d, 2H), 3.66 (s, 3H), 2.82 (s, 4H), 2.00-1.98 (6H), 1.60-1.57 (m, 1H), 0.86-0.80 (m, 4H).

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.021		¹ H NMR (400 MHz, Methanol-d ₄): δ = 6.83 (s, 1H), 6.76 (s, 1H), 4.28 (brs, 2H), 3.79 (brs, 2H), 3.66 (s, 3H), 2.79 (s, 4H), 1.99-1.98 (6H), 1.20 (s, 9H).
1.022		¹ H NMR (400 MHz, Methanol-d ₄): δ = 8.66-8.65 (d, 1H), 8.20-8.18 (d, 1H), 8.03-8.02 (t, 1H), 7.65-7.62 (m, 1H), 6.83 (s, 1H), 6.75 (s, 1H), 4.05-4.04 (m, 4H), 3.66 (s, 3H), 2.83 (s, 4H), 2.02-1.99 (6H).
1.023		¹ H NMR (400 MHz, Methanol-d ₄): δ = 6.80 (s, 1H), 6.70 (s, 1H), 4.04-4.02 (d, 2H), 3.82-3.81 (d, 2H), 3.63 (s, 3H), 2.66-2.64 (4H), 1.99-1.98 (6H).
1.024		¹ H NMR (400 MHz, Methanol-d ₄): δ = 6.84 (s, 1H), 6.77 (s, 1H), 6.29-6.03 (m, 1H), 4.25 (s, 2H), 3.94 (s, 2H), 3.66-3.65 (s, 3H), 2.86 (s, 4H), 1.99-1.98 (6H).

TABLE 1-continued

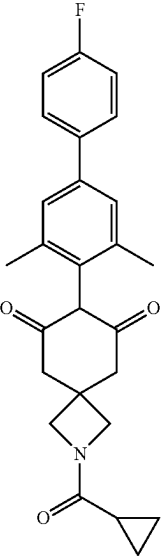
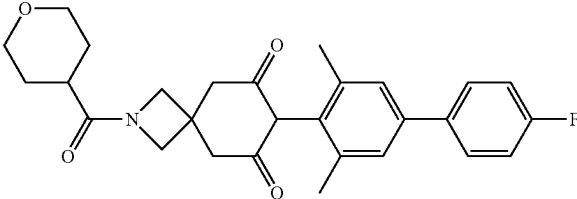
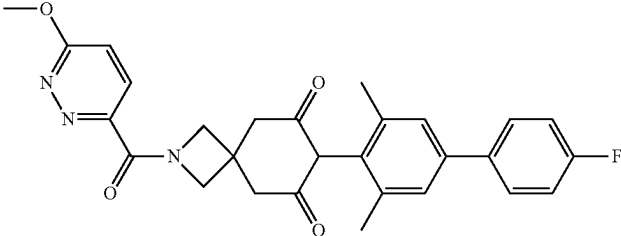
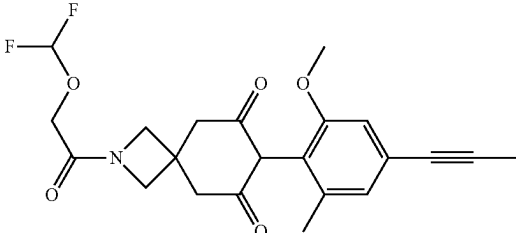
COM- POUND	STRUCTURE	NMR
1.025		^1H NMR (400 MHz, methanol- d_4) δ = 7.60 (dd, J = 5.3, 8.9 Hz, 2H), 7.26 (s, 2H), 7.13 (t, J = 8.9 Hz, 2H), 4.22 (s, 2H), 3.86 (s, 2H), 2.91 (d, J = 2.2 Hz, 4H), 2.10 (s, 3H), 2.09 (s, 3H), 1.65-1.57 (m, 1H), 0.91-0.81 (m, 4H)
1.026		^1H NMR (400 MHz, methanol) δ = 7.64-7.57 (m, 2H), 7.27 (s, 2H), 7.17-7.09 (m, 2H), 4.17 (s, 2H), 3.99-3.92 (m, 2H), 3.85 (s, 2H), 3.47 (dt, J = 2.1, 11.8 Hz, 2H), 2.90 (s, 4H), 2.61 (tt, J = 4.0, 11.4 Hz, 1H), 2.10 (d, J = 5.7 Hz, 6H), 1.81-1.69 (m, 2H), 1.67-1.58 (m, 2H)
1.027		^1H NMR (400 MHz, methanol) δ = 8.10 (d, J = 9.2 Hz, 1H), 7.61 (dd, J = 5.4, 8.7 Hz, 2H), 7.31-7.24 (m, 3H), 7.14 (t, J = 8.8 Hz, 2H), 4.64 (s, 2H), 4.19-4.15 (m, 3H), 4.12 (s, 2H), 2.97 (s, 4H), 2.12 (s, 6H)
1.028		^1H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 6.50 (dt, J = 3.4, 74.5 Hz, 1H), 4.44 (d, J = 2.8 Hz, 2H), 4.15 (d, J = 3.8 Hz, 2H), 3.88 (d, J = 2.1 Hz, 2H), 3.67 (d, J = 1.8 Hz, 3H), 2.85 (s, 4H), 2.00 (s, 3H), 1.98 (s, 3H)

TABLE 1-continued

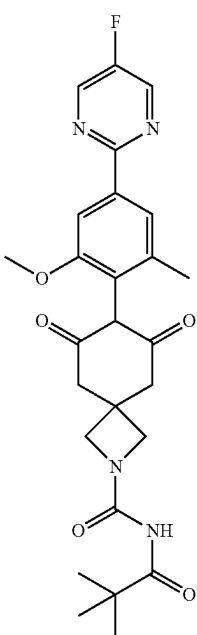
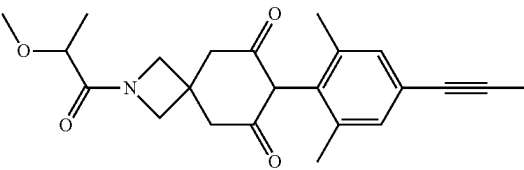
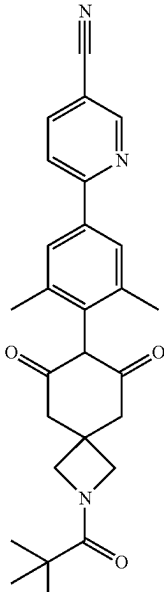
COM- POUND	STRUCTURE	NMR
1.029		^1H NMR (400 MHz, methanol) δ ppm 8.74 (s, 2 H) 7.88 (s, 1 H) 7.80 (s, 1 H) 3.94 (br s, 4 H) 3.78 (s, 3 H) 2.86 (s, 4 H) 2.11 (s, 3 H) 1.19-1.25 (m, 10 H)
1.030		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.27-4.13 (m, 2H), 3.96-3.83 (m, 3H), 3.33 (s, 3H), 2.88 (s, 4H), 1.99 (s, 3H), 1.98 (s, 6H), 1.32 (d, J = 6.7 Hz, 3H)
1.031		^1H NMR (400 MHz, methanol) δ = 8.93 (dd, J = 0.7, 2.2 Hz, 1H), 8.18 (dd, J = 2.2, 8.4 Hz, 1H), 8.04 (dd, J = 0.7, 8.4 Hz, 1H), 7.79 (s, 2H), 4.35 (br s, 2H), 3.85 (br s, 2H), 2.90 (s, 4H), 2.14 (s, 6H), 1.22 (s, 9H)

TABLE 1-continued

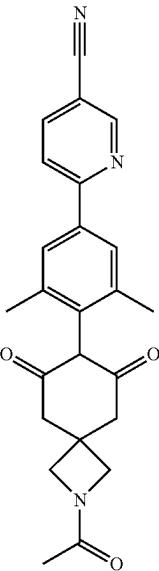
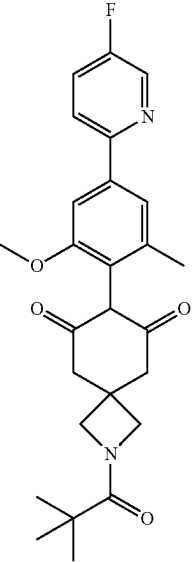
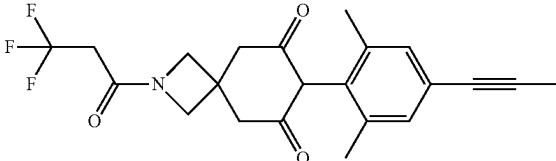
COM- POUND	STRUCTURE	NMR
1.032		^1H NMR (400 MHz, methanol) δ = 8.92 (dd, J = 0.7, 2.2 Hz, 1H), 8.17 (dd, J = 2.2, 8.4 Hz, 1H), 8.03 (dd, J = 0.6, 8.4 Hz, 1H), 7.78 (s, 2H), 4.11 (s, 2H), 3.86 (s, 2H), 2.91 (s, 4H), 2.13 (s, J = 3.1 Hz, 6H), 1.91 (s, 3H)
1.033		^1H NMR (400 MHz, methanol) δ = 8.49 (d, 1H), 7.97-7.87 (m, 1H), 7.71-7.62 (m, 1H), 7.40 (s, 2H), 4.38-4.27 (m, 2H), 3.87-3.80 (m, 2H), 3.79 (s, 3H), 2.84 (s, 4H), 2.12 (m, 3H), 1.22 (s, 9H)
1.034		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.13 (s, 2H), 3.88 (s, 2H), 3.23 (q, J = 10.8 Hz, 2H), 2.87 (s, 4H), 1.99 (s, 3H), 1.99 (s, 3H), 1.98 (s, 3H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.035		^1H NMR (400 MHz, methanol) δ = 8.47 (d, 1H), 7.91-7.84 (m, 1H), 7.68-7.62 (m, 1H), 7.60 (s, 2H), 4.15 (s, 2H), 4.02 (s, 2H), 3.90 (s, 2H), 3.39 (s, 3H), 2.86 (s, 4H), 2.12 (s, 6H)
1.036		^1H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.8 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.14-4.08 (m, 2H), 3.85 (d, J = 4.0 Hz, 2H), 3.77 (s, 3H), 3.65 (dt, J = 0.9, 6.0 Hz, 2H), 3.34 (d, J = 1.2 Hz, 3H), 2.86 (s, 4H), 2.41 (dt, J = 3.1, 6.0 Hz, 2H), 2.09 (d, J = 2.4 Hz, 3H)
1.037		^1H NMR (400 MHz, methanol) δ = 7.63-7.58 (m, 2H), 7.27 (s, 2H), 7.17-7.10 (m, 2H), 4.28-4.23 (m, 1H), 4.21-4.15 (m, 1H), 3.96-3.87 (m, 3H), 3.34 (s, 3H), 2.91 (s, 4H), 2.09 (s, 6H), 1.33 (d, J = 6.6 Hz, 3H)
1.038		
1.039		^1H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 5.14 (dq, J = 1.6, 6.6, 48.2 Hz, 1H), 4.22 (s, 2H), 3.88 (d, J = 2.3 Hz, 2H), 3.67 (d, J = 3.5 Hz, 3H), 2.85 (br s, 4H), 2.00 (s, 3H), 1.98 (s, 3H), 1.52 (ddd, J = 1.6, 6.7, 24.5 Hz, 3H)

TABLE 1-continued

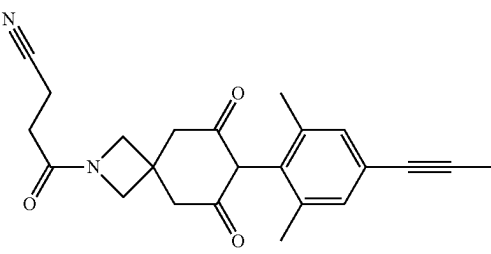
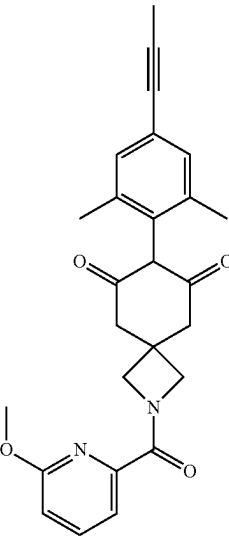
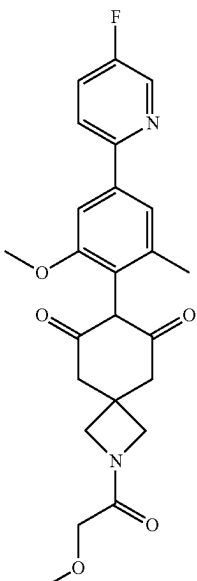
COM- POUND	STRUCTURE	NMR
1.040		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.10 (s, 2H), 3.87 (s, 2H), 2.88 (s, 4H), 2.70-2.64 (m, 2H), 2.59-2.52 (m, 2H), 2.00-1.97 (m, 9H)
1.041		^1H NMR (400 MHz, methanol) δ ppm 7.38 (t, J = 8.19 Hz, 1 H) 7.18-7.25 (m, 2 H) 7.06-7.12 (m, 1 H) 7.03 (d, J = 4.40 Hz, 2 H) 4.22 (s, 2 H) 4.03 (s, 2 H) 3.83 (s, 3 H) 2.88 (s, 4 H), 1.93- 2.03 (m, 9 H)
1.042		^1H NMR (400 MHz, methanol) δ = 8.50 (d, 1H), 7.92 (dd, 1H), 7.67 (td, 1H), 7.40 (s, 2H), 4.15 (d, 2H), 4.02 (d, 2H), 3.90 (d, 2H), 3.79 (s, 3H), 3.38 (d, 3H), 2.87 (s, 4H), 2.12 (s, 3H)

TABLE 1-continued

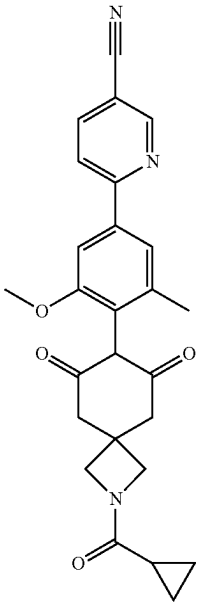
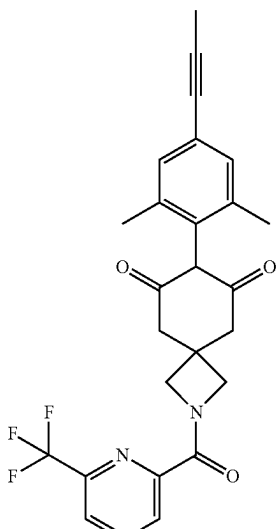
COM- POUND	STRUCTURE	NMR
1.043		¹ H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.57 (s, 2H), 4.21 (s, 2H), 3.85 (d, 2H), 3.81 (d, 3H), 2.89 (s, 4H), 2.14 (d, 3H), 1.61-1.59 (m, 1H), 0.94- 0.78 (m, 4H)
1.044		¹ H NMR (400 MHz, methanol) δ ppm 8.28-8.33 (m, 1 H) 8.18-8.24 (m, 1 H) 7.95 (dd, J = 7.76, 0.79 Hz, 1 H) 7.05 (s, 2 H) 4.66 (s, 2 H) 4.10 (s, 2 H) 2.93 (d, J = 1.59 Hz, 4 H) 1.97- 2.05 (m, 9 H)

TABLE 1-continued

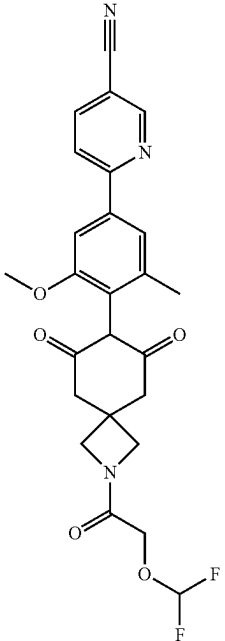
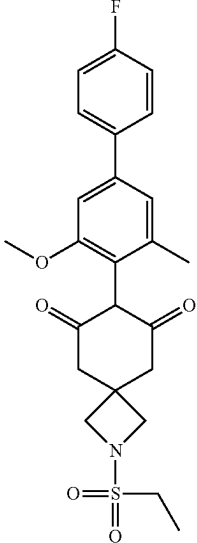
COM- POUND	STRUCTURE	NMR
1.045		¹ H NMR (400 MHz, Methanol) δ = 8.94 (d, 1 H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.57 (s, 2H), 6.51 (td, 1H), 4.45- 3.91 (m, 6H), 3.80 (d, 3H), 2.89 (s, 4H), 2.14 (s, 3H)
1.046		¹ H NMR (400 MHz, methanol) Shift = 7.68-7.58 (m, 2H), 7.15 (tt, J = 2.2, 8.8 Hz, 2H), 7.05 (d, J = 0.9 Hz, 1H), 6.97 (d, J = 1.0 Hz, 1H), 3.82 (d, J = 3.2 Hz, 4H), 3.76 (s, 3H), 3.10 (q, J = 7.4 Hz, 2H), 2.88 (s, 4H), 2.08 (s, 3H), 1.34 (t, J = 7.3 Hz, 3H)

TABLE 1-continued

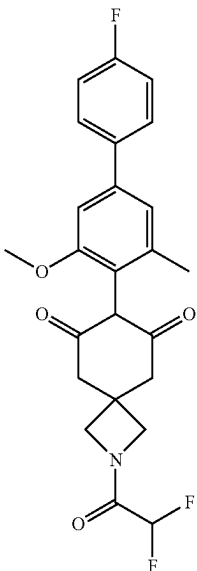
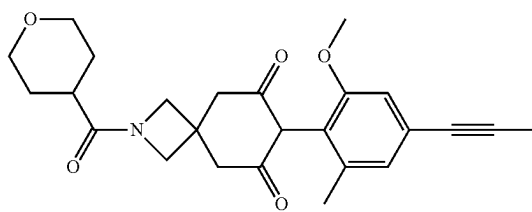
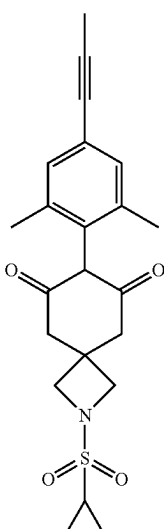
COM- POUND	STRUCTURE	NMR
1.047		¹ H NMR (400 MHz, methanol) δ = 7.64 (dd, J = 5.4, 8.8 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 6.18 (dt, J = 2.2, 53.3 Hz, 1H), 4.28 (s, 2H), 3.97 (s, 2H), 3.77 (d, J = 2.6 Hz, 3H), 2.90 (s, 4H), 2.09 (s, 3H)
1.048		¹ H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.12 (s, 2H), 3.99-3.91 (m, 2H), 3.81 (d, J = 3.4 Hz, 2H), 3.67 (d, J = 2.9 Hz, 3H), 3.46 (br t, J = 11.8 Hz, 2H), 2.84 (s, 4H), 2.64-2.52 (m, 1H), 2.03-1.97 (m, 6H), 1.80-1.68 (m, 2H), 1.66-1.55 (m, 2H)
1.049		¹ H NMR (400 MHz, methanol) δ = 7.03 (s, 2H), 3.85 (s, 4H), 2.89 (s, 4H), 2.67-2.58 (m, 1H), 1.99 (s, 3H), 1.97 (s, 6H), 1.10-1.04 (m, 4H)

TABLE 1-continued

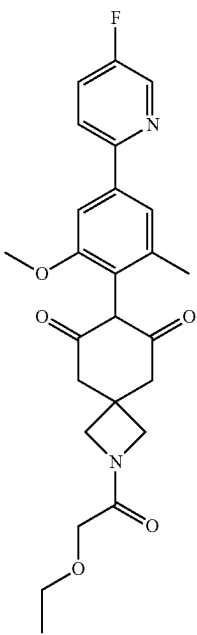
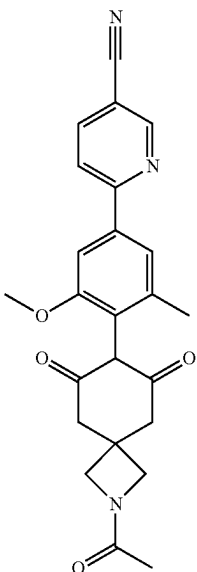
COM- POUND	STRUCTURE	NMR
1.050		¹ H NMR (400 MHz, methanol) δ = 8.47 (d, 1H), 7.91 (dd, 1H), 7.65 (dt, 1H), 7.38 (d, 2H), 4.17-4.11 (m, 2H), 4.06 (d, 2H), 3.90-3.84 (m, 2H), 3.78 (d, 3H), 3.59-3.50 (m, 2H), 2.83- 2.71 (m, 4H), 2.13 (d, 3H), 1.26-1.18 (m, 3H)
1.051		¹ H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.57 (s, 2H), 4.09 (s, 2H), 3.84 (d, 2H), 3.80 (s, 3H), 2.87 (s, 4H), 2.14 (d, 3H), 1.91 (d, 3H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.052		¹ H NMR (400 MHz, methanol) δ = 8.47 (d, 1H), 7.87 (dd, 1H), 7.65 (td, 1H), 7.60 (s, 2H), 6.19 (t, 1H), 4.29 (s, 2H), 3.98 (s, 2H), 2.88 (s, 4H), 2.12 (d, 6H)
1.053		¹ H NMR (400 MHz, Methanol) δ = 8.97-8.92 (m, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.56 (s, 2H), 3.85-3.75 (m, 7H), 3.10 (q, 2H), 2.88 (s, 4H), 2.11 (s, 3H), 1.34 (t, 3H)
1.054		¹ H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 6.50 (t, J = 74.5 Hz, 1H), 4.44 (s, 2H), 4.17 (s, 2H), 3.90 (s, 2H), 2.88 (br s, 4H), 1.99 (s, 3H), 1.98 (s, 6H)

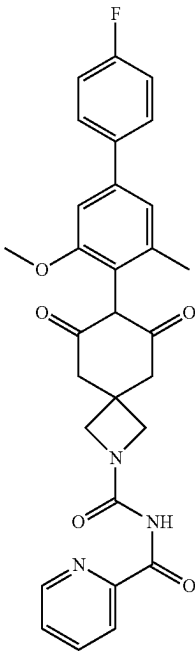
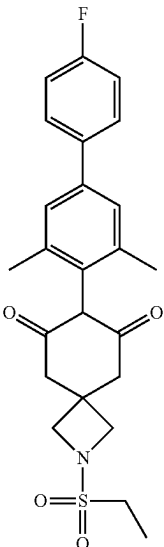
COM- POUND	STRUCTURE	NMR
1.055		^1H NMR (400 MHz, methanol) Shift = 8.68 (d, J = 4.6 Hz, 1H), 8.20 (d, J = 7.8 Hz, 1H), 8.03 (dt, J = 1.7, 7.8 Hz, 1H), 7.68-7.59 (m, 3H), 7.15 (t, J = 9.2 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.21-3.90 (m, 4H), 3.77 (s, 3H), 2.91 (s, 4H), 2.11 (s, 3H)
1.056		^1H NMR (400 MHz, methanol) δ = 7.63-7.58 (m, 2H), 7.26 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 3.84 (s, 4H), 3.10 (q, J = 7.3 Hz, 2H), 2.91 (s, 4H), 2.08 (s, 6H), 1.34 (t, J = 7.4 Hz, 3H)

TABLE 1-continued

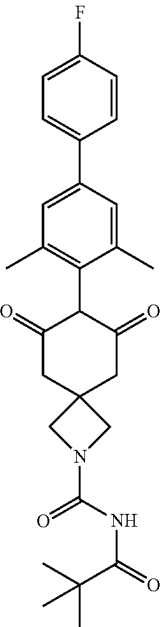
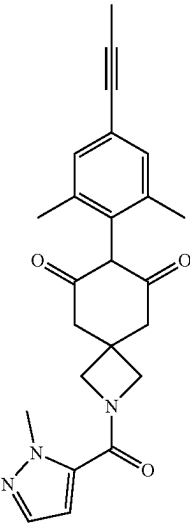
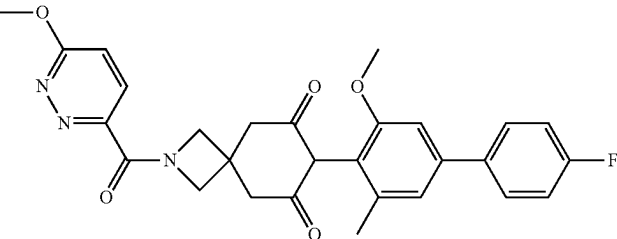
COM- POUND	STRUCTURE	NMR
1.057		^1H NMR (400 MHz, methanol) δ = 7.60 (dd, J = 5.4, 8.8 Hz, 2H), 7.26 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 3.96 (br s, 4H), 2.89 (s, 4H), 2.08 (s, 6H), 1.22 (s, 9H)
1.058		^1H NMR (400 MHz, methanol) δ ppm 7.49 (d, J = 2.08 Hz, 1 H) 7.04 (s, 2 H) 6.66 (d, J = 2.20 Hz, 1 H) 4.28 (s, 2 H) 4.09 (s, 3 H) 4.01 (s, 2 H) 2.91 (s, 4 H) 1.95-2.04 (m, 10 H)
1.059		^1H NMR (400 MHz, methanol) Shift = 8.10 (dd, J = 2.7, 9.2 Hz, 1H), 7.65-7.61 (m, 2H), 7.27 (d, J = 9.2 Hz, 1H), 7.15 (t, J = 8.9 Hz, 2H), 7.09-7.05 (m, 1H), 6.99 (s, 1H), 4.62 (s, 2H), 4.17 (s, 3H), 4.10 (br d, J = 2.9 Hz, 2H), 3.79 (s, 3H), 2.93 (s, 4H), 2.12 (s, 3H)

TABLE 1-continued

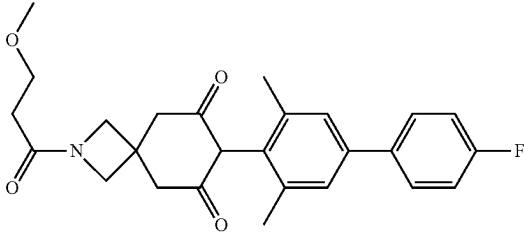
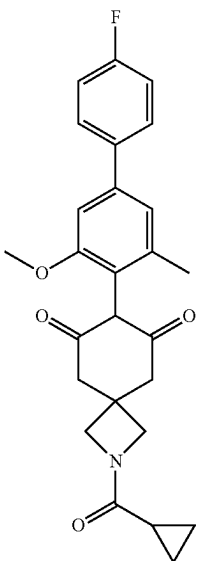
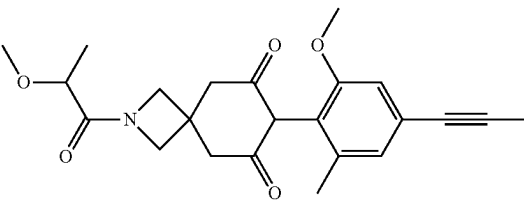
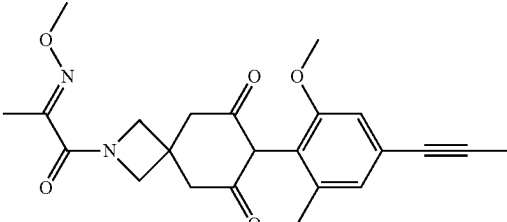
COM- POUND	STRUCTURE	NMR
1.060		^1H NMR (400 MHz, methanol) δ = 7.60 (dd, J = 5.4, 8.8 Hz, 2H), 7.26 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 4.12 (s, 2H), 3.86 (s, 2H), 3.64 (t, J = 6.0 Hz, 2H), 3.34 (s, 3H), 2.89 (s, 4H), 2.40 (t, J = 6.0 Hz, 2H), 2.09 (s, 3H), 2.09 (s, 3H)
1.061		^1H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.8 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.22-4.18 (m, 2H), 3.85 (d, J = 4.2 Hz, 2H), 3.77 (d, J = 2.0 Hz, 3H), 2.88 (s, 4H), 2.10 (d, J = 5.1 Hz, 3H), 1.64-1.57 (m, 1H), 0.90-0.80 (m, 4H)
1.062		^1H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.25-4.18 (m, 1H), 4.17-4.11 (m, 1H), 3.96-3.82 (m, 3H), 3.67 (s, 3H), 3.33 (d, J = 3.9 Hz, 3H), 2.84 (s, 4H), 2.01 (s, 3H), 1.99 (s, 3H), 1.32 (dd, J = 2.2, 6.7 Hz, 3H)
1.063		

TABLE 1-continued

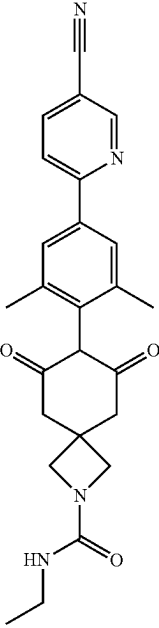
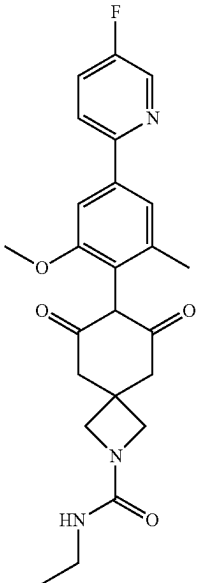
COM- POUND	STRUCTURE	NMR
1.064		¹ H NMR (400 MHz, methanol) δ = 8.93 (dd, J = 0.6, 2.0 Hz, 1H), 8.17 (dd, J = 2.2, 8.4 Hz, 1H), 8.04 (dd, J = 0.6, 8.4 Hz, 1H), 7.78 (s, 2H), 3.81 (s, 4H), 3.16 (q, J = 7.2 Hz, 2H), 2.88 (s, 4H), 2.13 (s, 6H), 1.10 (t, J = 7.2 Hz, 3H)
1.065		¹ H NMR (400 MHz, methanol) δ = 8.52-8.47 (m, 1H), 7.95-7.88 (m, 1H), 7.71-7.63 (m, 1H), 7.39 (s, 2H), 3.82-3.76 (m, 7H), 3.16 (q, 2H), 2.90- 2.77 (m, 4H), 2.12 (s, 3H), 1.10 (t, 3H)

TABLE 1-continued

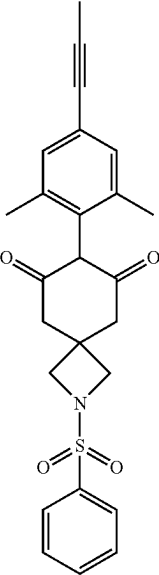
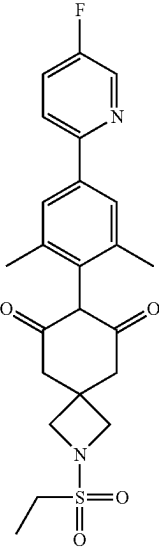
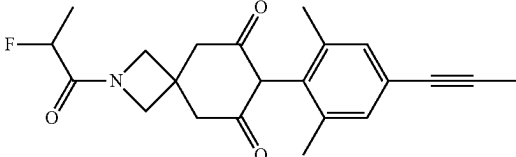
COM- POUND	STRUCTURE	NMR
1.066		^1H NMR (400 MHz, methanol) δ = 7.91-7.86 (m, 2H), 7.80-7.74 (m, 1H), 7.73-7.67 (m, 2H), 7.01 (s, 2H), 3.65 (s, 4H), 2.58 (s, 4H), 1.98 (s, 3H), 1.92 (s, 6H)
1.067		^1H NMR (400 MHz, methanol) δ = 8.47 (d, 1H), 7.87 (dd, 1H), 7.65 (td, 1H), 7.60 (s, 2H), 3.84 (s, 4H), 3.10 (q, 2H), 2.88 (s, 4H), 2.11 (s, 6H), 1.34 (t, 3H)
1.068		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 5.25-5.04 (m, 1H), 4.24 (d, J = 2.6 Hz, 2H), 3.90 (s, 2H), 2.89 (d, J = 3.1 Hz, 4H), 1.99 (s, 3H), 1.98 (s, 6H), 1.56-1.47 (m, 3H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.069		^1H NMR (400 MHz, d_6 -DMSO) δ 9.09-9.03 (m, 1H), 8.70 (d, 1H), 8.37 (dd, 1H), 8.21 (d, 1H), 8.15- 8.02 (m, 2H), 7.73-7.69 (m, 1H), 7.56 (d, 2H), 4.10-3.63 (m, 7H), 2.88-2.61 (m, 4H), 2.12-1.97 (m, 3H)
1.070		^1H NMR (400 MHz, methanol) Shift = 7.66-7.60 (m, 2H), 7.15 (tt, J = 2.0 8.7 Hz, 2H), 7.06 (d, J = 0.9 Hz, 1H), 6.97 (d, J = 1.3 Hz, 1H), 3.94 (br s, 4H), 3.76 (s, 3H), 2.86 (s, 4H), 2.09 (s, 3H), 1.23 (s, 9H)
1.071		^1H NMR (400 MHz, methanol) Shift = 7.68-7.60 (m, 2H), 7.15 (tt, J = 2.2, 9.0 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.14 (s, 2H), 3.96 (td, J = 2.3, 11.6 Hz, 2H), 3.84 (d, J = 3.7 Hz, 2H), 3.77 (d, J = 3.2 Hz, 3H), 3.47 (tt, J = 2.2, 11.9 Hz, 2H), 2.86 (s, 4H), 2.66-2.56 (m, 1H), 2.10 (d, J = 6.8 Hz, 3H), 1.78 (dd, J = 4.4, 13.3 Hz, 1H), 1.72 (dd, J = 4.5, 11.9 Hz, 1H), 1.67-1.58 (m, 2H)

TABLE 1-continued

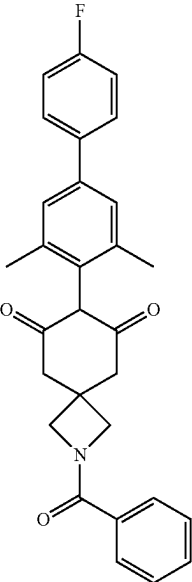
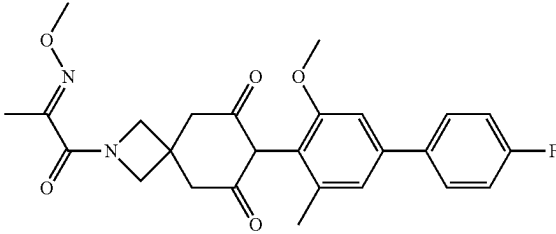
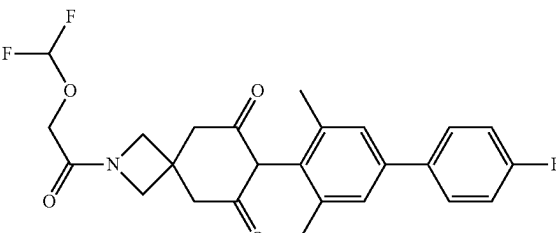
COM- POUND	STRUCTURE	NMR
1.072		¹ H NMR (400 MHz, methanol) δ = 7.72-7.64 (m, 2H), 7.60 (dd, J = 5.4, 8.8 Hz, 2H), 7.56-7.45 (m, 3H), 7.27 (br s, 1H), 7.25 (br s, 1H), 7.13 (t, J = 8.9 Hz, 2H), 4.25 (s, 2H), 4.07 (s, 2H), 2.93 (s, 4H), 2.11 (s, 3H), 2.06 (s, 3H)
1.073		
1.074		¹ H NMR (400 MHz, methanol) δ = 7.60 (dd, J = 5.4, 8.7 Hz, 2H), 7.27 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 6.50 (t, J = 74.6 Hz, 1H), 4.45 (s, 2H), 4.19 (s, 2H), 3.93 (s, 2H), 2.91 (s, 4H), 2.09 (s, 6H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.075		¹ H NMR (400 MHz, DMSO-d ₆) δ = 8.97 (s, 2 H) 7.80 (s, 1 H) 7.70 (s, 1 H) 3.97 (s, 2 H) 3.77 (d, J = 8.31 Hz, 2 H) 3.68-3.74 (m, 5 H) 2.76 (br d, J = 3.30 Hz, 4 H) 2.03 (s, 3 H)
1.076		¹ H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.07 (s, 2H), 3.85 (d, J = 3.2 Hz, 2H), 3.67 (d, J = 1.7 Hz, 3H), 2.84 (s, 4H), 2.70-2.63 (m, 2H), 2.56 (dd, J = 3.7, 6.6 Hz, 2H), 2.00 (s, 3H), 1.99 (d, J = 3.9 Hz, 3H)
1.077		¹ H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.10 (s, 2H), 3.85 (d, J = 2.6 Hz, 2H), 3.67 (d, J = 1.8 Hz, 3H), 3.22 (dq, J = 2.9, 10.7 Hz, 2H), 2.84 (s, 4H), 2.00 (s, 3H), 1.99 (d, J = 3.5 Hz, 3H)
1.078		¹ H NMR (400 MHz, methanol) δ = 8.49 (d, 1H), 7.92 (dd, 1H), 7.66 (td, 1H), 7.39 (s, 2H), 3.79 (s, 3H), 3.78-3.74 (m, 4H), 2.88-2.76 (m, 4H), 2.12 (s, 3H), 1.32 (s, 9H)

TABLE 1-continued

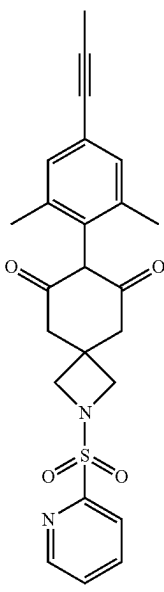
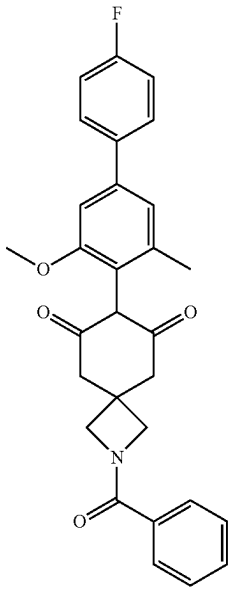
COM- POUND	STRUCTURE	NMR
1.079		¹ H NMR (400 MHz, methanol) δ = 8.84-8.80 (m, 1H), 8.16-8.10 (m, 1H), 8.02 (d, J = 7.8 Hz, 1H), 7.72 (ddd, J = 1.1, 4.7, 7.6 Hz, 1H), 7.01 (s, 2H), 3.95 (s, 4H), 2.71 (s, 4H), 1.98 (s, 3H), 1.93 (s, 6H)
1.080		¹ H NMR (400 MHz, methanol) δ = 7.68 (br d, J = 7.5 Hz, 2H), 7.63 (dd, J = 5.4, 8.8 Hz, 2H), 7.56-7.44 (m, 3H), 7.14 (t, J = 8.8 Hz, 2H), 7.05 (br d, J = 6.6 Hz, 1H), 6.97 (br d, J = 5.0 Hz, 1H), 4.23 (s, 2H), 4.05 (d, J = 3.2 Hz, 2H), 3.80-3.68 (m, 3H), 2.89 (s, 4H), 2.15-2.03 (m, 3H)

TABLE 1-continued

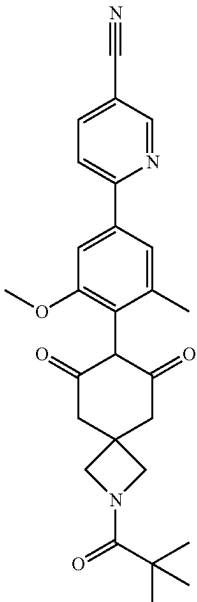
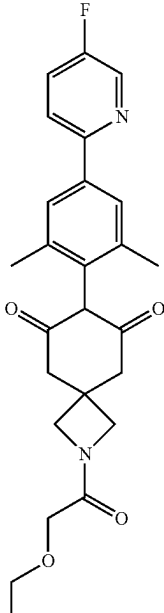
COM- POUND	STRUCTURE	NMR
1.081		^1H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.57 (s, 2H), 4.33 (s, 2H), 3.83 (s, 2H), 3.80 (s, 3H), 2.86 (s, 4H), 2.14 (s, 4H), 1.22 (s, 9H)
1.082		^1H NMR (400 MHz, methanol) δ = 8.47 (d, 1H), 7.90-7.84 (m, 1H), 7.68- 7.62 (m, 1H), 7.60 (s, 2H), 4.22- 4.15 (m, 2H), 4.08-4.04 (m, 2H), 3.92-3.87 (m, 2H), 3.54 (q, 2H), 2.85 (s, 4H), 2.12 (s, 6H), 1.22 (t, 3H)

TABLE 1-continued

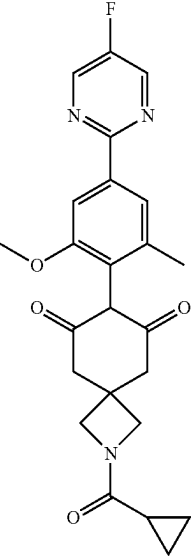
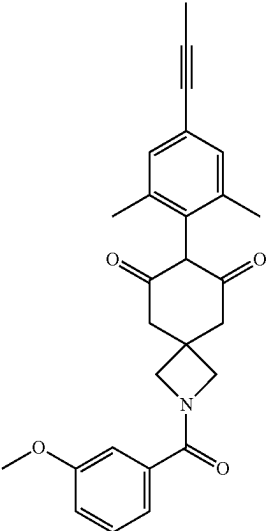
COM- POUND	STRUCTURE	NMR
1.083		¹ H NMR (400 MHz, DMSO-d ₆) δ ppm 8.97 (s, 2 H) 7.80 (s, 1 H) 7.70 (s, 1 H) 4.06 (s, 2 H) 3.71 (d, J = 2.93 Hz, 3 H) 3.66 (br s, 2 H) 2.66-2.92 (m, 4 H) 2.04 (d, J = 5.01 Hz, 3 H) 1.52-1.62 (m, 1 H) 0.66-0.76 (m, 4 H)
1.084		¹ H NMR (400 MHz, methanol) δ ppm 7.38 (t, J = 8.19 Hz, 1 H) 7.18-7.24 (m, 2 H) 7.06-7.11 (m, 1 H) 7.03 (d, J = 4.40 Hz, 2 H) 4.22 (s, 2 H) 4.03 (s, 2 H) 3.83 (s, 3 H) 2.88 (s, 4 H) 1.97-2.02 (m, 6 H) 1.95 (s, 3 H)

TABLE 1-continued

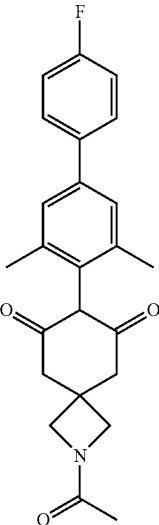
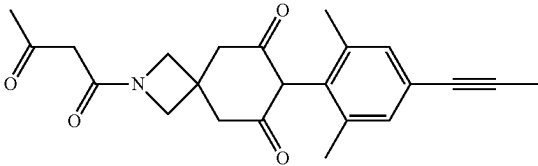
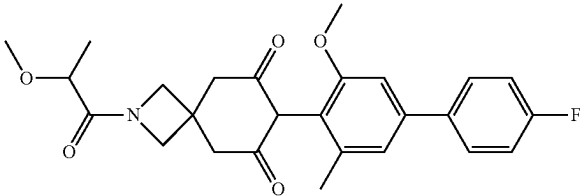
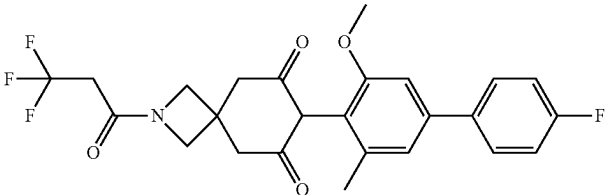
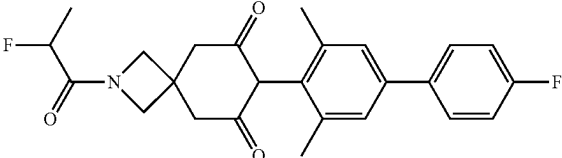
COM- POUND	STRUCTURE	NMR
1.085		^1H NMR (400 MHz, methanol) δ = 7.61 (dd, J = 5.5, 8.2 Hz, 2 H), 7.27 (s, 2H), 7.13 (t, J = 8.7 Hz, 2H), 4.11 (s, 2H), 3.86 (s, 2H), 2.92-2.85 (m, 4H), 2.10 (s, 3H), 2.09 (s, 3H), 1.91 (s, 3H)
1.086		
1.087		^1H NMR (400 MHz, methanol) δ = 7.64 (dd, J = 5.4, 8.8 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.26-4.21 (m, 1H), 4.16 (dd, J = 4.6, 9.7 Hz, 1H), 3.97-3.85 (m, 3H), 3.77 (s, 3H), 3.34 (d, J = 4.0 Hz, 3H), 2.87 (s, 4H), 2.10 (s, 3H), 1.33 (dd, J = 2.2, 6.7 Hz, 3H)
1.088		^1H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.8 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.14 (s, 2H), 3.88 (d, J = 2.9 Hz, 2H), 3.77 (d, J = 1.6 Hz, 3H), 3.24 (dq, J = 3.2, 10.7 Hz, 2H), 2.87 (s, 4H), 2.09 (d, J = 3.7 Hz, 3H)
1.089		^1H NMR (400 MHz, methanol) δ = 7.60 (dd, J = 5.4, 8.8 Hz, 2H), 7.27 (s, 2H), 7.13 (t, J = 8.9 Hz, 2H), 5.15 (qd, J = 6.7, 48.4 Hz, 1H), 4.27 (d, J = 2.4 Hz, 2H), 3.93 (s, 2H), 2.92 (d, J = 3.1 Hz, 4H), 2.09 (s, 6H), 1.53 (dd, J = 6.7, 24.5 Hz, 3H)

TABLE 1-continued

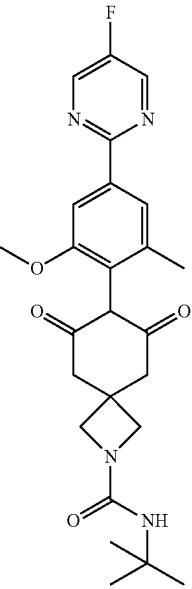
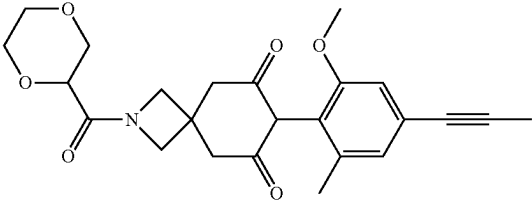
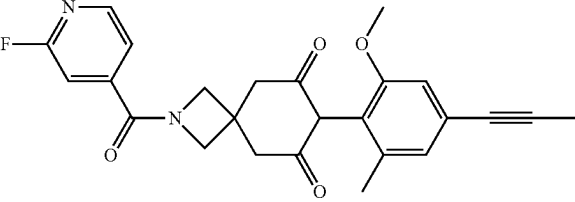
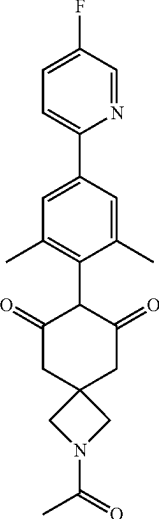
COM- POUND	STRUCTURE	NMR
1.090		^1H NMR (400 MHz, methanol) δ ppm 8.74 (s, 2 H) 7.87 (s, 1 H) 7.80 (s, 1 H) 3.78 (s, 3 H) 3.76 (d, J = 2.81 Hz, 4 H) 2.82 (d, J = 2.45 Hz, 4 H) 2.12 (s, 3 H) 1.32 (s, 9 H)
1.091		^1H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.29-4.19 (m, 3H), 3.92-3.78 (m, 4H), 3.74-3.57 (m, 7H), 2.83 (d, J = 5.1 Hz, 4H), 2.00 (s, 3H), 1.98 (s, 3H)
1.092		^1H NMR (400 MHz, methanol) δ = 8.37-8.31 (m, 1H), 7.55-7.49 (m, 1H), 7.32 (s, 1H), 6.84 (d, J = 5.5 Hz, 1H), 6.77 (d, J = 3.8 Hz, 1H), 4.23 (s, 2H), 4.03 (s, 2H), 3.65 (d, J = 21.3 Hz, 3H), 2.88 (s, 4H), 2.04-1.94 (m, 6H)
1.093		^1H NMR (400 MHz, DMSO- d_6) δ = 8.62 (d, 1H), 7.98 (dd, 1H), 7.85-7.75 (td, 1H), 7.66 (s, 2H), 3.96 (s, 2H), 3.65 (s, 2H), 2.88-2.71 (m, 4H), 2.03 (d, 6H), 1.78 (s, 3H)

TABLE 1-continued

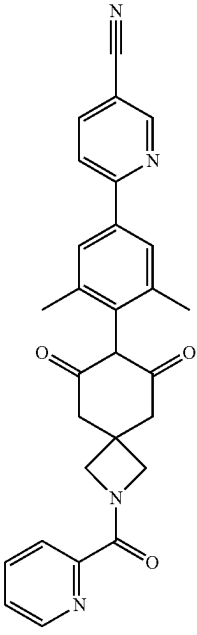
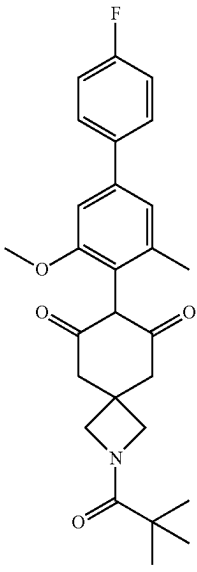
COM- POUND	STRUCTURE	NMR
1.094		¹ H NMR (400 MHz, methanol) Shift = 8.92 (d, J = 1.3 Hz, 1H), 8.63 (d, J = 4.6 Hz, 1H), 8.19-8.14 (m, 1H), 8.07- 8.01 (m, 2H), 7.98-7.91 (m, 1H), 7.79 (s, 2H), 7.54-7.47 (m, 1H), 4.65- 4.56 (m, 2H), 4.11-4.08 (m, 2H), 2.93 (s, 4H), 2.16 (d, J = 5.6 Hz, 6H)
1.095		¹ H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.8 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.97 (d, J = 1.0 Hz, 1H), 4.32 (br s, 2H), 3.82 (br s, 2H), 3.76 (s, 3H), 2.85 (s, 4H), 2.09 (s, 3H), 1.22 (s, 9H)

TABLE 1-continued

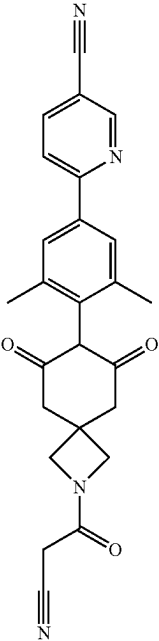
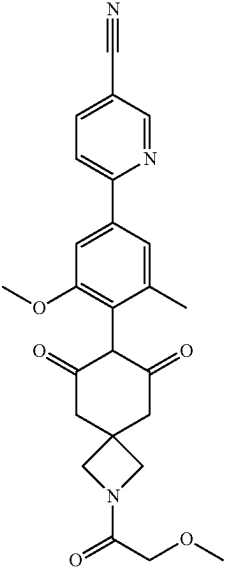
COM- POUND	STRUCTURE	NMR
1.096		¹ H NMR (400 MHz, methanol) δ = 8.93 (d, J = 1.5 Hz, 1H), 8.17 (dd, J = 2.2, 8.3 Hz, 1H), 8.04 (dd, J = 0.7, 8.4 Hz, 1H), 7.79 (s, 2H), 4.15 (s, 2H), 3.92 (s, 2H), 3.64 (s, 2H), 2.92 (s, 4H), 2.13 (s, 6H)
1.097		¹ H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.57 (s, 2H), 4.18-3.86 (m, 6H), 3.80 (m, 3H), 3.39 (d, 3H), 2.88 (s, 4H), 2.14 (s, 3H)

TABLE 1-continued

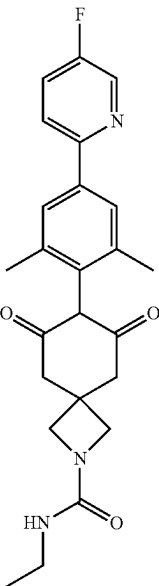
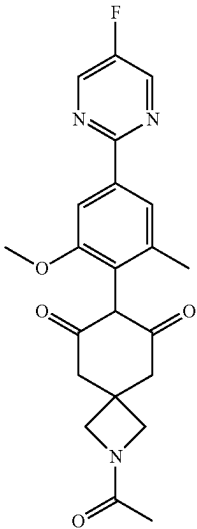
COM- POUND	STRUCTURE	NMR
1.098		^1H NMR (400 MHz, methanol) δ = 8.51-8.44 (m, 1H), 7.91-7.84 (m, 1H), 7.69-7.62 (m, 1H), 7.61 (s, 2H), 3.80 (s, 4H), 3.21-3.11 (q, 2H), 2.87 (s, 4H), 2.12 (s, 6H), 1.10 (t, 3H)
1.099		^1H NMR (400 MHz, DMSO- d_6) δ ppm 8.97 (s, 2 H) 7.80 (s, 1 H) 7.70 (s, 1 H) 3.93 (s, 2 H) 3.70 (d, J = 1.34 Hz, 3 H) 3.63 (br d, J = 2.57 Hz, 2 H) 2.65-2.88 (m, 4 H) 2.04 (d, J = 3.30 Hz, 3 H) 1.78 (d, J = 4.89 Hz, 3 H)

TABLE 1-continued

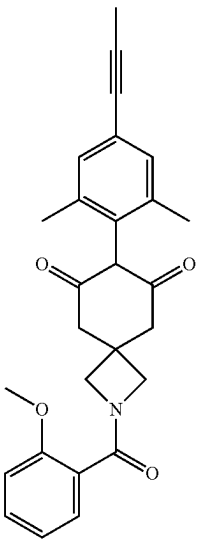
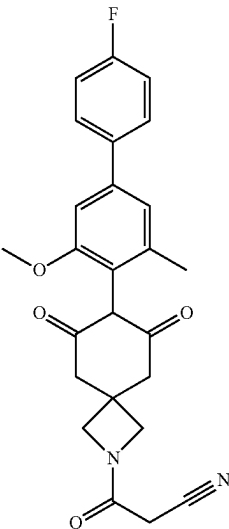
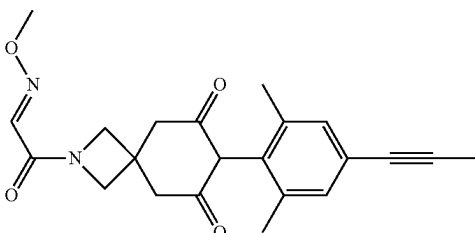
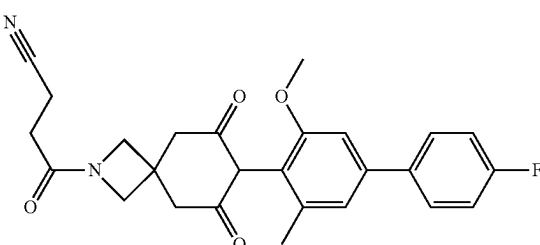
COM- POUND	STRUCTURE	NMR
1.100		¹ H NMR (400 MHz, methanol) δ ppm 7.45 (ddd, J = 8.59, 7.31, 1.71 Hz, 1 H) 7.33 (dd, J = 7.58, 1.71 Hz, 1 H) 7.10 (d, J = 8.44 Hz, 1 H) 6.99-7.06 (m, 3 H) 4.00 (s, 2 H) 3.90 (s, 3 H) 3.85 (s, 2 H) 2.87 (d, J = 6.11 Hz, 4 H) 1.99 (d, J = 2.32 Hz, 6 H) 1.90 (s, 3 H)
1.101		¹ H NMR (400 MHz, methanol) δ = 7.66-7.60 (m, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.12 (d, J = 3.8 Hz, 2H), 3.89 (d, J = 2.3 Hz, 2H), 3.76 (s, 3H), 3.64 (d, J = 4.5 Hz, 2H), 2.88 (s, 4H), 2.09 (d, J = 1.6 Hz, 3H)
1.102		¹ H NMR (400 MHz, methanol) δ = 7.49 (s, 1H), 7.04 (s, 2H), 4.30 (s, 2H), 3.97 (s, 3H), 3.96-3.92 (m, 2H), 2.89 (s, 4H), 1.99 (s, 3H), 1.98 (s, 6H)
1.103		¹ H NMR (400 MHz, methanol) Shift = 7.67-7.60 (m, 2H), 7.15 (tt, J = 3.1, 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.10 (s, 2H), 3.87 (d, J = 3.7 Hz, 2H), 3.77 (d, J = 1.8 Hz, 3H), 2.87 (s, 4H), 2.67 (t, J = 7.1 Hz, 2H), 2.61- 2.52 (m, 2H), 2.10 (d, J = 4.2 Hz, 3H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.104		¹ H NMR (400 MHz, methanol) Shift = 8.35 (t, J = 4.4 Hz, 1H), 7.69-7.59 (m, 2H), 7.54 (m, 1H), 7.33 (br s, 1H), 7.15 (tt, J = 2.7, 8.9 Hz, 2H), 7.06 (d, J = 5.9 Hz, 1H), 6.97 (d, J = 3.3 Hz, 1H), 4.26 (s, 2H), 4.06 (d, J = 2.2 Hz, 2H), 3.75 (d, J = 21.5 Hz, 3H), 2.91 (s, 4H), 2.09 (d, J = 15.5 Hz, 3H)
1.105		
1.106		¹ H NMR (400 MHz, DMSO-d ₆) δ ppm 8.97 (s, 2 H) 8.60-8.67 (m, 1 H) 7.95-8.01 (m, 2 H) 7.81 (s, 1 H) 7.71 (s, 1 H) 7.52-7.59 (m, 1 H) 4.44 (s, 2 H) 3.91 (s, 2 H) 3.72 (d, J = 2.81 Hz, 3 H) 2.06 (d, J = 7.58 Hz, 3 H)
1.107		¹ H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.13 (d, J = 2.8 Hz, 2H), 4.04 (d, J = 2.4 Hz, 2H), 3.85 (d, J = 3.4 Hz, 2H), 3.66 (d, J = 1.5 Hz, 3H), 3.53 (dq, J = 2.9, 7.0 Hz, 2H), 2.83 (s, 4H), 2.00 (s, 3H), 1.98 (s, 3H), 1.21 (q, J = 7.0 Hz, 3H)
1.108		¹ H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.43 (s, 2H), 3.90 (s, 2H), 3.67 (s, 3H), 2.86 (s, 4H), 2.00 (s, 3H), 1.99 (s, 3H), 1.56 (s, 6H)

TABLE 1-continued

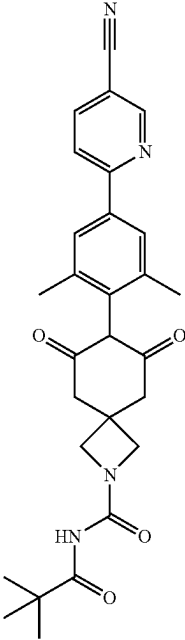
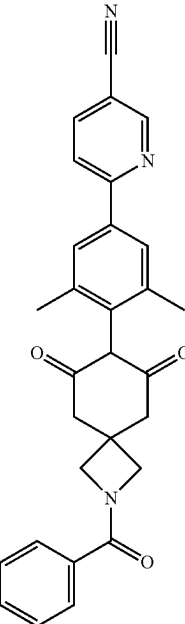
COM- POUND	STRUCTURE	NMR
1.109		^1H NMR (400 MHz, methanol) δ = 8.93 (dd, J = 0.7, 2.1 Hz, 1H), 8.17 (dd, J = 2.2, 8.4 Hz, 1H), 8.04 (dd, J = 0.7, 8.4 Hz, 1H), 7.78 (s, 2H), 3.97 (s, 4H), 2.91 (s, 4H), 2.12 (s, 6H), 1.23 (s, 9H)
1.110		^1H NMR (400 MHz, methanol) δ = 8.92 (dd, J = 0.9, 2.2 Hz, 1H), 8.17 (dd, J = 2.1, 8.4 Hz, 1H), 8.04 (dd, J = 0.7, 8.7 Hz, 1H), 7.78 (br d, J = 2.4 Hz, 2H), 7.71-7.66 (m, 2H), 7.55- 7.46 (m, 3H), 4.26 (s, 2H), 4.07 (s, 2H), 2.99-2.90 (s, 4H), 2.20-2.08 (s, 6H)

TABLE 1-continued

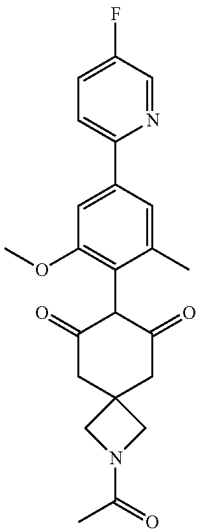
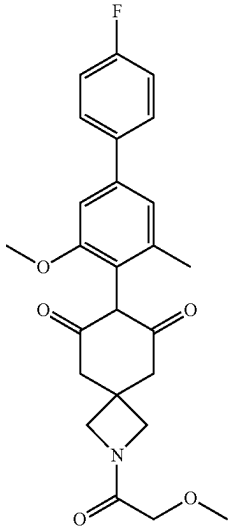
COM- POUND	STRUCTURE	NMR
1.111		^1H NMR (400 MHz, DMSO- d_6) δ = 8.65 (d, 1H), 8.11-8.01 (m, 1H), 7.87-7.75 (m, 1H), 7.46 (s, 1H), 7.41 (s, 1H), 3.99-3.88 (m, 2H), 3.75-3.67 (m, 3H), 3.66-3.59 (m, 2H), 2.90-2.66 (m, 4H), 2.07-1.98 (m, 3H), 1.83-1.72 (m, 3H)
1.112		^1H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.9 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.99-6.96 (m, 1H), 4.14 (d, J = 3.8 Hz, 2H), 4.02 (d, J = 3.2 Hz, 2H), 3.89 (d, J = 3.5 Hz, 2H), 3.77 (s, 3H), 3.39 (d, J = 3.4 Hz, 3H), 2.87 (s, 4H), 2.09 (s, 3H)

TABLE 1-continued

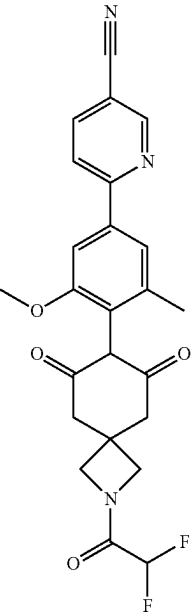
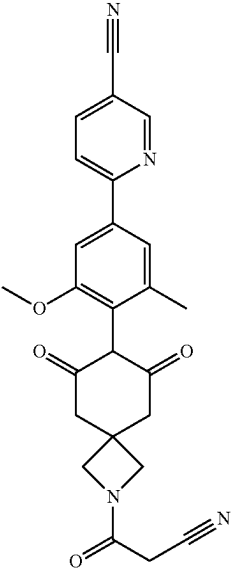
COM- POUND	STRUCTURE	NMR
1.113		^1H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.57 (s, 2H), 6.19 (td, 1H), 4.29 (s, 2H), 3.98 (s, 2H), 3.80 (d, 3H), 2.91 (s, 4H), 2.14 (s, 3H)
1.114		^1H NMR (400 MHz, d6-DMSO) δ = 9.09 (d, 1H), 8.38 (dd, 1H), 8.23 (d, 1H), 7.58 (d, 2H), 3.97 (s, 2H), 3.77 (d, 4H), 3.72 (s, 3H), 2.85-2.68 (m, 4H), 2.04 (s, 3H)

TABLE 1-continued

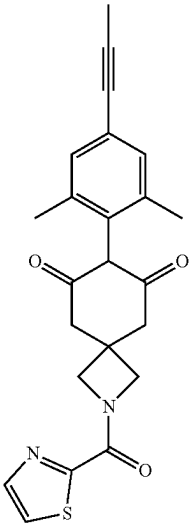
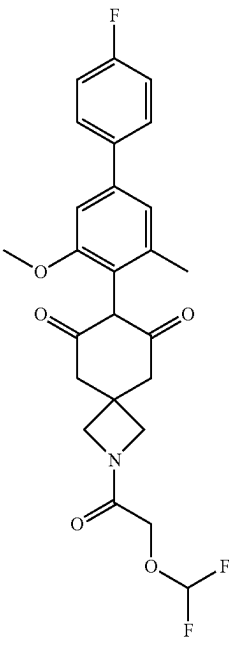
COM- POUND	STRUCTURE	NMR
1.115		¹ H NMR (400 MHz, DMSO-d ₆) δ ppm 8.04 (q, J = 3.10 Hz, 2 H) 7.02 (s, 2 H) 4.45 (s, 2 H) 3.93 (s, 2 H) 2.74-2.97 (m, 4 H) 2.02 (s, 3 H) 1.93 (s, 6 H)
1.116		¹ H NMR (400 MHz, methanol) Shift = 7.68-7.57 (m, 2H), 7.15 (tt, J = 3.2, 8.9 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 6.51 (dt, J = 3.5, 74.5 Hz, 1H), 4.45 (d, J = 2.7 Hz, 2H), 4.17 (d, J = 4.3 Hz, 2H), 3.91 (d, J = 2.8 Hz, 2H), 3.76 (d, J = 1.3 Hz, 3H), 2.88 (s, 4H), 2.09 (s, 3H)

TABLE 1-continued

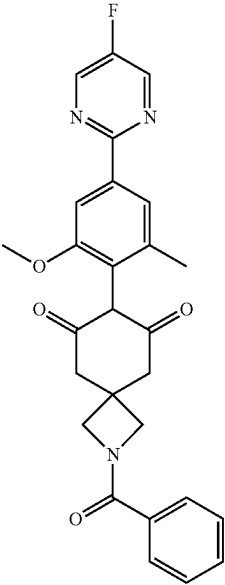
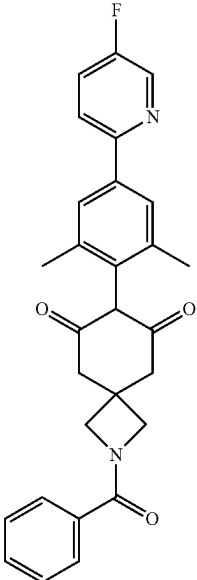
COM- POUND	STRUCTURE	NMR
1.117		¹ H NMR (400 MHz, DMSO-d ₆) δ ppm 8.97 (s, 2 H) 7.80 (br d, J = 6.85 Hz, 1 H) 7.64-7.72 (m, 3 H) 7.43-7.57 (m, 3 H) 4.13 (br s, 2 H) 3.88 (s, 2 H) 3.63-3.75 (m, 3 H) 2.79 (br s, 4 H) 1.97-2.08 (m, 3 H)
1.118		¹ H NMR (400 MHz, methanol) δ = 8.47 (d, 1H), 7.91-7.83 (m, 1H), 7.72-7.58 (m, 5H), 7.56-7.44 (m, 3H), 4.25 (s, 2H), 4.06 (s, 2H), 2.90 (s, 4H), 2.14 (s, 3H), 2.09 (s, 3H)

TABLE 1-continued

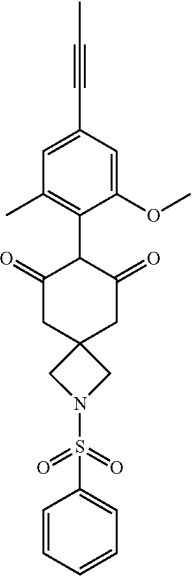
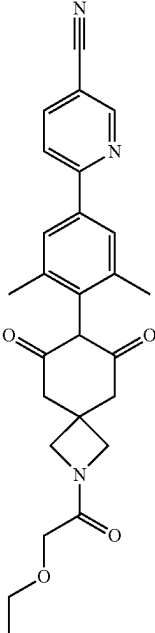
COM- POUND	STRUCTURE	NMR
1.119		^1H NMR (400 MHz, methanol) δ = 7.92-7.85 (m, 2H), 7.81-7.74 (m, 1H), 7.73-6.67 (m, 2H), 6.82 (s, 1H), 6.74 (s, 1H), 3.67-3.59 (m, 7H), 2.54 (s, 4H), 1.99 (s, 3H), 1.92 (s, 3H)
1.120		^1H NMR (400 MHz, methanol) δ = 8.92 (dd, J = 0.6, 2.1 Hz, 1H), 8.17 (dd, J = 2.2, 8.4 Hz, 1H), 8.04 (dd, J = 0.7, 8.4 Hz, 1H), 7.78 (s, 2H), 4.20 (s, 2H), 4.07 (s, 2H), 3.91 (s, 2H), 3.55 (q, J = 7.1 Hz, 2H), 2.91 (s, 4H), 2.13 (s, 6H), 1.22 (t, J = 7.0 Hz, 3H)

TABLE 1-continued

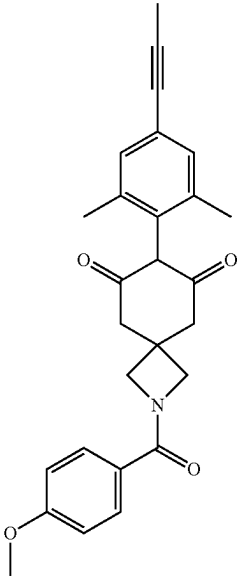
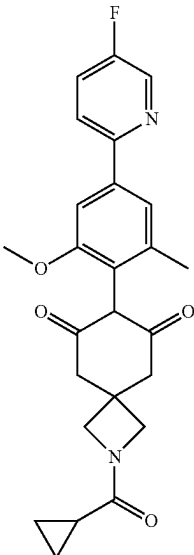
COM- POUND	STRUCTURE	NMR
1.121		¹ H NMR (400 MHz, methanol) δ ppm 7.63-7.69 (m, 2 H) 6.97-7.06 (m, 4 H) 4.26 (s, 2 H) 4.02 (s, 2 H) 3.85 (s, 3 H) 2.89 (s, 4 H) 1.93-2.02 (m, 9 H)
1.122		¹ H NMR (400 MHz, methanol) δ = 8.52-8.46 (m, 1H), 7.95-7.87 (m, 1H), 7.70-7.61 (m, 1H), 7.39 (s, 2H), 4.24-4.16 (m, 2H), 3.87-3.82 (m, 2H), 3.79 (s, 3H), 2.92-2.81 (m, 4H), 2.13 (d, 3H), 1.66-1.55 (m, 1H), 0.93-0.76 (m, 4H)

TABLE 1-continued

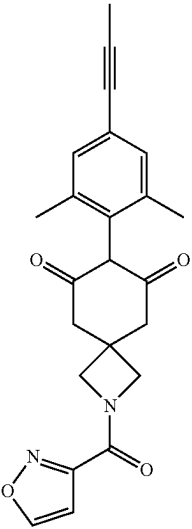
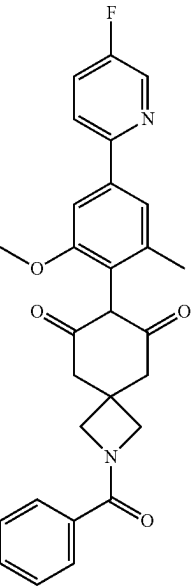
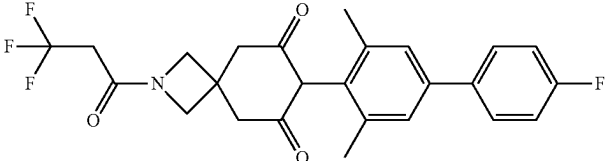
COM- POUND	STRUCTURE	NMR
1.123		^1H NMR (400 MHz, methanol) δ ppm 8.79 (d, J = 1.71 Hz, 1 H) 7.04 (s, 2 H) 6.81 (d, J = 1.59 Hz, 1 H) 4.44 (s, 2 H) 4.05 (s, 2 H) 2.91 (s, 4 H) 2.00 (d, J = 2.32 Hz, 9 H)
1.124		^1H NMR (400 MHz, methanol) δ = 8.49 (d, 1H), 7.95-7.88 (m, 1H), 7.72- 7.62 (m, 3H), 7.58-7.44 (m, 3H), 7.38 (br s, 2H), 4.28-4.18 (m, 2H), 4.09-4.02 (m, 2H), 3.85-3.71 (m, 3H), 2.88 (s, 4H), 2.17-2.06 (m, 3H)
1.125		^1H NMR (400 MHz, methanol) δ = 7.60 (dd, J = 5.4, 8.8 Hz, 2H), 7.27 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 4.16 (s, 2H), 3.90 (s, 2H), 3.24 (q, J = 10.7 Hz, 2H), 2.90 (s, 4H), 2.10 (s, 3H), 2.09 (s, 3H)

TABLE 1-continued

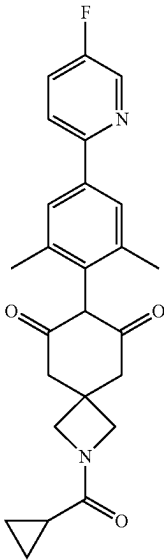
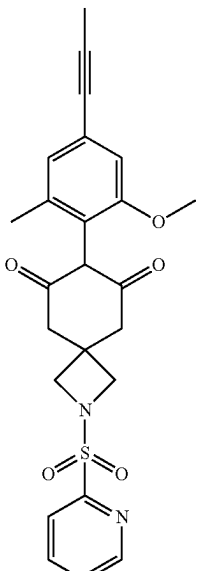
COM- POUND	STRUCTURE	NMR
1.126		^1H NMR (400 MHz, methanol) δ = 8.50-8.45 (m, 1H), 7.88 (dd, 1H), 7.68-7.62 (m, 1H), 7.61 (s, 2H), 4.27- 4.18 (m, 2H), 3.90-3.81 (m, 2H), 2.89 (s, 4H), 2.13 (d, 6H), 1.67-1.56 (m, 1H), 0.94-0.78 (m, 4H)
1.127		^1H NMR (400 MHz, methanol) δ = 8.85-8.78 (m, 1H), 8.17-8.10 (m, 1H), 8.04-8.00 (m, 1H), 7.76-7.69 (m, 1H), 6.83 (s, 1H), 6.75 (s, 1H), 3.92 (s, 4H), 3.64 (s, 3H), 2.67 (s, 4H), 2.00 (s, 3H), 1.94 (s, 3H)

TABLE 1-continued

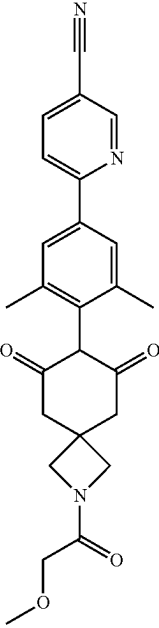
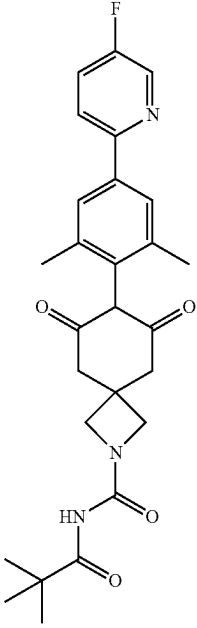
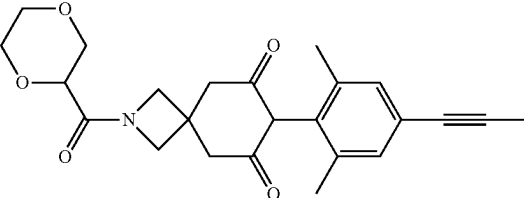
COM- POUND	STRUCTURE	NMR
1.128		^1H NMR (400 MHz, methanol) δ = 8.93 (dd, J = 0.7, 2.1 Hz, 1H), 8.17 (dd, J = 2.2, 8.4 Hz, 1H), 8.04 (dd, J = 0.7, 8.4 Hz, 1H), 7.78 (s, 2H), 4.17 (s, 2H), 4.02 (s, 2H), 3.92 (s, 2H), 3.39 (s, 3H), 2.92 (s, 4H), 2.13 (s, 6H)
1.129		^1H NMR (400 MHz, methanol) δ = 8.46 (d, 1H), 7.87 (dd, 1H), 7.64 (td, 1H), 7.59 (s, 2H), 3.94 (s, 4H), 2.82 (s, 4H), 2.12 (s, 6H), 1.22 (s, 9H)
1.130		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.30-4.21 (m, 3H), 3.93-3.78 (m, 4H), 3.74-3.66 (m, 2H), 3.65-3.57 (m, 2H), 2.87 (d, J = 5.5 Hz, 4H), 1.99 (s, 3H), 1.98 (s, 6H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.131		^1H NMR (400 MHz, methanol) δ = 8.34 (d, J = 5.1 Hz, 1H), 7.53 (td, J = 1.5, 5.1 Hz, 1H), 7.32 (s, 1H), 7.04 (d, J = 3.8 Hz, 2H), 4.26 (s, 2H), 4.06 (s, 2H), 2.92 (s, 4H), 1.99 (d, J = 1.5 Hz, 6H) 1.95 (s, 3H)
1.132		^1H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.8 Hz, 2H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 5.15 (dq, J = 1.8, 6.8, 48.4 Hz, 1H), 4.25 (br s, 2H), 3.91 (d, J = 3.1 Hz, 2H), 3.76 (d, J = 3.3 Hz, 3H), 2.88 (d, J = 2.8 Hz, 4H), 2.09 (s, 3H), 1.53 (ddd, J = 1.6, 6.7, 24.5 Hz, 3H)
1.133		^1H NMR (400 MHz, methanol) δ = 7.63-7.58 (m, 2H), 7.27 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 4.12 (s, 2H), 3.89 (s, 2H), 2.91 (s, 4H), 2.71-2.65 (m, 2H), 2.60-2.54 (m, 2H), 2.09 (d, J = 3.3 Hz, 6H)
1.134		
1.135		^1H NMR (400 MHz, methanol) δ = 8.67-8.60 (m, 1H), 8.49 (d, 1H), 8.05-8.00 (m, 1H), 7.98-7.88 (m, 2H), 7.70-7.62 (m, 1H), 7.54-7.47 (m, 1H), 7.40 (s, 2H), 4.58 (d, 2H), 4.08 (d, 2H), 3.80 (d, 3H), 2.88 (s, 4H), 2.15 (d, 3H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.136		^1H NMR (400 MHz, methanol) δ = 8.35 (d, J = 5.3 Hz, 1H), 7.63-7.56 (m, 2H), 7.56-7.52 (m, 1H), 7.33 (s, 1H), 7.28-7.22 (m, 2H), 7.16-7.09 (m, 2H), 4.30-4.22 (m, 2H), 4.07 (br s, 2H), 2.97-2.84 (m, 4H), 2.11 (s, 3H), 2.07 (s, 3H)
1.137		^1H NMR (400 MHz, methanol) δ = 8.68 (d, J = 5.1 Hz, 1H), 8.20 (d, J = 7.7 Hz, 1H), 8.03 (dt, J = 1.6, 7.7 Hz, 1H), 7.68-7.58 (m, 3H), 7.27 (s, 2H), 7.14 (t, J = 8.8 Hz, 2H), 4.26-3.94 (m, 4H), 2.95 (s, 4H), 2.10 (s, 6H)
1.138		^1H NMR (400 MHz, methanol) δ = 8.46 (d, 1H), 7.87 (dd, 1H), 7.67-7.61 (m, 1H), 7.60 (s, 2H), 4.12 (s, 2H), 3.89 (s, 2H), 2.82 (s, 4H), 2.13 (s, 6H). Deuteration of alpha-CN protons observed.

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.139		¹ H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.56 (s, 2H), 3.95 (s, 4H), 3.79 (s, 3H), 2.87 (s, 4H), 2.13 (s, 3H), 1.23 (s, 9H)
1.140		¹ H NMR (400 MHz, methanol) Shift = 7.67-7.60 (m, 2H), 7.15 (tt, J = 2.0, 8.8 Hz, 2H), 7.06 (d, J = 0.9 Hz, 1H) 6.97 (d, J = 1.6 Hz, 1H), 3.78 (d, J = 4.5 Hz, 4H), 3.77 (s, 3H), 3.16 (q, J = 7.3 Hz, 2H), 2.83 (d, J = 1.2 Hz, 4H), 2.09 (s, 3H), 1.10 (t, J = 7.2 Hz, 3H)

TABLE 1-continued

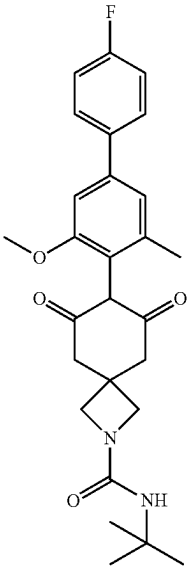
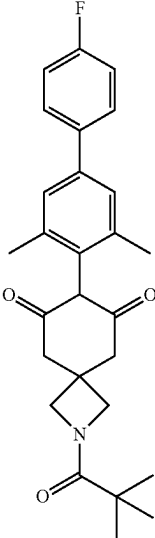
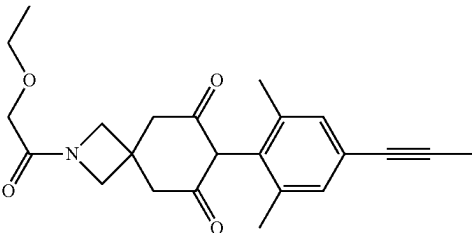
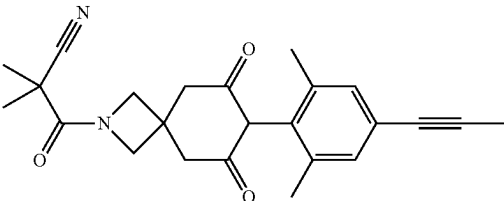
COM- POUND	STRUCTURE	NMR
1.141		^1H NMR (400 MHz, methanol) Shift = 7.67-7.59 (m, 2H), 7.15 (tt, J = 2.8, 8.9 Hz, 2H), 7.05 (d, J = 0.7 Hz, 1H), 6.97 (d, J = 1.1 Hz, 1H), 3.76 (s, 7H), 2.82 (d, J = 2.6 Hz, 4H), 2.09 (s, 3H), 1.32 (s, 9H)
1.142		^1H NMR (400 MHz, methanol) δ = 7.63-7.58 (m, 2H), 7.27 (s, 2H), 7.16- 7.10 (m, 2H), 4.34 (br s, 2H), 3.84 (br s, 2H), 2.88 (s, 4H), 2.09 (s, 6H), 1.22 (s, 9H)
1.143		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.17 (s, 2H), 4.05 (s, 2H), 3.88 (s, 2H), 3.53 (q, J = 7.0 Hz, 2H), 2.87 (s, 4H), 1.99 (s, 3H), 1.98 (s, 6H), 1.21 (t, J = 7.0 Hz, 3H)
1.144		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.45 (s, 2H), 3.92 (s, 2H), 2.90 (d, J = 1.5 Hz, 4H), 1.99 (s, 3H), 1.98 (s, 6H), 1.56 (s, 6H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.145		
1.146		^1H NMR (400 MHz, methanol) δ = 7.64-7.57 (m, 2H), 7.27 (s, 2H), 7.18- 7.10 (m, 2H), 4.33-4.23 (m, 3H), 3.93-3.87 (m, 3H), 3.85-3.79 (m, 1H), 3.75-3.67 (m, 2H), 3.66-3.57 (m, 2H), 2.89 (d, J = 5.4 Hz, 4H), 2.09 (s, 6H)
1.147		^1H NMR (400 MHz, DMSO- d_6) δ ppm 8.97 (s, 2 H) 7.80 (s, 1 H) 7.70 (s, 1 H) 6.72 (d, J = 6.36 Hz, 1 H) 4.42 (d, J = 4.77 Hz, 2 H) 4.00 (s, 2 H) 3.67- 3.78 (m, 5 H) 2.77 (br s, 4 H) 2.04 (s, 3 H)
1.148		^1H NMR (400 MHz, methanol) δ = 7.49 (d, J = 1.0 Hz, 1 H) 6.85 (s, 1H), 6.77 (s, 1H), 4.28 (s, 2H), 3.98 (d, J = 7.5 Hz, 3H), 3.92 (d, J = 2.6 Hz, 2H), 3.67 (d, J = 1.1 Hz, 3H), 2.85 (s, 4H), 2.01 (s, 3H), 1.99 (s, 3H)

TABLE 1-continued

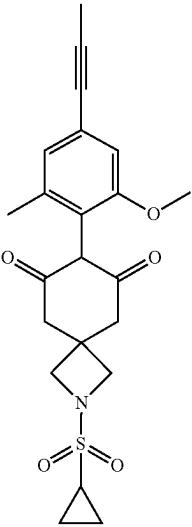
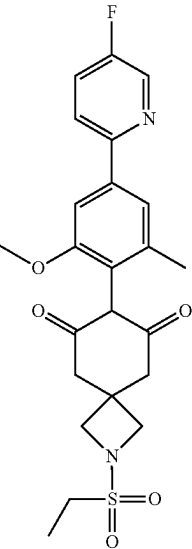
COM- POUND	STRUCTURE	NMR
1.149		¹ H NMR (400 MHz, methanol) δ = 6.84 (s, 1H), 6.77 (s, 1H), 3.86-3.80 (m, 4H), 3.66 (s, 3H), 2.86 (s, 4H), 2.67-2.57 (m, 1H), 2.00 (s, 3H), 1.98 (s, 3H), 1.11-1.03 (m, 4H)
1.150		¹ H NMR (400 MHz, methanol) δ = 8.53-8.46 (m, 1H), 7.95-7.88 (m, 1H), 7.67 (dd, 1H), 7.39 (s, 2H), 3.85- 3.80 (m, 4H), 3.78 (s, 3H), 3.10 (q, 2H), 2.87 (s, 4H), 2.11 (s, 3H), 1.34 (t, 3H)

TABLE 1-continued

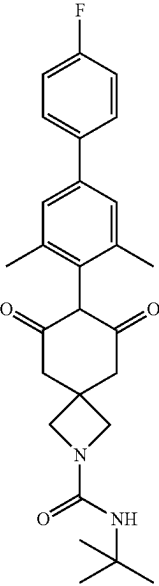
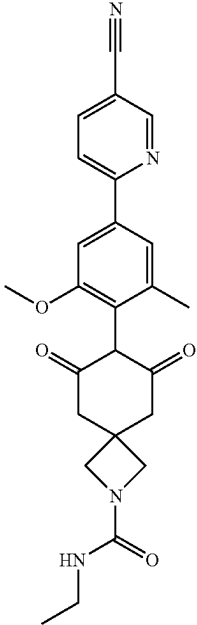
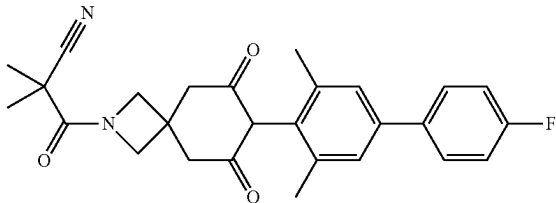
COM- POUND	STRUCTURE	NMR
1.151		^1H NMR (400 MHz, methanol) δ = 7.63-7.57 (m, 2H), 7.26 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 3.78 (s, 4H), 2.85 (s, 4H), 2.09 (s, 6H), 1.32 (s, 9H)
1.152		^1H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.56 (s, 2H), 3.82-3.75 (m, 7H), 3.16 (q, 2H), 2.91-2.78 (m, 4H), 2.13 (s, 3H), 1.11 (t, 3H)
1.153		^1H NMR (400 MHz, methanol) δ = 7.61 (dd, J = 5.4, 8.7 Hz, 2H), 7.27 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 4.48 (s, 2H), 3.95 (s, 2H), 2.99-2.88 (m, 4H), 2.09 (s, 6H), 1.57 (s, 6H)

TABLE 1-continued

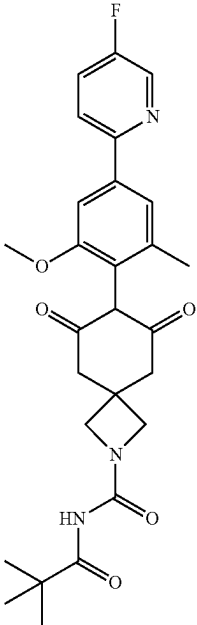
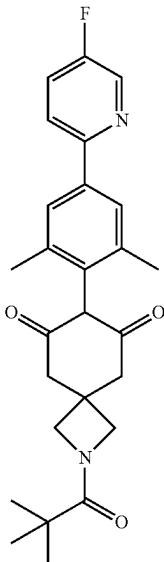
COM- POUND	STRUCTURE	NMR
1.154		^1H NMR (400 MHz, methanol) δ = 8.49 (d, 1H), 7.96-7.88 (m, 1H), 7.71- 7.62 (m, 1H), 7.39 (s, 2H), 4.01- 3.89 (m, 4H), 3.78 (s, 3H), 2.87 (s, 4H), 2.12 (s, 3H), 1.23 (s, 9H)
1.155		^1H NMR (400 MHz, methanol) δ = 8.47 (d, 1H), 7.87 (dd, 1H), 7.65 (td, 1H), 7.60 (s, 2H), 4.33 (s, 2H), 3.84 (s, 2H), 2.85 (s, 4H), 2.12 (s, 6H), 1.22 (s, 9H)

TABLE 1-continued

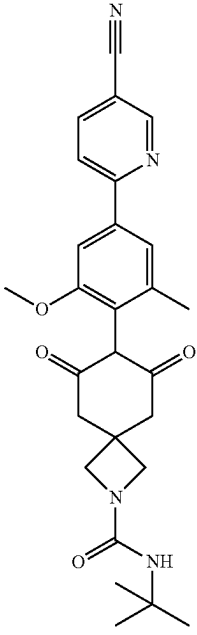
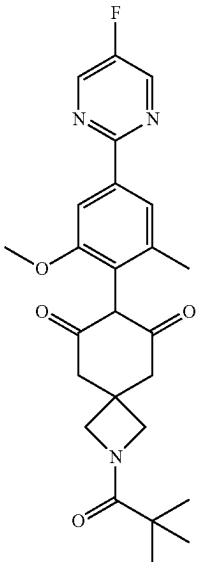
COM- POUND	STRUCTURE	NMR
1.156		¹ H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.56 (s, 2H), 3.80 (s, 3H), 3.77 (d, 4H), 2.89-2.76 (m, 4H), 2.13 (s, 3H), 1.33 (s, 9H)
1.157		¹ H NMR (400 MHz, DMSO-d ₆) δ ppm 8.97 (s, 2 H) 7.80 (d, J = 0.73 Hz, 1 H) 7.70 (s, 1 H) 4.16 (br s, 2 H) 3.70 (s, 3 H) 3.60-3.68 (m, 1 H) 3.65 (br dd, J = 9.84, 4.22 Hz, 1 H) 2.67-2.87 (m, 4 H) 2.04 (s, 3 H) 1.12 (s, 9 H)

TABLE 1-continued

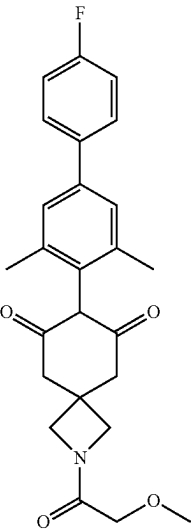
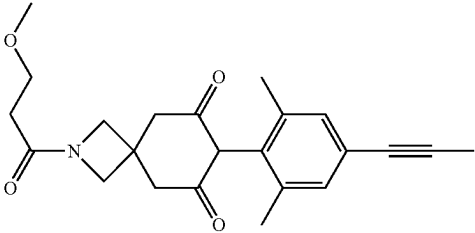
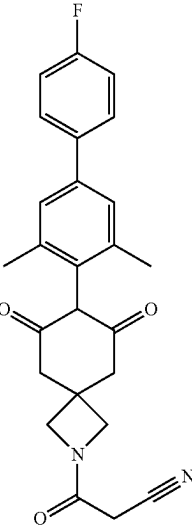
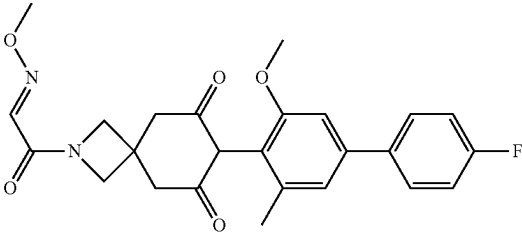
COM- POUND	STRUCTURE	NMR
1.158		^1H NMR (400 MHz, methanol) δ = 7.60 (dd, J = 5.4, 8.7 Hz, 2H), 7.26 (s, 2H), 7.13 (t, J = 8.9 Hz, 2H), 4.16 (s, 2H), 4.02 (s, 2H), 3.91 (s, 2H), 3.39 (s, 3H), 2.90 (br s, 4H), 2.09 (s, 6H)
1.159		^1H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.11 (s, 2H), 3.84 (s, 2H), 3.64 (t, J = 6.0 Hz, 2H), 3.33 (s, 3H), 2.86 (s, 4H), 2.39 (t, J = 6.0 Hz, 2H), 1.99 (s, 3H), 1.99-1.97 (m, 6H)
1.160		^1H NMR (400 MHz, methanol) δ = 7.63-7.58 (m, 2H), 7.27 (s, 2H), 7.16-7.10 (m, 2H), 4.14 (s, 2H)*, 3.91 (s, 2H), 3.65-3.62 (m, 2H), 2.91 (s, 4H), 2.09 (s, 6H)
1.161		^1H NMR (400 MHz, methanol) δ = 7.64 (dd, J = 5.4, 8.8 Hz, 2H), 7.50 (d, J = 0.7 Hz, 1H), 7.15 (t, J = 8.8 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.30 (d, J = 2.6 Hz, 2H), 3.99 (d, J = 7.7 Hz, 3H), 3.95 (d, J = 3.5 Hz, 2H), 3.77 (d, J = 1.1 Hz, 3H), 2.88 (s, 4H), 2.10 (s, 3H)

TABLE 1-continued

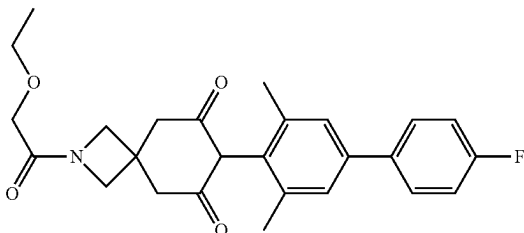
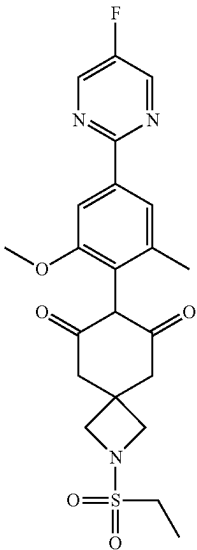
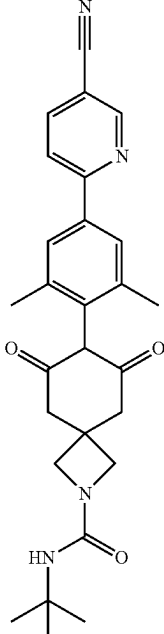
COM- POUND	STRUCTURE	NMR
1.162		^1H NMR (400 MHz, methanol) δ = 7.63-7.58 (m, 2H), 7.27 (s, 2H), 7.17-7.09 (m, 2H), 4.19 (s, 2H), 4.06 (s, 2H), 3.91 (s, 2H), 3.54 (q, J = 7.0 Hz, 2H), 2.90 (s, 4H), 2.09 (s, 6H), 1.22 (t, J = 7.0 Hz, 3H)
1.163		^1H NMR (400 MHz, methanol) δ ppm 8.75 (s, 2 H) 7.88 (s, 1 H) 7.81 (d, J = 0.86 Hz, 1 H) 3.82 (d, J = 2.93 Hz, 4 H) 3.78 (s, 3 H) 3.09 (d, J = 7.34 Hz, 2 H) 2.87 (s, 4 H) 2.11 (s, 3 H) 1.33 (t, J = 7.40 Hz, 3 H)
1.164		^1H NMR (400 MHz, methanol) δ = 8.92 (dd, J = 0.6, 2.1 Hz, 1H), 8.17 (dd, J = 2.2, 8.4 Hz, 1H), 8.04 (dd, J = 0.6, 8.4 Hz, 1H), 7.78 (s, 2H), 3.79 (s, 4H), 2.86 (s, 4H), 2.13 (s, 6H), 1.33 (s, 9H)

TABLE 1-continued

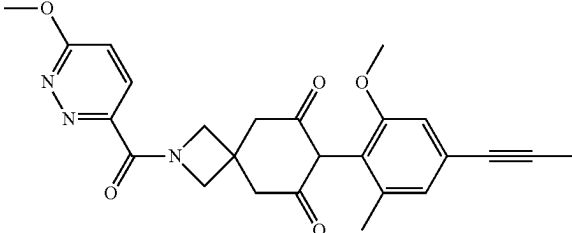
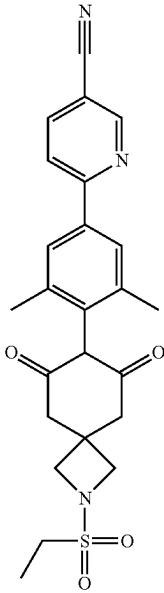
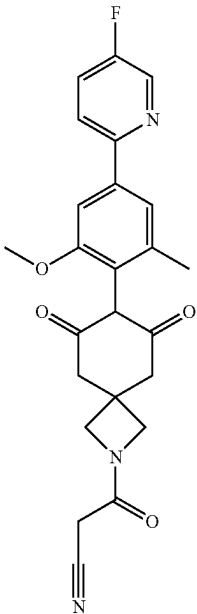
COM- POUND	STRUCTURE	NMR
1.165		
1.166		^1H NMR (400 MHz, methanol) δ = 8.92 (d, J = 1.6 Hz, 1H), 8.17 (dd, J = 2.2, 8.3 Hz, 1H), 8.04 (d, J = 8.7 Hz, 1H), 7.78 (s, 2H), 3.85 (s, 4H), 3.15 (q, J = 7.3 Hz, 2H), 2.91 (s, 4H), 2.12 (s, 6H), 1.34 (t, J = 7.7 Hz, 3H)
1.167		^1H NMR (400 MHz, methanol) δ = 8.51-8.47 (m, 1H), 7.95-7.89 (m, 1H), 7.70-7.62 (m, 1H), 7.39 (s, 2H), 4.15-4.09 (m, 2H), 3.92-3.86 (m, 2H), 3.79 (s, 3H), 2.86 (s, 4H), 2.12 (s, 3H)

TABLE 1-continued

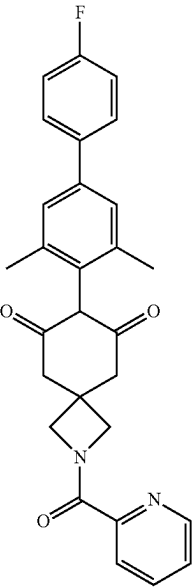
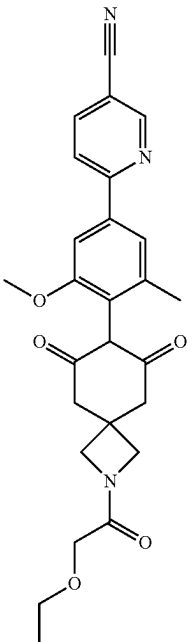
COM- POUND	STRUCTURE	NMR
1.168		¹ H NMR (400 MHz, methanol) δ = 8.64-8.61 (m, 1H), 8.04-8.01 (m, 1H), 7.94 (dt, J = 1.8, 7.7 Hz, 1H), 7.61 (dd, J = 5.5, 8.7 Hz, 2H), 7.51 (ddd, J = 1.2, 4.8, 7.6 Hz, 1H), 7.27 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 4.61 (s, 2H), 4.09 (s, 2H), 2.97-2.90 (m, 4H), 2.12 (s, 3H), 2.11 (s, 3H)
1.169		¹ H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.19 (dd, 1H), 8.08 (d, 1H), 7.57 (s, 2H), 4.18-3.89 (m, 6H), 3.81 (d, 3H), 3.58-3.52 (m, 2H), 2.88 (s, 4H), 2.13 (s, 3H), 1.23 (q, 3H)

TABLE 1-continued

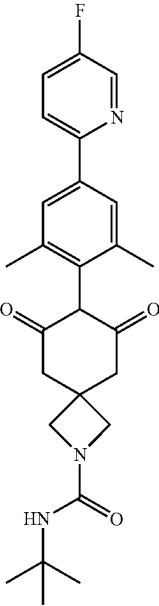
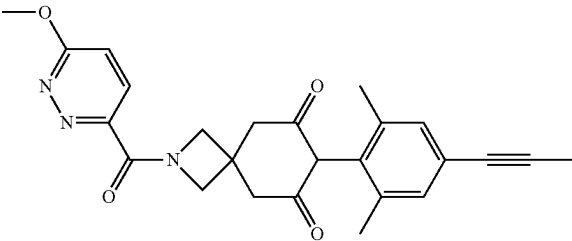
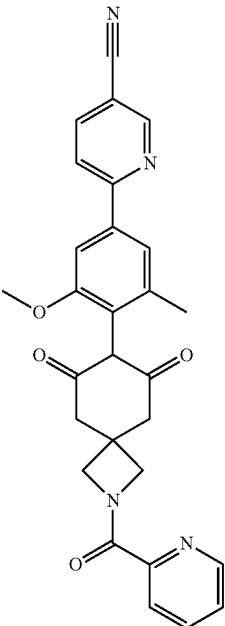
COM- POUND	STRUCTURE	NMR
1.170		^1H NMR (400 MHz, methanol) δ = 8.48 (d, 1H), 7.91-7.84 (m, 1H), 7.69-7.62 (m, 1H), 7.60 (s, 2H), 3.78 (s, 4H), 2.85 (s, 4H), 2.12 (s, 6H), 1.32 (s, 9H)
1.171		^1H NMR (400 MHz, methanol) δ = 8.09 (d, J = 9.3 Hz, 1H), 7.27 (d, J = 9.2 Hz, 1H), 7.05 (s, 2H), 4.62 (s, 2H), 4.16 (s, 3H), 4.09 (s, 2H), 2.94 (s, 4H), 2.01 (s, 6H), 2.00 (s, 3H)
1.172		^1H NMR (400 MHz, d_6 -DMSO) δ = 9.09 (d, 1H), 8.66-8.61 (m, 1H), 8.38 (dd, 1H), 8.23 (d, 1H), 8.00-7.93 (m, 2H), 7.64-7.59 (m, 1H), 7.59-7.50 (m, 2H), 4.43 (d, 2H), 3.90 (s, 2H), 3.73 (d, 3H), 2.79 (s, 4H), 2.06 (d, 3H)

TABLE 1-continued

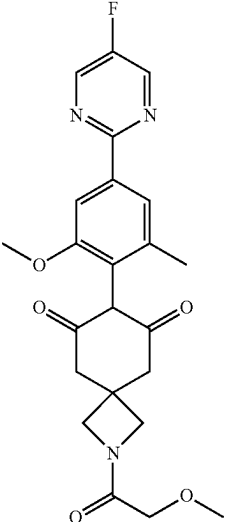
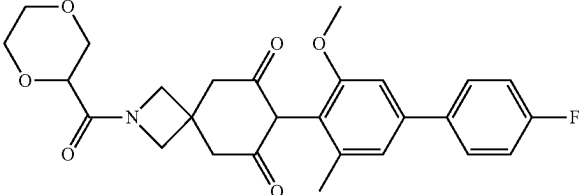
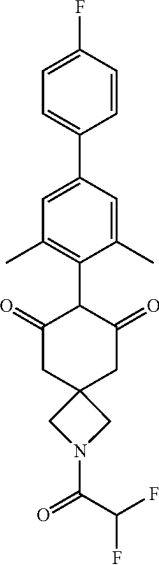
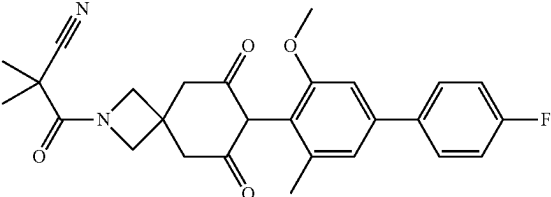
COM- POUND	STRUCTURE	NMR
1.173		¹ H NMR (400 MHz, DMSO-d ₆) δ ppm 8.97 (s, 2 H) 7.80 (s, 1 H) 7.70 (s, 1 H) 3.97 (br d, J = 2.93 Hz, 2 H) 3.93 (d, J = 5.14 Hz, 2 H) 3.70 (s, 5 H) 3.28 (d, J = 5.75 Hz, 3 H) 2.76 (br s, 4 H) 2.04 (s, 3 H)
1.174		¹ H NMR (400 MHz, methanol) Shift = 7.69-7.57 (m, 2H), 7.21-7.09 (m, 2H), 7.06 (s, 1H), 6.97 (s, 1H), 4.27-4.20 (m, 2H), 3.94-3.79 (m, 4H), 3.76 (s, 3H), 3.74-3.57 (m, 4H), 2.85 (d, J = 5.3 Hz, 4H), 2.09 (d, J = 1.3 Hz, 3H)
1.175		¹ H NMR (400 MHz, methanol) δ = 7.60 (dd, J = 5.4, 8.8 Hz, 2H), 7.27 (s, 2H), 7.13 (t, J = 8.8 Hz, 2H), 6.18 (t, J = 52.8 Hz, 1H), 4.30 (s, 2H), 3.99 (s, 2H), 2.93 (s, 4H), 2.09 (s, 6H)
1.176		¹ H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.8 Hz, 2H), 7.14 (t, J = 8.9 Hz, 2H), 7.06 (s, 1H), 6.97 (s, 1H), 4.50-4.41 (m, 2H), 3.92 (s, 2H), 3.76 (s, 3H), 2.89 (s, 4H), 2.10 (s, 3H), 1.57 (s, 6H)

TABLE 1-continued

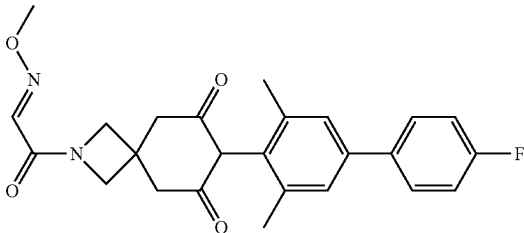
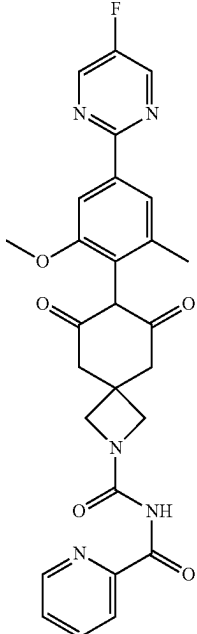
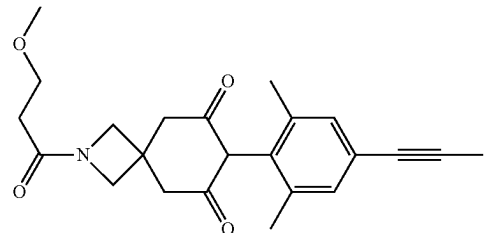
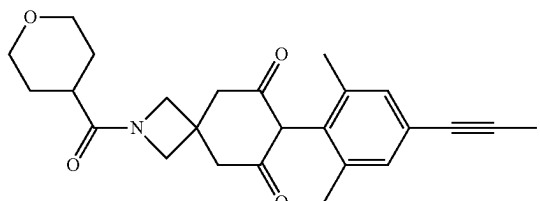
COM- POUND	STRUCTURE	NMR
1.177		¹ H NMR (400 MHz, methanol) Shift = 7.65-7.57 (m, 2H), 7.51 (s, 1H), 7.27 (s, 2H), 7.18-7.09 (m, 2H), 4.32 (s, 2H), 3.98 (s, 3H), 3.97 (s, 2H), 2.92 (s, 4H), 2.10 (s, 6H)
1.178		¹ H NMR (400 MHz, DMSO-d ₆) δ ppm 10.70 (br d, J = 2.57 Hz, 1 H) 10.08 (s, 1 H) 8.97 (s, 2 H) 8.71 (d, J = 4.52 Hz, 1 H) 8.04-8.16 (m, 2 H) 7.81 (s, 1 H) 7.67-7.76 (m, 2 H) 3.75-4.17 (m, 4 H) 3.71 (s, 3 H) 2.59-3.05 (m, 4 H) 2.04 (s, 3 H)
1.179		¹ H NMR (400 MHz, methanol) δ = 6.85 (s, 1H), 6.77 (s, 1H), 4.08 (s, 2H), 3.82 (d, J = 3.7 Hz, 2H), 3.67 (s, 3H), 3.64 (dt, J = 0.9, 6.0 Hz, 2H), 3.33 (d, J = 1.3 Hz, 3H), 2.83 (s, 4H), 2.39 (dt, J = 2.9, 6.0 Hz, 2H), 2.00 (s, 3H), 1.99 (d, J = 2.3 Hz, 3H)
1.180		¹ H NMR (400 MHz, methanol) δ = 7.04 (s, 2H), 4.14 (s, 2H), 3.99-3.92 (m, 2H), 3.83 (s, 2H), 3.50-3.42 (m, 2H), 2.87 (s, 4H), 2.65-2.54 (m, 1H), 1.99 (s, 6H), 1.98 (s, 3H), 1.80-1.68 (m, 2H), 1.66-1.58 (m, 2H)

TABLE 1-continued

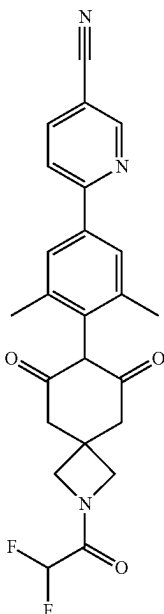
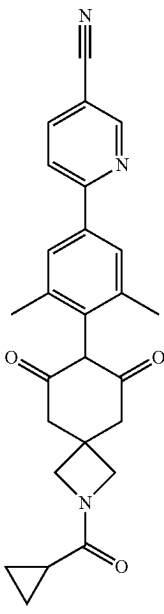
COM- POUND	STRUCTURE	NMR
1.181		¹ H NMR (400 MHz, methanol) δ = 8.92 (dd, J = 0.7, 2.1 Hz, 1H), 8.17 (dd, J = 2.2, 8.4 Hz, 1H), 8.04 (dd, J = 0.6, 8.3 Hz, 1H), 7.78 (s, 2H), 6.38-6.02 (t, 1H), 4.31 (s, 2H), 4.00 (s, 2H), 2.95 (s, 4H), 2.13 (s, 6H)
1.182		¹ H NMR (400 MHz, methanol) δ = 8.92 (dd, J = 0.9, 2.2 Hz, 1H), 8.17 (dd, J = 2.3, 8.4 Hz, 1H), 8.06-8.00 (m, 1H), 7.78 (s, 2H), 4.23 (s, 2H), 3.87 (s, 2H), 2.96-2.88 (m, 4H), 2.15 (s, 3H), 2.13 (br s, 3H), 1.65-1.57 (m, 1H), 0.95-0.78 (m, 4H)

TABLE 1-continued

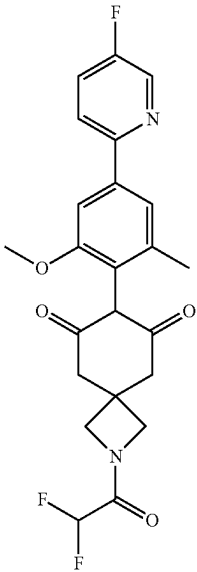
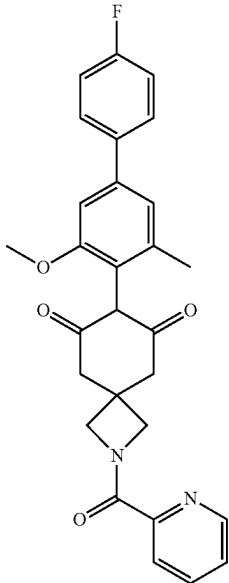
COM- POUND	STRUCTURE	NMR
1.183		¹ H NMR (400 MHz, methanol) δ = 8.52-8.47 (m, 1H), 7.95-7.88 (m, 1H), 7.72-7.62 (m, 1H), 7.40 (s, 2H), 6.35-6.01 (m, 1H), 4.31-4.26 (m, 2H), 4.00-3.94 (m, 2H), 3.81-3.77 (m, 3H), 2.88 (s, 4H), 2.12 (s, 3H)
1.184		¹ H NMR (400 MHz, methanol) δ = 8.64 (t, J = 5.0 Hz, 1H), 8.07-8.01 (m, 1H), 7.99-7.93 (m, 1H), 7.64 (dd, J = 5.4, 8.8 Hz, 2H), 7.55-7.50 (m, 1H), 7.15 (t, J = 8.8 Hz, 2H), 7.07 (s, 1H), 6.98 (s, 1H), 4.59 (d, J = 3.3 Hz, 2H), 4.08 (d, J = 3.2 Hz, 2H), 3.78 (d, J = 4.8 Hz, 3H), 2.91 (s, 4H), 2.11 (d, J = 5.6 Hz, 3H)

TABLE 1-continued

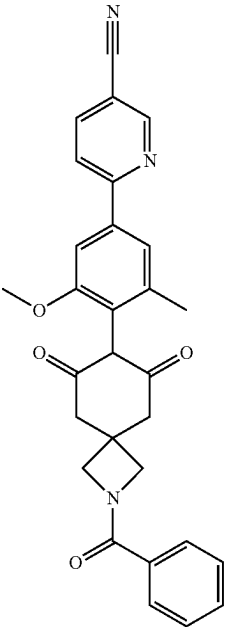
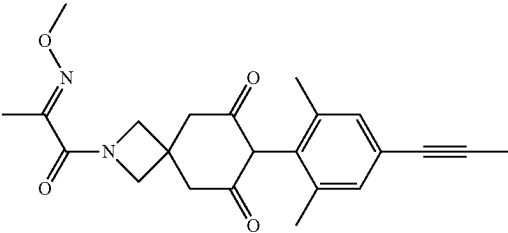
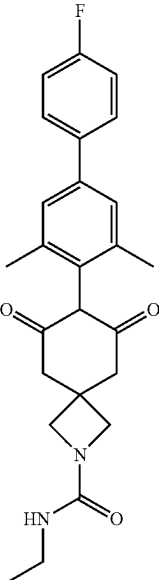
COM- POUND	STRUCTURE	NMR
1.185		^1H NMR (400 MHz, Methanol) δ = 8.94 (d, 1H), 8.18 (dd, 1H), 8.08 (d, 1H), 7.68 (d, 2H), 7.62-7.42 (m, 5H), 4.24 (s, 2H), 4.06 (d, 2H), 3.79 (d, 3H), 2.91 (s, 4H), 2.13 (d, 3H)
1.186		
1.187		^1H NMR (400 MHz, methanol) δ = 7.63-7.57 (m, 2H), 7.26 (s, 2H), 7.17-7.10 (m, 2H), 3.80 (s, 4H), 3.16 (q, J = 7.2 Hz, 2H), 2.87 (s, 4H), 2.09 (s, 6H), 1.10 (t, J = 7.2 Hz, 3H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.188		^1H NMR (400 MHz, methanol) δ = 8.66-8.61 (m, 1H), 8.50-8.46 (m, 1H), 8.05-8.01 (m, 1H), 7.97-7.91 (m, 1H), 7.91-7.86 (m, 1H), 7.69-7.63 (m, 1H), 7.62 (s, 2H), 7.54-7.48 (m, 1H), 4.61 (s, 2H), 4.09 (s, 2H), 2.94 (s, 4H), 2.14 (d, 6H)
1.189		^1H NMR (400 MHz, methanol) δ ppm 8.76 (s, 2 H) 7.90 (s, 1 H) 7.83 (s, 1 H) 6.04-6.34 (m, 1 H) 4.28 (s, 2 H) 3.97 (s, 2 H) 3.79 (d, J = 2.57 Hz, 3 H) 2.89 (s, 4 H) 2.13 (s, 3 H)
1.190		^1H NMR (400 MHz, methanol) δ = 7.63 (dd, J = 5.4, 8.9 Hz, 2H), 7.15 (t, J = 8.9 Hz, 2H), 7.06 (s, 1H), 6.98 (s, 1H), 4.17 (d, J = 3.5 Hz, 2H), 4.06 (d, J = 2.8 Hz, 2H), 3.89 (d, J = 3.9 Hz, 2H), 3.76 (d, J = 1.6 Hz, 3H), 3.55 (dq, J = 3.4, 7.0 Hz, 2H), 2.87 (s, 4H), 2.09 (s, 3H), 1.22 (q, J = 6.9 Hz, 3H)

TABLE 1-continued

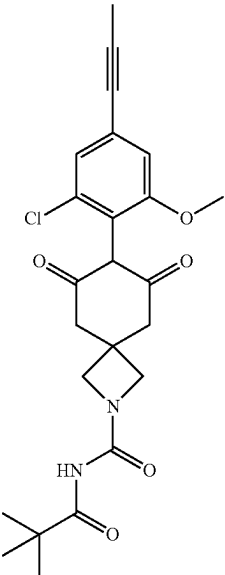
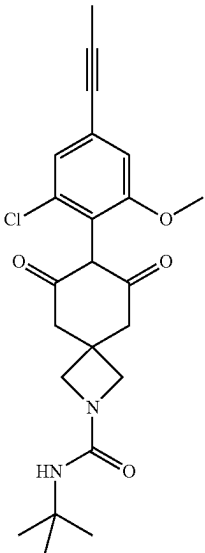
COM- POUND	STRUCTURE	NMR
1.191		^1H NMR (400 MHz, DMSO- d_6) δ = 10.90 (bs, 1H), 9.30 (s, 1H), 7.01 (s, 1H), 6.90 (s, 1H), 3.73-3.71 (m, 4H), 3.64 (s, 3H), 2.83 (m, 4H), 2.05 (s, 3H), 1.14-1.12 (m, 9H)
1.192		^1H NMR (400 MHz, DMSO- d_6) δ = 10.86 (br s, 1H), 7.01 (s, 1H), 6.91 (s, 1H), 5.68 (s, 1H), 3.65 (s, 3H), 3.55 (s, 4H), 2.74-2.56 (m, 4H), 2.05 (s, 3H), 1.22 (s, 9H)

TABLE 1-continued

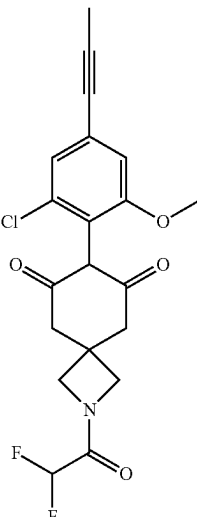
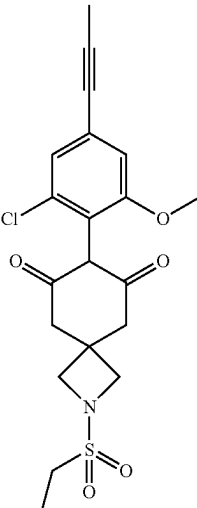
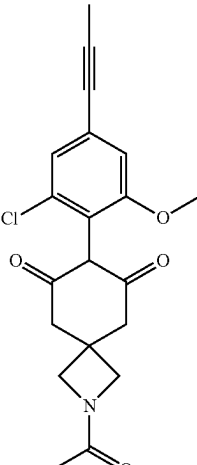
COM- POUND	STRUCTURE	NMR
1.193		^1H NMR (400 MHz, methanol) δ = 7.00 (s, 1H), 6.87 (s, 1H), 6.16 (t, 1H), 4.25 (s, 2H), 3.94 (s, 2H), 3.70 (s, 3H), 2.86 (s, 4H), 2.02 (s, 3H)
1.194		^1H NMR (400 MHz, DMSO- d_6) δ = 10.93 (br s, 1H), 7.00 (s, 1H), 6.91 (s, 1H), 3.69-3.64 (m, 7H), 3.17-3.15 (m, 2H), 2.89 (m, 2H), 2.62 (m, 2H), 2.05 (s, 3H), 1.21 (m, 3H)
1.195		^1H NMR (400 MHz, d_6 -DMSO, 100° C., VT) δ = 10.30 (bs, 1H), 6.99 (s, 1H), 6.90 (s, 1H), 3.85 (br, 4H), 3.68 (s, 3H), 2.74 (s, 4H), 2.05 (s, 3H), 1.78 (s, 3H)

TABLE 1-continued

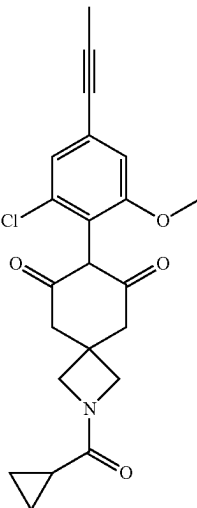
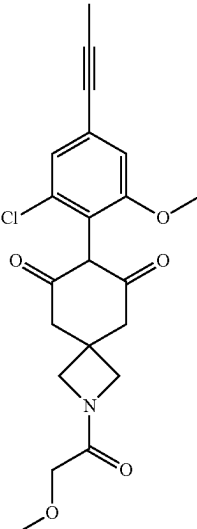
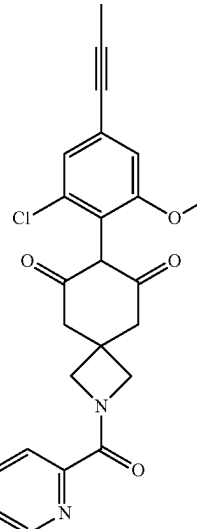
COM- POUND	STRUCTURE	NMR
1.196		^1H NMR (400 MHz, methanol) δ = 7.00 (s, 1H), 6.87 (s, 1H), 4.16 (s, 2H), 3.81 (s, 2H), 3.70 (s, 3H), 2.89-2.80 (m, 4H), 2.02 (s, 3H), 1.60-1.55 (m, 1H), 0.86-0.81 (m, 4H)
1.197		^1H NMR (400 MHz, methanol) δ = 7.00 (s, 1H), 6.87 (s, 1H), 4.11 (br s, 2H), 4.00-3.99 (m, 2H), 3.86 (s, 2H), 3.70 (s, 3H), 3.38-3.36 (m, 3H), 2.83 (s, 4H), 2.02 (s, 3H)
1.198		^1H NMR (400 MHz, d6-DMSO, VT, 100° C.) δ = 8.60 (m, 1H), 7.95-7.93 (m, 2H), 7.50 (m, 1H), 7.00 (s, 1H), 6.90 (s, 1H), 4.39 (m, 2H), 3.90 (m, 2H), 3.69 (s, 3H), 2.78 (s, 4H), 2.06-2.04 (m, 3H)

TABLE 1-continued

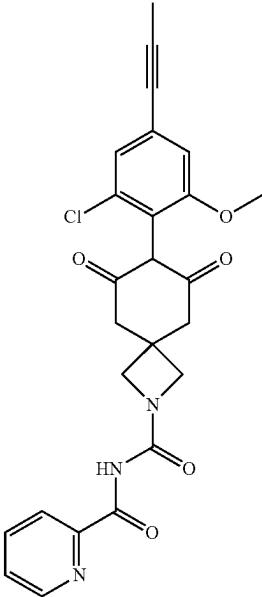
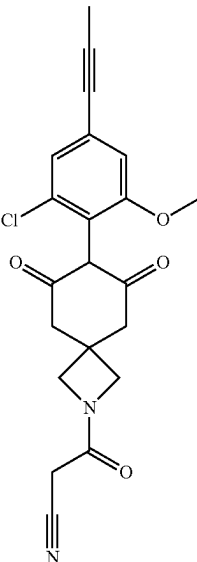
COM- POUND	STRUCTURE	NMR
1.199		¹ HNMR (400 MHz, d ₆ -DMSO, VT, 100° C.) δ 9.38 (bs, 1H), 8.68-8.61 (m, 1H), 8.11-8.03 (m, 2H), 7.68-7.65 (m, 1H), 6.99 (s, 1H), 6.89 (s, 1H), 3.93-3.92 (m, 4H), 3.67 (s, 3H), 2.78 (s, 4H), 2.05 (s, 3H)
1.200		¹ HNMR (400 MHz, d ₆ -DMSO, VT, 100° C.) δ 10.33 (bs, 1H), 6.99 (s, 1H), 6.90 (s, 1H), 4.06-3.93 (m, 2H), 3.86-3.78 (m, 2H), 3.68 (s, 3H), 3.63 (s, 2H), 2.76 (bs, 4H), 2.06 (s, 3H).

TABLE 1-continued

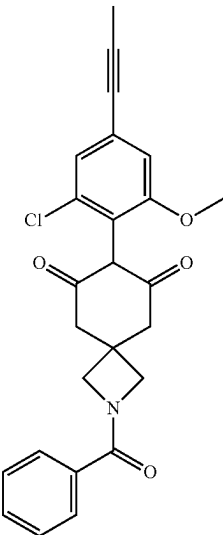
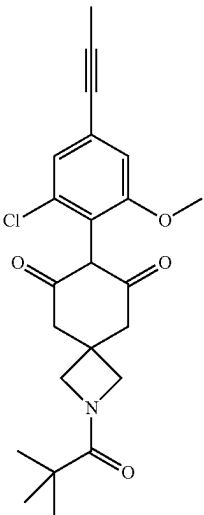
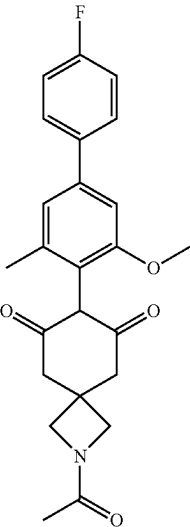
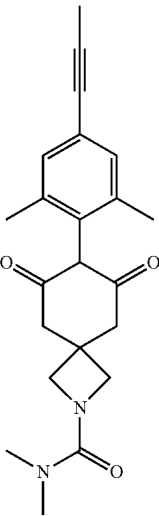
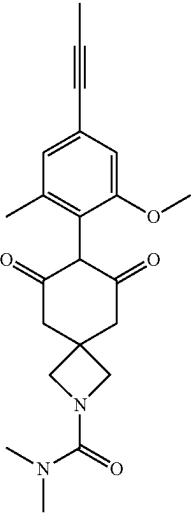
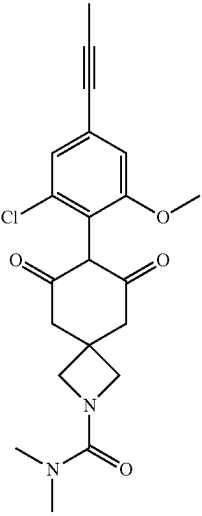
COM- POUND	STRUCTURE	NMR
1.201		^1H NMR (400 MHz, methanol) δ = 7.66 (br s, 2H), 7.52-7.45 (m, 3H), 7.01-6.99 (m, 1H), 6.87 (br s, 1H), 4.20-4.19 (m, 2H), 4.02 (s, 2H), 3.71-3.66 (m, 3H), 2.86 (s, 4H), 2.02 (s, 3H)
1.202		^1H NMR (400 MHz, methanol) δ = 7.00 (s, 1H), 6.87 (s, 1H), 4.28 (br s, 2H), 3.80 (br s, 2H), 3.70 (s, 3H), 2.86-2.77 (m, 4H), 2.02 (s, 3H), 1.35-1.19 (m, 9H)
1.203		^1H NMR (400 MHz, methanol) δ = 7.64-7.61 (m, 2H), 7.14 (t, 2H), 7.05 (s, 1H), 6.96 (s, 1H), 4.07 (s, 2H), 3.83-3.82 (m, 2H), 3.76 (s, 3H), 2.85-2.81 (m, 4H), 2.09-2.08 (m, 3H), 1.90-1.89 (m, 3H)

TABLE 1-continued

COM- POUND	STRUCTURE	NMR
1.204		
1.205		
1.206		

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TABLE 2

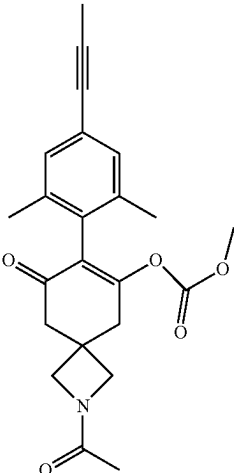
Compound	Structure	NMR
2.001		¹ H NMR (400 MHz, CDCl ₃): δ = 7.08 (s, 2H), 4.01 (d, J = 8.44 Hz, 2H), 3.87-3.99 (m, 2H), 3.71 (s, 3H), 3.07 (s, 2H), 2.87 (d, J = 2.08 Hz, 2H), 2.03 (s, 3H), 1.98 (s, 6H), 1.90 (s, 3H).

TABLE 3

Prior art comparator.	
Compound	Structure
C1 (Compound A-38 from WO2014/096289)	

Biological Examples

Seeds of a variety of test species are sown in standard soil in pots (*Lolium perenne* (LOLPE), *Setaria faberi* (SETFA), *Echinochloa crus-galli* (ECHCG), *Avena fatua* (AVEFA)). After cultivation for one day (pre-emergence) or after 8 days cultivation (post-emergence) under controlled conditions in a glasshouse (at 24/16° C., day/night; 14 hours light; 65% humidity), the plants are sprayed with an aqueous spray solution derived from the formulation of the technical active ingredient in acetone/water (50:50) solution containing 0.5% Tween 20 (polyoxyethelyene sorbitan monolaurate, CAS RN 9005-64-5). Compounds are applied at 250 g/h. The test plants are then grown in a glasshouse under controlled conditions in a glasshouse (at 24/16° C., day/night; 14 hours light; 65% humidity) and watered twice daily. After 13 days for pre and post-emergence, the test is evaluated for the percentage damage caused to the plant. The biological activities are shown in the following table on a five point scale (5=80-100%; 4=60-79%; 3=40-59%; 2=20-39%; 1=0-19%).

TABLE B1

Application pre-emergence				
COMPOUND	SETFA	LOLPE	AVEFA	ECHCG
1.001	5	5	5	2
1.002	5	5	5	5
1.003	5	5	5	5
1.004	5	5	5	5
1.005	5	5	5	5
1.006	5	5	2	5
1.007	5	5	3	5
1.008	5	5	5	5

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TABLE B1-continued

Application pre-emergence				
COMPOUND	SETFA	LOLPE	AVEFA	ECHCG
1.009	5	5	5	5
1.010	5	5	5	5
1.011	3	2	1	2
1.012	5	5	5	5
1.013	5	5	5	5
1.014	5	5	5	5
1.015	5	5	5	5
1.016	5	5	5	5
1.017	5	5	5	5
1.018	5	5	5	5
1.019	4	4	4	5
1.020	5	5	5	5
1.021	5	5	5	5
1.022	5	3	2	4
1.023	3	3	2	1
1.024	5	5	5	5
1.193	5	5	5	5
1.195	5	5	5	5
1.196	5	5	5	5
1.201	5	5	5	5
1.202	5	5	5	5
1.203	5	4	5	5
2.001	5	5	5	5

TABLE B2

Application post-emergence				
COMPOUND	ECHCG	SETFA	LOLPE	AVEFA
1.001	5	5	5	5
1.002	5	5	5	5
1.003	5	5	5	5
1.004	5	5	5	5
1.005	5	5	5	5
1.006	5	5	5	5
1.007	5	4	5	4
1.008	5	5	5	5
1.009	5	5	5	5
1.010	5	5	5	5
1.011	3	5	3	3
1.012	5	5	5	4
1.013	5	5	5	5
1.014	5	5	5	5
1.015	5	5	5	5
1.016	5	5	5	5
1.017	4	5	5	5
1.018	5	5	5	5
1.019	4	5	5	5
1.020	5	5	5	5
1.021	5	5	5	5
1.022	5	5	5	5
1.023	5	5	5	5
1.024	5	5	5	5
1.193	5	5	5	5
1.195	5	5	5	5
1.196	5	5	5	5
1.201	5	5	5	5
1.202	5	5	5	5
1.203	5	5	5	5
2.001	5	5	5	5

TABLE B3

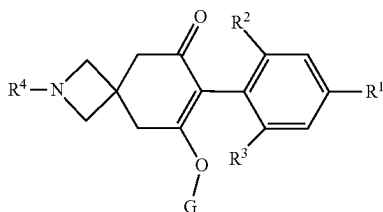
Prior Art Comparison.				
Using procedures outlines above, wheat and barley crop plants and a representative weed species <i>Panicum miliaceum</i> (PANMI) are treated post-emergence with compound 1.001 of the present invention or comparator compound C1 (Compound A-38 from WO2014/096289) at the application rates indicated. The compounds were also applied to wheat in conjunction with the safener compound cloquintocet-mexyl (CQC) at 50 g/ha.				
Compound	Rate	Wheat	Barley	PANMI
C1	60	70	90	100
	125	80	90	100
1.001	60 + CQC	60	NT	NT
	60	0	0	100
	125	0	0	100
	60 + CQC	0	NT	NT

NT = not tested.

These results demonstrate that the compounds of the present invention, exemplified using compound 1.001 exhibit significantly improved crop safety vis-à-vis structurally similar prior art compounds e.g C1 whilst retaining good overall weed control.

The invention claimed is:

1. A compound of Formula (I)



wherein

R¹ is selected from the group consisting of methyl, ethynyl, 1-propynyl, phenyl and a 5 or 6 membered heteroaryl which comprises one or two nitrogen heteroatoms, said phenyl and heteroaryl optionally substituted by one or two R¹⁵ substituents;

R² is selected from the group consisting of methyl, ethyl, methoxy and chloro;

R³ is selected from the group consisting of methyl, ethyl, methoxy and chloro;

R⁴ is selected from the group consisting of C₁-C₄alkyl, C₁-C₄alkoxy-, C₁-C₄haloalkyl, —C(O)C₁-C₄alkyl, —C(O)C₁-C₄haloalkyl, —S(O)_nC₁-C₆alkyl, —S(O)_nC₁-C₆haloalkyl, —S(O)_n—(CH₂)_n—C₃-C₆cycloalkyl, —S(O)_nC(R¹¹)R¹²R¹³, —C(O)H, —C(O)—(CH₂)_n—C₃-C₆cycloalkyl, —C(O)C(R¹¹)R¹²R¹³, —C(O)C₂-C₄alkenyl, —C(O)(CR⁹R¹⁰)CN, —C(O)(CR⁹R¹⁰)(CR⁹R¹⁰)CN, —C(O)CH₂C(O)—C₁-C₆alkyl, —C(O)CH₂OC(O)—C₁-C₆alkyl, —C(O)OC₁-C₆alkyl, —C(O)OC₁-C₆haloalkyl, —C(O)(CH₂)_nS(O)_nC₁-C₆alkyl, —C(O)C₁-C₃alkoxyC₁-C₆alkyl, —C(O)C₁-C₃alkoxyC₂-C₆alkenyl, —C(O)C₁-C₃alkoxyC₂-C₆alkynyl, —C(O)C₁-C₃alkoxyC₁-C₆haloalkyl, —C(O)C₁-C₃alkoxyC₃-C₆cycloalkyl, —C(O)OC₁-C₃alkoxyC₁-C₆alkyl, —C(O)C₁-C₃alkoxyC₁-C₆alkyl, —C(O)(CH₂)_nNR⁵R⁶, —C(O)—(CH₂)_n—NR⁷C(O)R⁸, —C(O)—(CH₂)_n—O—N=CR⁵R⁵, —CN, —S(O)₂NR¹⁶R¹⁷, —S(O)(=NR¹⁸)R¹⁹, —C(O)C(O)R²⁰, —C(O)C(R²³)=N—O—R²⁴, —C(O)C(R²³)=N—NR²⁵R²⁶, —(CH₂)_n-phenyl,

—C(O)—(CH₂)_n-phenyl, —S(O)_n—(CH₂)_n-phenyl, -heterocyclyl, —C(O)—(CH₂)_n-heterocyclyl, —S(O)_n—(CH₂)_n-heterocyclyl, wherein each heterocyclyl is a 5- or 6-membered heterocyclyl which may be aromatic, saturated or partially saturated and can contain from 1 to 4 heteroatoms each independently selected from the group consisting of oxygen, nitrogen and sulphur, and wherein said heterocyclyl or phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R⁵ is selected from the group consisting of hydrogen and C₁-C₆ alkyl;

R⁶ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₂-C₆alkenyl, C₂-C₆alkynyl, C₁-C₆haloalkyl, hydroxyl-, C₁-C₆alkoxy, C₃-C₆cycloalkyl, —C₁-C₄alkoxyC₁-C₆alkyl, —C₁-C₃alkoxyC₁-C₆haloalkyl, —(CR⁹R¹⁰)C₁-C₆haloalkyl, —(CR⁹R¹⁰)C(O)NR⁵R⁵, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro; or

R⁵ and R⁶ together form —CH₂CH₂OCH₂CH₂—; and

R⁷ is selected from the group consisting of hydrogen and C₁-C₆ alkyl;

R⁸ is selected from the group consisting of hydrogen, C₁-C₆ alkyl, C₁-C₆ alkoxy, C₃-C₆ cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro;

R⁹ is hydrogen or methyl;

R¹⁰ is hydrogen or methyl; or

R⁹ and R¹⁰ together form —CH₂CH₂—; and

R¹¹ is hydrogen or methyl;

R¹² is selected from the group consisting of hydrogen, C₁-C₆ alkyl, hydroxyl and C₁-C₆ alkoxy-;

R¹³ is selected from the group consisting of hydrogen, C₁-C₆ alkyl, hydroxyl and C₁-C₆ alkoxy; or

R¹² and R¹³ together form —CH₂—X—CH₂—; and

X is selected from the group consisting of O, S and N—R¹⁴;

R¹⁴ is selected from the group consisting of hydrogen, C₁-C₃ alkyl and C₁-C₃ alkoxy-;

R¹⁵ is independently selected from the group consisting of C₁-C₄ alkyl, C₁-C₄ haloalkyl, cyano and halogen;

R¹⁶ is hydrogen or C₁-C₆alkyl; and

R¹⁷ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₃-C₆cycloalkyl, C₁-C₆ alkoxy-C₁-C₃alkyl-, —C(O)C₁-C₆alkyl, —C(O)OC₁-C₆alkyl and CH₂CN; or

R¹⁶ and R¹⁷ together form —CH₂CH₂OCH₂CH₂—, —CH₂CH₂S(O)₂CH₂CH₂—;

R¹⁸ is hydrogen or C₁-C₆alkyl;

R¹⁹ is selected from the group consisting of hydrogen, C₁-C₆ alkyl, C₁-C₆ alkoxy, C₃-C₆cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro;

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R²⁰ is selected from the group consisting of C₁-C₆alkyl, C₁-C₆haloalkyl, C₁-C₆alkoxy-, C₁-C₆haloalkoxy, —NR²¹R²², phenyl and -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro;

R²¹ is selected from the group consisting of hydrogen, C₁-C₆ alkyl, C₁-C₆ alkoxy, C₁-C₆alkoxyC₁-C₃alkyl-, C₃-C₆ cycloalkyl, C₁-C₆haloalkyl- and C₁-C₆haloalkoxy-, —C(O)C₁-C₆alkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro;

R²² is hydrogen or C₁-C₆alkyl; or

R²¹ and R²² together form —CH₂CH₂OCH₂CH₂—;

R²³ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₁-C₆haloalkyl, C₁-C₆alkoxy- and C₁-C₆haloalkoxy-;

R²⁴ is selected from the group consisting of hydrogen, C₁-C₆alkyl, C₁-C₆alkoxyC₁-C₃alkyl-, C₃-C₆ cycloalkyl, —CH₂CN, tetrahydropyranyl-, phenyl and -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R²⁵ is hydrogen or C₁-C₆ alkyl;

R²⁶ is hydrogen or C₁-C₆ alkyl;

G is selected from the group consisting of hydrogen, —(CH₂)_n—R^a, —C(O)—R^a, —C(O)—(CR^cR^d)_n—O—R^b, —C(O)NR^aR^a, —S(O)₂—R^a and C₁-C₈alkoxy-C₁-C₃alkyl-;

R^a is independently selected from the group consisting of hydrogen, C₁-C₈alkyl, C₁-C₃haloalkyl, C₂-C₈alkenyl, C₂-C₈alkynyl, C₃-C₆cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R^b is selected from the group consisting of C₁-C₈alkyl, C₁-C₃haloalkyl, C₂-C₈alkenyl, C₂-C₈alkynyl, C₃-C₆ cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R^c is hydrogen or C₁-C₃ alkyl;

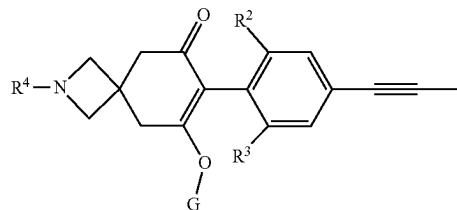
R^d is hydrogen or C₁-C₃ alkyl; and

n is independently 0, 1 or 2;

or an agriculturally acceptable salt thereof.

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2. The compound according to claim 1, which is a compound of Formula (Ia)



wherein

R² is methyl or methoxy;

R³ is methyl or methoxy;

R⁴ is selected from the group consisting of C₁-C₄alkyl, C₁-C₄alkoxy-, C₁-C₄haloalkyl, —C(O)C₁-C₄alkyl, —C(O)C₁-C₄haloalkyl, —S(O)_nC₁-C₆alkyl, —S(O)_nC₁-C₆haloalkyl, —S(O)_n—(CH₂)_n—C₃-C₆cycloalkyl, —S(O)_nC(R¹¹)R¹²R¹³, —C(O)H, —C(O)—(CH₂)_n—C₃-C₆cycloalkyl, —C(O)C(R¹¹)R¹²R¹³, —C(O)C₂-C₄alkenyl, —C(O)(CR⁹R¹⁰)CN, —C(O)OC₁-C₆alkyl, —C(O)OC₁-C₆haloalkyl, —C(O)(CH₂)_nS(O)_nC₁-C₆alkyl, —C(O)C₁-C₃alkoxyC₁-C₆alkyl, —C(O)NR⁵R⁶, —C(O)—(CH₂)_n—NR⁷C(O)R⁸, —CN, —(CH₂)_n-phenyl, —C(O)—(CH₂)_n-phenyl, —S(O)_n—(CH₂)_n-phenyl, -heterocyclyl, —C(O)—(CH₂)_n-heterocyclyl, —S(O)_n—(CH₂)_n-heterocyclyl, wherein each heterocyclyl is a 5- or 6-membered heterocyclyl which may be aromatic, saturated or partially saturated and can contain from 1 to 4 heteroatoms each independently selected from the group consisting of oxygen, nitrogen and sulphur, and wherein said heterocyclyl or phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃alkyl, C₁-C₃haloalkyl, C₁-C₃alkoxy, C₂-C₃alkenyl, C₂-C₃alkynyl, halogen, cyano and nitro;

R⁵ is selected from the group consisting of hydrogen and C₁-C₆ alkyl;

R⁶ is selected from the group consisting of hydrogen, C₁-C₆ alkyl, C₁-C₆ alkoxy, C₃-C₆ cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro; or

R⁵ and R⁶ together form —CH₂CH₂OCH₂CH₂—; and

R⁷ is selected from the group consisting of hydrogen and C₁-C₆ alkyl;

R⁸ is selected from the group consisting of hydrogen, C₁-C₆ alkyl, C₁-C₆ alkoxy, C₃-C₆ cycloalkyl, phenyl, -pyridyl, wherein the phenyl and pyridyl are optionally substituted by one, two or three substituents independently selected from the group consisting of C₁-C₃ alkyl, C₁-C₃ haloalkyl, C₁-C₃ alkoxy, C₂-C₃ alkenyl, C₂-C₃ alkynyl, halogen, cyano and nitro;

R⁹ is hydrogen or methyl;

R¹⁰ is hydrogen or methyl; or

R⁹ and R¹⁰ together form —CH₂CH₂—; and

R¹¹ is hydrogen or methyl;

R¹² and R¹³ together form —CH₂—X—CH₂—;

X is selected from the group consisting of O, S and N—R¹⁴;

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R^{14} is selected from the group consisting of hydrogen, C_1 - C_3 alkyl and C_1 - C_3 alkoxy-;

G is selected from the group consisting of hydrogen, $-(CH_2)_n-R^a$, $-C(O)-R^a$, $-C(O)-(CR^cR^d)_n-$, $O-R^b$, $-C(O)NR^aR^a$, $-S(O)_2-R^a$ and C_1 - C_8 alkoxy- C_1 - C_3 alkyl-;

R^a is independently selected from the group consisting of hydrogen, C_1 - C_8 alkyl, C_1 - C_3 haloalkyl, C_2 - C_8 alkenyl, C_2 - C_8 alkynyl, C_3 - C_6 cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C_1 - C_3 alkyl, C_1 - C_3 haloalkyl, C_1 - C_3 alkoxy, C_2 - C_3 alkenyl, C_2 - C_3 alkynyl, halogen, cyano and nitro;

R^b is selected from the group consisting of C_1 - C_8 alkyl, C_1 - C_3 haloalkyl, C_2 - C_8 alkenyl, C_2 - C_8 alkynyl, C_3 - C_6 cycloalkyl, heterocyclyl and phenyl wherein said heterocyclyl and phenyl groups are optionally substituted by one, two or three substituents independently selected from the group consisting of C_1 - C_3 alkyl, C_1 - C_3 haloalkyl, C_1 - C_3 alkoxy, C_2 - C_3 alkenyl, C_2 - C_3 alkynyl, halogen, cyano and nitro;

R^c is hydrogen or C_1 - C_3 alkyl;

R^d is hydrogen or C_1 - C_3 alkyl; and

n is independently 0, 1 or 2;

or an agriculturally acceptable salt thereof.

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3. The compound according to claim 1, wherein R^2 is methyl.

4. The compound according to claim 1, wherein R^3 is methyl.

5. The compound according to claim 1, wherein R^3 is methoxy.

6. The compound according to claim 1, wherein R^4 is $-C(O)OC_1$ - C_6 alkyl.

7. The compound according to claim 1, wherein R^4 is $-C(O)NR^5R^6$.

8. The compound according to claim 1, wherein G is hydrogen.

9. The compound according to claim 1, wherein G is $-C(O)C_1$ - C_6 alkyl.

10. The compound according to claim 1, wherein G is $-C(O)-O-C_1$ - C_6 alkyl.

11. A herbicidal composition comprising a compound of Formula (I) according to claim 1 and an agriculturally acceptable formulation adjuvant.

12. The herbicidal composition according to claim 11, further comprising at least one pesticide.

13. The herbicidal composition according to claim 12, wherein the pesticide is an herbicide or herbicide safener.

14. A method of controlling weeds at a locus comprising application to the locus of a weed controlling amount of a composition according to claim 11.

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